Lecture Notes on

STOCHASTIC PROCESSES (F79SP)

Senior Lecturer:
Prof. Sergey Foss (Edinburgh)

Lecturers:
Dr. Kai Lin Ong (Malaysia)
Mr. Dominic Miller (Dubai)

$*$

1 Introduction I

A **stochastic process** is a collection of random variables ordered by time.
Time could be **discrete**: $\{X_n\}_{n=0}^{\infty}$.
Time could be **continuous**: $\{X(t)\}_{t\geq 0}$.
In general, $t = 0, n = 0$ means **now**, and $t > 0, n > 0$ are **times in the future**.
At each future time $X(t)$ (or X_n) is a random variable with some distribution.

1.1 Examples of SP's in actuarial work

Example 1.
Let X_0 be a current surplus of an unsurer (known).
Let P be a premium income collected every year in the future (assumed fixed).
Let Z_n be a claims total paid in year $(n-1, n]$.
Define $Y_n = P - Z_n \equiv$ *profit in year* n and $X_n = X_{n-1} + Y_n \equiv$ *surplus at end of year* n.
We model Z_n (and, therefore, Y_n and X_n) as real-valued random variables. Thus, $\{X_n\}_{n=0}^{\infty}$ (and $\{Y_n\}_{n=1}^{\infty}$, $\{Z_n\}_{n=1}^{\infty}$) is a SP.
Comments:
(i) Note the simplifications in the model.
(ii) an interesting problem is:
Calculate the probability of **ruin**

$$\mathbf{P}(X_n < 0 \quad \text{for some} \quad n \geq 1).$$

(iii) We could assume $\{Z_n\}_{n=1}^{\infty}$ is a sequence of independent random variables. Then $\{X_n\}_{n=0}^{\infty}$ is a **random walk** (see next Section).
(iv) Further, if each Z_n has a discrete (or continuous) distribution, then each X_n has also a discrete (or continuous) distribution.

Example 2. Consider an insurance policy issued to a life aged 30 and ceasing at age 65 (or on earlier death).
The policyholder pays premiums continuously at the rate of P p.a. to the insurance company while (s)he is healthy.
The insurance company pays benefit continuously at the rate B p.a. to the policyholder while (s)he is sick.
Let $X(t)$ denote the state of the policyholder at age t. Then $X(t)$ may take one of three values: H(ealthy), S(ick), and D(ead). Each $X(t)$ has a discrete distribution (modelled as a random variable).
Thus, $\{X(t)\}_{30\leq t\leq 65}$ is a continuous time stochastic process, each $X(t)$ having a discrete distribution.
Interesting problem: Use data to estimate the distributions of $X(t)$. Then use these distributions to calculate the value of P for a given benefit level B.

Example 3.
Let I_n be an inflation rate in year $(n-1, n]$, $n \geq 0$.
Let P_n be a price at time n of portfolio of equities.
Let C_n be a yield in year $(n-1, n]$ on long-dated government bonds.
Model I_n, P_n, C_n as real valued random variables. Then we get a discrete time, continuous state space stochastic process. Here I_n depends on $(I_0, \ldots, I_{n-2}, I_{n-1})$;
P_n depends on $(P_0, P_1, \ldots, P_{n-1})$ and $(I_0, \ldots, I_{n-2}, I_{n-1})$;
C_n depends on $(C_0, C_1, \ldots, C_{n-1})$, $(P_0, P_1, \ldots, P_{n-1})$, and $(I_0, \ldots, I_{n-2}, I_{n-1})$.
This is **the Wilkie Model** for investments (see in Transactions of the Faculty of Actiaries, Vol 39, 1986).

2 Introduction II: Basic facts from probability theory

2.1 Discrete probability: rapid revision

2.1.1 Definition and basic properties

A random variable X has a discrete distribution if there is a finite or countable set $\{x_i\}$ such that

$$\mathbf{P}(X = x_i) > 0, \quad \text{for all} \quad i \quad \text{and} \quad \sum_i \mathbf{P}(X = x_i) = 1.$$

For a discrete r.v. X and for any function g, if $\mathbf{E}|g(X)|$ is finite, then

$$\mathbf{E}g(X) = \sum_i g(x_i)\mathbf{P}(X = x_i).$$

In particular,

$$\mathbf{E}X = \sum_i x_i \mathbf{P}(X = x_i) \quad \text{and} \quad \mathbf{E}(X^2) = \sum_i x_i^2 \mathbf{P}(X = x_i).$$

Then

$$\mathbf{V}arX = \mathbf{E}(X - \mathbf{E}X)^2 = \mathbf{E}(X^2) - (\mathbf{E}X)^2 = \sum_i x_i^2 \mathbf{P}(X = x_i) - \left(\sum_i x_i \mathbf{P}(X = x_i)\right)^2.$$

2.1.2 Standard Examples of Discrete Random Variables.

Bernoulli distribution. Let $0 < p < 1$. Random variable X has a *Bernoulli distribution with parameter* p if it may take only two values, 0 and 1, with probabilities $\mathbf{P}(X = 1) = p$ and $\mathbf{P}(X = 0) = 1 - p$. Then

$$\mathbf{E}X = p \cdot 1 + (1 - p) \cdot 0 = p; \quad \mathbf{E}(X^2) \quad = p \cdot 1^2 + (1 - p) \cdot 0^2 = p; \quad \mathbf{V}arX = p - p^2 = p(1 - p).$$

Binomial distribution. Let $0 < p < 1$ and $n = 1, 2, \ldots$. Random variable X has a *binomial distribution with parameters* p and n if

$$\mathbf{P}(X = k) = \frac{n!}{k!(n - k)!} p^k (1 - p)^{n-k}, \quad \text{for} \quad k = 0, 1, \ldots, n. \tag{1}$$

Here $0! = 1! = 1$ and, for $k = 2, 3, \ldots$,

$$k! = 1 \cdot 2 \cdot \ldots \cdot k.$$

Then

$$\mathbf{E}X = np \quad \text{and} \quad \mathbf{V}arX = npq.$$

Geometric distribution. Let $0 < p < 1$. Random variable X has a *geometric distribution with parameter* p if

$$\mathbf{P}(X = k) = p(1 - p)^{k-1}, \quad \text{for} \quad k = 1, 2, \ldots.$$

Then

$$\mathbf{E}X = 1/p \quad \text{and} \quad \mathbf{V}arX = (1 - p)/p^2.$$

Poisson distribution. Let $\lambda > 0$. Random variable ξ has a *Poisson distribution with parameter* λ and write $\xi \sim \Pi(\lambda)$ if ξ takes a non-negative integer value k with probability

$$\mathbf{P}(\xi = k) = \frac{\lambda^k}{k!} e^{-\lambda}.$$

Lemma 2.1.1 Mean and variance of a Poisson r.v. with parameter λ are equal to λ.

Proof: We need to recall the Taylor expansion for exponential function: for any number x,

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!} \equiv 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \dots$$

Then

$$\mathbf{E}\xi = \sum_{k=0}^{\infty} k \cdot \frac{\lambda^k}{k!} e^{-\lambda} = \sum_{k=1}^{\infty} k \cdot \frac{\lambda^k}{k!} e^{-\lambda} = \sum_{k=1}^{\infty} \frac{\lambda^k}{(k-1)!} e^{-\lambda}$$
$$= \sum_{l=0}^{\infty} \frac{\lambda^{l+1}}{l!} e^{-\lambda} = \lambda e^{-\lambda} \sum_{l=0}^{\infty} \frac{\lambda^l}{l!} = \lambda e^{\lambda} e^{-\lambda} = \lambda.$$

Similarly,

$$\mathbf{E}\left(\xi^2\right) = \sum_{k=0}^{\infty} k^2 \cdot \frac{\lambda^k}{k!} e^{-\lambda} = \sum_{k=1}^{\infty} k^2 \cdot \frac{\lambda^k}{k!} e^{-\lambda} = \sum_{k=1}^{\infty} k \cdot \frac{\lambda^k}{(k-1)!} e^{-\lambda} = \sum_{l=0}^{\infty} (l+1) \cdot \frac{\lambda^{l+1}}{l!} e^{-\lambda}$$
$$= \lambda e^{-\lambda} \left(\sum_{l=0}^{\infty} l \cdot \frac{\lambda^l}{l!} + \sum_{l=0}^{\infty} \frac{\lambda^l}{l!} \right) = \lambda \mathbf{E}\xi + \lambda e^{\lambda} e^{-\lambda} = \lambda^2 + \lambda.$$

Then

$$\mathbf{Var}(\xi) = \mathbf{E}\left(\xi^2\right) - (\mathbf{E}\xi)^2 = \lambda^2 + \lambda - \lambda^2 = \lambda.$$

□

2.2 Continuous random variables: rapid revision

2.2.1 Definitions and basic properties

A random variable X has a *continuous* distribution if there exists a function $f(x) \equiv f_X(x)$ such that, for any set A,

$$\mathbf{P}(X \in A) = \int_A f(x)dx$$

Then a function f is a *density function* for X.

Properties of density. there are two properties:
(1) $f(x) \geq 0$ for all x;
(ii) $\int_{-\infty}^{\infty} f(x)dx = 1$.

Comment. If an r.v. X has a continuous distribution, then $\mathbf{P}(X = x) = 0$ for all x !

Distribution function. A function

$$F(x) \equiv F_X(x) = \mathbf{P}(X \leq x), \quad -\infty < x < \infty$$

is a *distribution function* of an r.v. X. A distribution function of a continuous r.v. has the following properties:
(1) it is non-decreasing: if $x_1 < x_2$, then $F(x_1) \leq F(x_2)$;
(2) $0 \leq F(x) \leq 1$ for all x;
(3) $\lim_{x \downarrow -\infty} F(x) = 0$ and $\lim_{x \uparrow \infty} F(x) = 1$;
(4) F is *continuous*, i.e., for any x,

$$\lim_{y \to x} F(y) = F(x).$$

Relations between distribution and density functions.
(1) given a density f, a distribution function is defined as

$$F(x) = \int_{-\infty}^{x} f(t)dt.$$

(2) given a distribution function F which is differentiable at 'almost all" points,

$$f(x) = F^{'}(x).$$

(More comments are required!)

Expectation and variance.
(1) $\mathbf{E}X = \int_{\infty}^{\infty} xf(x)dx$;
(2) more generally, for any function g,

$$\mathbf{E}g(X) = \int_{\infty}^{\infty} g(x)f(x)dx;$$

(3) $\mathbf{Var}X = \mathbf{E}\left(X - \mathbf{E}X\right)^2 = \mathbf{E}X^2 - \left(\mathbf{E}X\right)^2 = \int_{\infty}^{\infty} x^2 f(x)dx - \left(\int_{\infty}^{\infty} xf(x)dx\right)^2$.

2.2.2 Standard examples of continuous distributions

Uniform distribution. Let $a < b$ be two numbers. An r.v. X has a *uniform* distribution in the interval (a, b), $X \sim U(a, b)$, if

$$f(x) = \frac{1}{b-a} \quad \text{for} \quad x \in (a, b) \quad \text{and} \quad f(x) = 0 \quad \text{otherwise.}$$

A *standard* uniform distribution is $U(0, 1)$.
If $X \sim U(a, b)$, then

$$F(x) = \frac{x-a}{b-a} \quad \text{for} \quad x \in (a, b); \quad F(x) = 0 \quad \text{for} \quad x \le a; \quad \text{and} F(x) = 1 \quad \text{for} \quad x \ge b;$$

$$\mathbf{E}X = \int_a^b x\frac{1}{b-a}dx = \frac{a+b}{2},$$

$$\mathbf{E}(X^2) = \int_a^b x^2\frac{1}{b-a}dx = \frac{a^2+ab+b^2}{3},$$

and

$$\mathbf{Var}X = \frac{(b-a)^2}{12}$$

Exponential distribution. Let $\alpha > 0$. An r.v. X has an *exponential* distribution with parameter α, $X \sim Exp(\alpha)$, if

$$f(x) = \alpha e^{-\alpha x} \quad \text{for} \quad x > 0 \quad \text{and} \quad f(x) = 0 \quad \text{otherwise.}$$

A *standard* exponential distribution is $Exp(1)$.
If $X \sim Exp(\alpha)$, then

$$F(x) = 1 - e^{-\alpha x} \quad \text{for} \quad x > 0 \quad \text{and} \quad F(x) = 0 \quad \text{otherwise;}$$

$$\mathbf{E}X = \int_0^{\infty} x\alpha e^{-\alpha x}dx = \frac{1}{\alpha},$$

$$\mathbf{E}(X^2) = \int_0^{\infty} x^2\alpha e^{-\alpha x}dx = \frac{2}{\alpha^2},$$

and

$$\mathbf{Var}X = \frac{1}{\alpha^2}.$$

Normal distribution. Let c and $\sigma > 0$ be two numbers. An r.v. X has a *normal* distribution with parameters c and σ^2, $X \sim N(c, \sigma^2)$, if, for any x,

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-c)^2}{2\sigma^2}}.$$

A *standard* normal distribution is $N(0,1)$. Its density is denoted by $\varphi(x)$ and distribution function by $\Phi(x)$.

If $X \sim N(c, \sigma^2))$, then

$$F(x) = \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{x} e^{-\frac{(t-c)^2}{2\sigma^2}} dt;$$

$$\mathbf{E}X = c,$$

$$\mathbf{E}(X^2) = c^2 + \sigma^2,$$

and

$$\mathbf{Var}X = \sigma^2.$$

2.2.3 Multivariate distributions.

We consider joint distributions two r.v.'s. A general case is treated similarly.

Two r.v.'s X and Y have a *joint continuous* distribution if there exists a function $f_{X,Y}(x,y) \equiv f(x,y)$ of two variables x and y such that, for any two-dimensional set A,

$$\mathbf{P}((X,Y) \in A) = \left(\int \int \right)_A f(x,y) dx dy.$$

Then $f(x,y)$ is a *joint density* function of random variables X and Y, or a *density* function of a random *vector* (X,Y).

For instance, for any u and v,

$$\mathbf{P}(X \le u; Y \le v) = \int_{-\infty}^{u} \left(\int_{-\infty}^{v} f(x,y) dy \right) dx$$

and

$$\mathbf{P}(X \ge u; Y \ge v) = \int_{u}^{\infty} \left(\int_{v}^{\infty} f(x,y) dy \right) dx.$$

Properties of joint densities. There are two properties:

(1) $f(x,y) \ge 0$;

(2) $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x,y) dx dy = 1$.

Joint distribution function. A function

$$F(x,y) \equiv F_{X,Y}(x,y) = \mathbf{P}(X \le x; Y \le y), \quad -\infty < x, y < \infty$$

is a *joint distribution function* of r.v.'s X and Y. A joint distribution function of continuous r.v.'s has properties which are similar to that of a single r.v.'s: monotonicity, continuity, and limits are either 0 or 1.

Relations between distribution and density functions.

(1) given a density $f(x,y)$, a distribution function is defined as

$$F(x,y) = \int_{-\infty}^{x} \int_{-\infty}^{y} f(u,v) du dv.$$

(2) given a distribution function F which is differentiable at 'almost all" points,

$$f(x,y) = \frac{\partial^2}{\partial x \partial y} F(x,y).$$

(More comments are required!)

Given a joint density, find marginals. Given a joint density $f(x,y)$ of two r.v.'s X and Y, *marginal* densities of X and of Y are

$$f_X(x) = \int_{-\infty}^{\infty} f(x,y)dy \quad \text{and} \quad f_Y(y) = \int_{-\infty}^{\infty} f(x,y)dx.$$

2.3 Independence

Two random variables X and Y are independent, if, for any one-dimensional sets A and B,

$$\mathbf{P}(X \in A; Y \in B) = \mathbf{P}(X \in A) \cdot \mathbf{P}(Y \in B).$$

Theorem 2.3.1 The following are equivalent:
(1) two r.v.'s X and Y are independent;
(2) for any x and y, $F_{X,Y}(x,y) = F_X(x) \cdot F_Y(y)$.
For continuous random variables, any of (1) or (2) is equivalent to
(3) for any x and y, $f_{X,Y}(x,y) = f_X(x) \cdot f_Y(y)$.

2.4 Main properties of expectation and variance.

(1) If X is a constant (i.e. $\mathbf{P}(X = c) = 1$), then $\mathbf{E}X = c$. In short: $\mathbf{E}c = c$.
(2) $\mathbf{E}(X + Y) = \mathbf{E}X + \mathbf{E}Y$;
(3) for any constant c, $\mathbf{E}(cX) = c\mathbf{E}X$;
(4) if X and Y are independent, then $\mathbf{E}(XY) = \mathbf{E}X \cdot \mathbf{E}Y$.
(5) $\mathbf{Var}c = 0$;
(6) $\mathbf{Var}(cX) = c^2\mathbf{Var}X$;
(7) if X and Y are independent, then $\mathbf{Var}(X + Y) = \mathbf{Var}(X - Y) = \mathbf{Var}X + \mathbf{Var}Y$.

2.5 Limit Theorems

2.5.1 Law of Large Numbers (LLN)

Theorem 2.5.1 Let $X_1, X_2, \ldots$ be a sequence of i.i.d. r.v.'s with a finite mean $\mu = \mathbf{E}X_1$ and a finite variance $\sigma^2 = \mathbf{Var}X_1$. Let $S_n = X_1 + X_2 + \ldots + X_n$. Then, for any $a > 0$, as $n \to \infty$,

$$\mathbf{P}\left(\left|\frac{S_n}{n} - \mu\right| > a\right) \to 0.$$

Short notation: $S_n/n \to_p \mu$, or $S_n/n \to \mu$ in probability.

Proof: Denote $Y = \frac{S_n}{n} - \mu$. Then

$$\mathbf{E}Y = \frac{\mathbf{E}S_n}{n} - \mu = \frac{\mathbf{E}X_1 + \ldots + \mathbf{E}X_n}{n} - \mu = \mu - \mu = 0$$

and, therefore,

$$\mathbf{E}\left(Y^2\right) = \mathbf{Var}Y = \mathbf{Var}\left(\frac{S_n}{n}\right) = \frac{\mathbf{Var}S_n}{n^2} = \frac{n\mathbf{Var}X_1}{n^2} = \frac{\sigma^2}{n}.$$

From the Chebyshev inequality,

$$\mathbf{P}\left(\left|\frac{S_n}{n} - \mu\right| > a\right) = \mathbf{P}(|Y| > a) \le \frac{\mathbf{E}\left(Y^2\right)}{a^2} = \frac{\sigma^2}{a^2 n} \to 0$$

as $n \to \infty$. □

2.5.2 Strong Law of Large Numbers (SLLN)

A more general statement is also valid:

Theorem 2.5.2 Let $X_1, X_2, \ldots$ be a sequence of i.i.d. r.v.'s with a finite mean $\mu = \mathbf{E}X_1$. Let $S_n = X_1 + X_2 + \ldots + X_n$. Then, as $n \to \infty$,

$$\mathbf{P}\left(\left|\frac{S_n}{n} - \mu\right| \to 0\right) = 1.$$

Short notation: $S_n/n \to_{a.s.} \mu$, or $S_n/n \to \mu$ a.s. (almost surely).

Proof: Omitted. □

General comment. Both LLN and SLLN say that "mean in time converges to mean in space". Indeed, assume that X_i is an outcome of an experiment at time i. Then $\frac{S_n}{n}$ is an average, or a mean of all experiments results.

2.5.3 Central Limit Theorem (CLT)

Recall that $\varphi(x)$ is a standard normal density,

$$\varphi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2},$$

and $\Phi(x)$ is a distribution function of a standard normal random variable,

$$\Phi(x) = \int_{-\infty}^{x} \varphi(t)dt = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} e^{-t^2/2} dt.$$

If $Y \sim N(0,1)$ (i.e. a r.v. Y has a standard normal distribution), then, for any $-\infty \le a < b \le \infty$,

$$\mathbf{P}(a \le Y \le b) = \mathbf{P}(a \le Y < b) = \mathbf{P}(a < Y \le b) = \mathbf{P}(a < Y < b)$$

$$= \int_a^b \varphi(t)dt = \frac{1}{\sqrt{2\pi}} \int_a^b e^{-t^2} dt = \Phi(b) - \Phi(a).$$

Here is the Central Limit Theorem.

Theorem 2.5.3 Let $X_1, X_2, \ldots$ be a sequence of i.i.d. r.v.'s with a finite mean $\mu = \mathbf{E}X_1$ and a finite variance $\sigma^2 = \mathbf{Var}X_1$. Let $S_n = X_1 + X_2 + \ldots + X_n$. Let Y be a standard normal random variable. Then, as $n \to \infty$

$$\frac{S_n - n\mu}{\sqrt{n\sigma^2}} \to Y$$

in distribution. More precisely, for any $-\infty \le a < b \le \infty$, as $n \to \infty$,

$$\mathbf{P}\left(a \le \frac{S_n - n\mu}{\sqrt{n\sigma^2}} \le b\right) \to \Phi(b) - \Phi(a).$$

General Comments.
Comment 1. CLT has also another name: an "invariance principle" (the most known "invariance principle" you've (possibly) met in your life is: "all roads lead to Rome"). If we are interested in a distribution of accumulative sum S_n for large n, we have to know only μ and σ since a distribution of of S_n is approximately normal with parameters $n\mu$

and $n\sigma^2$.

Comment 2. CLT is an extension of LLN (but not of SLLN). Here is a rough explanation of that.

LLN says that $\frac{S_n}{n} \to \mu$, or $\frac{S_n}{n} - \mu \to 0$, or

$$\frac{S_n - n\mu}{n} \to 0.$$

CLT says that

$$\frac{S_n - n\mu}{n^{1/2}\sigma} \to Y$$

Therefore, for any power $d > 1/2$,

$$\frac{S_n - n\mu}{n^d} = \frac{S_n - n\mu}{n^{1/2}\sigma} \cdot \frac{\sigma}{n^{d-1/2}} \approx Y \cdot \frac{\sigma}{n^{d-1/2}} \to 0.$$

In particular, for $d = 1$ we get LLN.

3 Random walks

3.1 Introduction and general properties

Definition. Let a be a constant and let $\xi_1.\xi_2, \ldots$ be a sequence of independent identically distributed random variables. Define the stochastic process $\{S_n\}_{n\geq 0}$ by

$$S_0 = a, \qquad S_n = a + \sum_{i=1}^{n} \xi_i, \quad n \geq 1.$$

Then $\{S_n\}_{n\geq 0}$ is a **random walk**, and this is a stochastic process in discrete time $n = 0, 1, 2, \ldots$ with independent and identically distributed (i.i.d.) increments.

Note. The random walk above starts at $S_0 = a$. It is easy to see that it is in fact sufficient to understand the behaviour of random walks which start at $S_0 = 0$ (*why?*).

Motivation Insurance, finance, the ruin problem. Also many non-financial applications.

3.1.1 General properties – repetition!

Let

$$\mu = \mathbf{E}\,\xi_i \qquad \textbf{mean of increments } \xi_i$$
$$\sigma^2 = \mathbf{Var}\xi_i \qquad \textbf{variance of increments } \xi_i.$$

We shall assume that both the above quantities exist and are finite.

1. We have

$$\begin{aligned} \mathbf{E}\,S_n &= a + \sum_{i=1}^{n} \mathbf{E}\,\xi_i \\ &= a + n\mu, \\ \mathbf{Var}S_n &= \sum_{i=1}^{n} \mathbf{Var}\xi_i \qquad (\text{since } \xi_1, \xi_2, \ldots \text{ independent}) \\ &= n\sigma^2. \end{aligned}$$

   Further, by the Central Limit Theorem (CLT), for moderate to large n, S_n is approximately normally distributed with mean and variance as given above, i.e.

$$S_n \sim N(a + n\mu,\, n\sigma^2) \qquad \text{approximately.}$$

2. Consider now the "average" process $\{S_n/n\}_{n\geq 0}$, and assume, for simplicity, that $a = 0$, i.e. $S_n = \sum_{i=1}^{n} \xi_i$. Note that

$$\mathbf{E}\left(\frac{S_n}{n}\right) = \frac{n\mu}{n} = \mu \qquad \text{for all } n$$
$$\mathbf{Var}\left(\frac{S_n}{n}\right) = \frac{\mathbf{Var}S_n}{n^2} = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n} \qquad \to 0 \quad \text{as } n \to \infty.$$

We also have, from Chebyshev's Inequality, that, for any $a > 0$,

$$\begin{aligned}\mathbf{P}\left(\left|\frac{S_n}{n} - \mu\right| \geq a\right) &\leq \frac{\mathbf{Var}(S_n/n)}{a^2} \\ &= \frac{\sigma^2}{na^2} \qquad \to 0 \quad \text{as } n \to \infty,\end{aligned}$$

i.e. as $n \to \infty$, the process $\{S_n/n\}_{n\geq 1}$ tends to cluster ever closer around the constant μ.

Indeed the Strong Law of Large Numbers (SLLN) states that

$$\frac{S_n}{n} = \frac{1}{n}\sum_{i=1}^{n} \xi_i \to \mu \quad \text{almost surely}$$

as $n \to \infty$, i.e.

$$\mathbf{P}\left(\frac{S_n}{n} \to \mu\right) = 1.$$

We say that the process $\{S_n\}$ has **drift** μ.

3.1.2 The ruin problem

Suppose now that $a > 0$. We shall say that the event R, or **ruin** occurs if there exists some $n > 0$ such that $S_n \leq 0$.

Define also the random variable

$$T_R = \begin{cases} \text{the first } \ n > 0 \text{ such that } S_n \leq 0 \\ \infty, \quad \text{if } S_n > 0 \text{ for all } n, \end{cases}$$

i.e. T_R is the **time to ruin**.

We are interested in $\mathbf{P}(R)$ (the **probability of ruin**) and in the distribution of T_R.

Notes

1. We might also be interested in the probability that ruin occurs by some fixed time N, i.e.
$$\mathbf{P}(\text{there exists } n \leq N \text{ with } S_n \leq 0) \quad [\equiv \mathbf{P}(T_R \leq N)] \,.$$
This can be difficult.

2. Many authors require $S_n < 0$ (instead of $S_n \leq 0$) for ruin.

Consider the 3 cases: $\mu < 0$, $\mu = 0$ and $\mu > 0$.

1. $\mu < 0$ (process drifts downwards). Then
$$\mathbf{P}(R) = 1 \qquad \text{(ruin certain)}.$$
This is immediate from the SLLN (see above). For a large, we can also use the CLT to estimate the probability of ruin by time N: if Φ is the (cumulative) distribution function of the $N(0,\,1)$ distribution, then
$$\begin{aligned} \mathbf{P}(\text{ruin by time } N) &\geq \mathbf{P}(S_N \leq 0) \\ &= \mathbf{P}\left(\frac{S_N - (a + N\mu)}{\sigma\sqrt{N}} \leq -\frac{a + N\mu}{\sigma\sqrt{N}}\right) \\ &\approx \Phi\left(-\frac{a + N\mu}{\sigma\sqrt{N}}\right) \\ &\to 1 \quad \text{as } N \to \infty, \ \text{ since } \mu < 0. \end{aligned}$$
(In fact when a is moderate to large, the above approximation is good—despite the inequality.)

One can show that we have $\mathbf{E}\,T_R < \infty$ in this case. (Intuitively, for large a, $T_R \approx a/(-\mu)$.)

2. $\mu = 0$ (process has zero drift). Then one can show that
$$\begin{aligned} &\mathbf{P}(R) = 1 \qquad \text{(ruin again certain)} \\ &\mathbf{E}\,T_R = \infty \qquad \text{(expected time to ruin is nevertheless infinite)}. \end{aligned}$$

3. $\mu > 0$ (process drifts upwards). Then
$$\mathbf{P}(R) < 1 \qquad \text{(ruin not certain)}.$$

In fact it can be shown that, for "most" distributions of ξ_i, the probability of ruin declines exponentially as the starting point a of the random walk increases—indeed there exists a constant $\gamma > 0$ such that, for all a,

$$\mathbf{P}(R) < e^{-\gamma a}.$$

Note that, in this case,

$$\begin{aligned} \mathbf{P}(T_R = \infty) &= \mathbf{P}(\text{never ruined}) \\ &= 1 - \mathbf{P}(R) \\ &> 0, \end{aligned}$$

and so $\mathbf{E}\, T_R = \infty$.

Let us summarise
Conclusions on $\mathbf{P}(R)$ and $\mathbf{E}\, T_R$ for a general random walk:

if $\mu < 0$, then $\mathbf{P}(R) = 1$ and $\mathbf{E}\, T_R < \infty$;
if $\mu = 0$, then $\mathbf{P}(R) = 1$ and $\mathbf{E}\, T_R = \infty$;
if $\mu > 0$, then $\mathbf{P}(R) < 1$ and $\mathbf{E}\, T_R = \infty$.

We turn now to an important particular example of random walks – the *simple random walk* – and provide in this case a detailed analysis (with proofs!) of general conclusions we've made here.

3.2 The simple random walk

Definition. The **simple random walk** $\{S_n\}_{n\geq 0}$ is defined on the integer state space $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ by $S_0 = a$ (where a in an integer) and, for $n \geq 1$, $S_n = a + \sum_{i=1}^{n} \xi_i$, where $\xi_1, \xi_2 \dots$ are independent identically distributed random variables with

$$\begin{aligned} \mathbf{P}(\xi_i = 1) &= p \qquad\qquad (0 < p < 1) \\ \mathbf{P}(\xi_i = -1) &= q = 1 - p \\ (\Rightarrow \mathbf{P}(\xi_i = k) &= 0 \quad \text{for all other integers } k) \end{aligned}$$

Then

$$\begin{aligned} \mu &\equiv \mathbf{E}\,\xi_i = p - q \\ \sigma^2 &\equiv \mathbf{Var}\xi_i = \mathbf{E}\,\xi_i^2 - \mu^2 = 1 - (p-q)^2 = 4pq \leq 1, \end{aligned}$$

and note that $4pq = 1$ if and only if $p = q = 1/2$, and $4pq < 1$, otherwise. Recall that we then have

$$\begin{aligned} \mathbf{E}\,S_n &= a + n\mu, \\ \mathbf{Var}S_n &= n\sigma^2 \end{aligned}$$

as in the general case, where μ and σ^2 are now given as above.

Motivation.

1. Exact calculations are possible for this model.
2. For many distributions of the increments ξ_i, the behaviour of a general random walk, as previously considered, is very similar to that of the simple random walk, and so the latter process provides very good insight into the general behaviour of random walks.

3.2.1 Simple exact calculations

Assume, again for simplicity of exposition, that $a = 0$, i.e. consider the simple random walk starting at $S_0 = 0$. Note that then, for each integer $n \geq 0$, S_{2n} is always even and S_{2n+1} is always odd. Further, from consideration of the **binomial distribution**,

$$\begin{aligned} \mathbf{P}(S_{2n} = 2m) &= \binom{2n}{n+m} p^{n+m} q^{n-m}, \qquad -n \leq m \leq n \\ \mathbf{P}(S_{2n+1} = 2m+1) &= \binom{2n+1}{n+m+1} p^{n+m+1} q^{n-m}, \qquad -n-1 \leq m \leq n. \end{aligned}$$

Let us show, for example, the first of these results. Let X denote the number of $(+1)$'s in the first $2n$ steps and Y the number of (-1)'s. Clearly, X is the number of successes in $2n$ trials, with the probability of success p and the probability of fail $q = 1 - p$. Then X has the binomial distribution with parameters $2n$ and p. Further, $X + Y = 2n$ and $S_{2n} = X - Y = X - (2n - X) = 2(X - n)$. Then $\mathbf{P}(S_{2n} = 2m) = \mathbf{P}(2(X-n) = 2m) = \mathbf{P}(X - n = m) = \mathbf{P}(X = n + m)$, and the formula (1) from Section 2.1.2 may be applied, with $2n$ in place of n and $n + m$ in place of k.

Note that these are exact probabilities. For n moderate to large, the probability that S_n lies within a given interval (not too small) is probably best given by the normal approximation given earlier.

Example. Consider the **symmetric simple random walk**, defined by $p = q = 1/2$ and starting at $S_0 = 0$.

1. Calculate the exact probability that, for any $n > 0$, $S_{2n} = 0$. We have

$$\mathbf{P}(S_{2n} = 0) = \binom{2n}{n} \frac{1}{2^{2n}} = \frac{(2n)!}{(n!)^2} \frac{1}{2^{2n}}.$$

   Now use **Stirling's approximation**: $n! \sim e^{-n} n^n \sqrt{2\pi n}$ as $n \to \infty$ to deduce

$$\begin{aligned} \mathbf{P}(S_{2n} = 0) &\sim \frac{e^{-2n}(2n)^{2n}\sqrt{4\pi n}}{e^{-2n} n^{2n} (2\pi n)} \frac{1}{2^{2n}} \\ &= \frac{1}{\sqrt{\pi n}}, \end{aligned}$$

   valid for all large n.

   Note that this probability declines in proportion to $1/\sqrt{n}$, exactly as we would expect (see below).

2. For n moderate to large (and odd or even), find a reasonable approximation to the entire distribution of S_n. We have

$$\mathbf{E}\, S_n = 0, \qquad \mathbf{Var} S_n = n,$$

   and so the distribution of S_n is approximately normal with this mean and variance (i.e. with standard deviation $\sqrt{n}$).

3. Assume now that $p \neq q$ (this means that either $0 < p < 1/2$ or $1/2 < p < 1$). Then

$$\mathbf{P}(S_{2n} = 0) = \binom{2n}{n} p^n q^n = \binom{2n}{n} \frac{1}{2^{2n}} \cdot (4pq)^n = \frac{(2n)!}{(n!)^2} \frac{1}{2^{2n}} \cdot (4pq)^n.$$

   Use again **Stirling's approximation**, to deduce

$$\mathbf{P}(S_{2n} = 0) \sim \frac{e^{-2n}(2n)^{2n}\sqrt{4\pi n}}{e^{-2n} n^{2n} (2\pi n)} \frac{1}{2^{2n}} \cdot (4pq)^n = \frac{1}{\sqrt{\pi n}} \cdot (4pq)^n, \tag{2}$$

   so now the probabilities decay exponentially fast!

3.2.2 The ruin problem

Suppose now that $a \geq 0$. Recall that the event R, or **ruin** occurs if there exists some $n > 0$ such that $S_n \leq 0$.

Before calculating $\mathbf{P}(R)$ we consider a more general problem. Take integer $b \geq a \geq 0$ and define

$$\alpha_{b,a} = \mathbf{P}(\text{process } \{S_n\} \text{ hits state 0 before state } b \mid S_0 = a).$$

(We can think of states 0 and b as being **absorbing barriers** for the process $\{S_n\}$. We start in some state a between these two barriers, and are interested in the probability of hitting the lower barrier before the upper one.)

Theorem 3.2.1 We have

$$\alpha_{b,a} = \begin{cases} \frac{(q/p)^a - (q/p)^b}{1-(q/p)^b} & p \neq \frac{1}{2} \\ 1 - \frac{a}{b} & p = \frac{1}{2}. \end{cases}$$

Proof: Consider 0 and b as fixed and a varying. Clearly, $\alpha_{b,0} = 1$ and $\alpha_{b,b} = 0$, that agrees with the formula above. For a such that $0 < a < b$, we have

$$\begin{aligned} \alpha_{b,a} &= \mathbf{P}(\text{process } \{S_n\} \text{ hits state 0 before state } b \mid S_0 = a) \\ &= p\mathbf{P}(\text{process } \{S_n\} \text{ hits state 0 before state } b \mid S_1 = a+1) \\ &\quad + q\mathbf{P}(\text{process } \{S_n\} \text{ hits state 0 before state } b \mid S_1 = a-1) \\ &= p\mathbf{P}(\text{process } \{S_n\} \text{ hits state 0 before state } b \mid S_0 = a+1) \\ &\quad + q\mathbf{P}(\text{process } \{S_n\} \text{ hits state 0 before state } b \mid S_0 = a-1) \\ &= p\alpha_{b,a+1} + q\alpha_{b,a-1}. \end{aligned}$$

We have $b-1$ equations for $a = 1, 2, \ldots, b-1$. We will solve them now.

One can write

$$\alpha_{b,a} = p\alpha_{b,a} + q\alpha_{b,a} = p\alpha_{b,a+1} + q\alpha_{b,a-1}.$$

Therefore

$$\alpha_{b,a} - \alpha_{b,a+1} = \frac{q}{p}\left(\alpha_{b,a-1} - \alpha_{b,a}\right),$$

and

$$\begin{aligned} \alpha_{b,a} - \alpha_{b,a+1} &= \frac{q}{p}\left(\alpha_{b,a-1} - \alpha_{b,a}\right) = \left(\frac{q}{p}\right)^2 \left(\alpha_{b,a-2} - \alpha_{b,a-1}\right) = \ldots \\ &= \left(\frac{q}{p}\right)^a \left(\alpha_{b,0} - \alpha_{b,1}\right) = \left(\frac{q}{p}\right)^a \left(1 - \alpha_{b,1}\right), \end{aligned}$$

where the last step is obtained by noticing that $\alpha_{b,0} = 1$. Further,

$$\begin{aligned} \alpha_{b,0} - \alpha_{b,a} &= \alpha_{b,a-1} - \alpha_{b,a} + \alpha_{b,a-2} - \alpha_{b,a-1} + \ldots + \alpha_{b,0} - \alpha_{b,1} \\ &= \left(\left(\frac{q}{p}\right)^{a-1} + \left(\frac{q}{p}\right)^{a-2} + \ldots \left(\frac{q}{p}\right) + 1\right)\left(1 - \alpha_{b,1}\right). \end{aligned} \tag{3}$$

If $p = q = \frac{1}{2}$ then

$$1 - \alpha_{b,a} = a\left(1 - \alpha_{b,1}\right) \iff \alpha_{b,a} = 1 - a\left(1 - \alpha_{b,1}\right).$$

But $\alpha_{b,b} = 0$ so $1 - \alpha_{b,1} = \frac{1}{b}$ and finally

$$\alpha_{b,a} = 1 - \frac{a}{b}.$$

Now $p \neq q$ and formula (3) can be transformed into

$$1 - \alpha_{b,a} = \frac{1 - \left(\frac{q}{p}\right)^a}{1 - \left(\frac{q}{p}\right)} (1 - \alpha_{b,1}) .$$

By letting $a = b$ and using $\alpha_{b,b} = 0$ we have

$$1 = \frac{1 - \left(\frac{q}{p}\right)^b}{1 - \left(\frac{q}{p}\right)} (1 - \alpha_{b,1}) \iff \frac{1 - \left(\frac{q}{p}\right)}{1 - \left(\frac{q}{p}\right)^b} = (1 - \alpha_{b,1})$$

Finally,

$$1 - \alpha_{b,a} = \frac{1 - \left(\frac{q}{p}\right)^a}{1 - \left(\frac{q}{p}\right)} (1 - \alpha_{b,1}) \iff$$

$$\alpha_{b,a} = 1 - \frac{1 - \left(\frac{q}{p}\right)^a}{1 - \left(\frac{q}{p}\right)} \times \frac{1 - \left(\frac{q}{p}\right)}{1 - \left(\frac{q}{p}\right)^b} = 1 - \frac{1 - \left(\frac{q}{p}\right)^a}{1 - \left(\frac{q}{p}\right)^b} = \frac{\left(\frac{q}{p}\right)^a - \left(\frac{q}{p}\right)^b}{1 - \left(\frac{q}{p}\right)^b},$$

which has been required. □

To find

$$\beta_{b,a} = \mathbf{P}(\text{process } \{S_n\} \text{ hits state } b \text{ before state } 0 \mid S_0 = a),$$

one should consider the formula for $\alpha_{b,a}$, swap there upper and lower bounds, p and q and write $b - a$ instead of a (by symmetry). Then one gets

$$\beta_{b,a} = \begin{cases} \frac{(p/q)^{b-a} - (p/q)^b}{1 - (p/q)^b} & p \neq \frac{1}{2} \\ 1 - \frac{b-a}{b} & p = \frac{1}{2}. \end{cases}$$

One can check that $\beta_{b,a} = 1 - \alpha_{b,a}$ (this is a good exercise). The result is not trivial on its own. It proves that the process will necessarily hit at least one of the boundaries.

Now note that, for fixed a, the events

$$E_{b,a} := \{\text{process } \{S_n\} \text{ hits state } 0 \text{ before state } b \mid S_0 = a\}$$

are increasing in b and their probabilities $\alpha_{b,a} = \mathbf{P}(E_{b,a})$ are increasing in b, too. Define the **ruin probability**

$$\alpha_a = \mathbf{P}(\text{process } \{S_n\} \text{ ever hits state } 0 \mid S_0 = a) \equiv \mathbf{P}(R).$$

Then, $R = \lim_{b\to\infty} E_{b,a}$ and, by the property of the *continuity of probability*, $\mathbf{P}(R) = \alpha_a = \lim_{b\to\infty} \alpha_{b,a}$. We thus obtain the following corollary to Theorem 3.2.1.

Corollary 3.2.2

$$\mathbf{P}(R) = \alpha_a = \begin{cases} \left(\frac{q}{p}\right)^a & p > \frac{1}{2} \\ 1 & p \leq \frac{1}{2}. \end{cases}$$

Note that we can rewrite this result as $\mathbf{P}(R) = \min(1, (q/p)^a)$ (and the latter formula may be more useful, because it helps to remember that the probability cannot be bigger than 1).

Note also that this result is in agreement with our earlier results for a general random walk.

Example. Let $\{S_n\}$ be a simple random walk starting at $S_0 = 0$ ($a = 0$) with $p = 2/3$. Find the probability that $\{S_n\}$ is ever negative (< 0).

This is the same as the probability of ruin ($\mathbf{P}$(there exists some $n > 0$ such that $S_n \leq 0$)) for a simple random walk with the same value of p but starting at $a = 1$. Hence the required probability is simply $q/p = (1 - p)/p) = 1/2$.

3.2.3 The expected time to ruin.

Now consider further the case $p < \frac{1}{2}$ (otherwise the expected time to ruin is infinite). Let $T_a \equiv T_{a,0}$ be the **time to ruin** when $S_0 = a$.

Theorem 3.2.3 Suppose that $p < \frac{1}{2}$ and $a > 0$. Then

$$\mathbf{E}\, T_a = \frac{a}{q-p} = \frac{a}{1-2p}.$$

Proof: Denote by $T_{i,i-1}$ the first time to get to level $i-1$ for a random walk that starts from i. Then the distribution of $T_{i,i-1}$ does not depend on i and, therefore,

$$\mathbf{E}\, T_{i,i-1} = \mathbf{E}\, T_{1,0} \equiv ET_1.$$

Clearly,

$$T_a \equiv T_{a,0} = T_{a,a-1} + T_{a-1,a-2} + \ldots + T_{1,0}$$

and, therefore,

$$\mathbf{E}\, T_a = \sum_{i=1}^{a} \mathbf{E}\, T_{i,i-1} = a\mathbf{E}\, T_1.$$

But also, by conditioning on the first jump, as in the proof of Theorem 3.2.1, we have

$$\mathbf{E}\, T_1 = p\mathbf{E}\,(T_1 \mid \xi_1 = 1) + q\mathbf{E}\,(T_1 \mid \xi_1 = -1) = p \cdot (1 + \mathbf{E}\, T_2) + q \cdot 1 = 1 + p\mathbf{E}\, T_2$$

(the term 1 is the time for the first step, after which we either have $S_1 = 0$, with probability $q = 1-p$ and are already ruined, or we have $S_1 = 2$, with probability p, and so can consider the process as starting again in state 2).
Combining the two results above we obtain the required conclusion. □

Notes.

1. This result agrees with that stated earlier for the general random walk.
2. For fixed $a > 0$, $\mathbf{E}\, T_a \to \infty$ as $p \to \frac{1}{2}$ from below. A comparison argument shows that we may indeed conclude that $\mathbf{E}\, T_a = \infty$ in the case $p = \frac{1}{2}$.

Therefore, we arrive at a more precise version of **Conclusion on $\mathbf{P}(R)$ and $\mathbf{E}\, T_R$ for a general random walk** from Section 3.1.2:

Conclusion on ruin for a simple random walk with initial value a:
if $\mu = p - q < 0$, then $\mathbf{P}(R) = 1$ and $\mathbf{E}\, T_R = \frac{a}{q-p}$;
if $\mu = p - q = 0$, then $\mathbf{P}(R) = 1$ and $\mathbf{E}\, T_R = \infty$;
if $\mu = p - q > 0$, then $\mathbf{P}(R) = (q/p)^a < 1$ and $\mathbf{E}\, T_R = \infty$.

3.3 Other random walks

3.3.1 The simple random walk with the possibility of zero jumps.

Consider the random walk $\{S_n\}_{n\geq 0}$ defined on the integers $\mathbb{Z}$ by $S_0 = a$ and, for $n \geq 1$, $S_n = a + \sum_{i=1}^{n} \xi_i$, where $\xi_1, \xi_2 \ldots$ are independent identically distributed random variables with

$$\begin{aligned} \mathbf{P}(\xi_i = 1) &= p \\ \mathbf{P}(\xi_i = 0) &= 1 - (p+q) \\ \mathbf{P}(\xi_i = -1) &= q \\ (\Rightarrow \mathbf{P}(\xi_i = k) &= 0 \quad \text{for all other integers } k) \end{aligned}$$

where $p > 0$, $q > 0$, $p + q < 1$.

We may study this random walk by considering a modified process in which we sample at just those times at which a (nonzero) jump occurs. This gives a new random walk, which is a *simple* random walk as considered earlier, with up and down transition probabilities given by p' and q' respectively, where these conditional probabilities are defined as

$$p' = \mathbf{P}(A \mid A \cup B) = \frac{\mathbf{P}(A \cap (A \cup B))}{\mathbf{P}(A \cup B)} = \frac{\mathbf{P}(A)}{\mathbf{P}(A) + \mathbf{P}(B)} = \frac{p}{p+q} \quad \text{and}$$

$$q' = \mathbf{P}(B \mid A \cup B) = \frac{\mathbf{P}(B \cap (A \cup B))}{\mathbf{P}(A \cup B)} = \frac{\mathbf{P}(B)}{\mathbf{P}(A) + \mathbf{P}(B)} = \frac{q}{p+q}.$$

Here $A = \{\text{jump up}\}$ and $B = \{\text{jump down}\}$.

Example. For $a > 0$ find the probability $\mathbf{P}(R)$ of ruin for this process.

Note that ruin occurs if and only if it occurs in the modified process. Hence by Corollary 3.2.2 we have

$$\begin{aligned} \mathbf{P}(R) &= \min(1,\, (q'/p')^a) \\ &= \min(1,\, (q/p)^a) \end{aligned}$$

Exercise. Assume that $0 < p < q$ and $p + q < 1$. Show that the formula

$$\mathbf{E}\, T_R = \frac{a}{q-p}$$

continues to hold. However, $\frac{a}{q-p} \neq \frac{a}{1-2p}$ in this case.

3.3.2 Coupling arguments

For a more general random walk it is sometimes possible to obtain bounds by comparison with a simple random walk.

Example. An insurance company has initial capital a. Each year it receives premium income of 1 and pays claims. The total claim paid in each year i is Y_i where $\{Y_i\}_{i\geq 1}$ is a sequence of independent identically distributed random variables, each with the following distribution:

$$\mathbf{P}(Y_i = 0) = 0.40; \quad \mathbf{P}(Y_i = 1) = 0.40; \quad \mathbf{P}(Y_i = 2) = 0.19; \quad \mathbf{P}(Y_i = 3) = 0.01.$$

Find a *lower* bound on the probability of ruin of the insurance company.

The fortune of the insurance company is a random walk $S_n = a + \sum_{i=1}^{n} \xi_i$ where each $\xi_i = 1 - Y_i$ and so has distribution

$$\mathbf{P}(\xi_i = 1) = 0.40; \quad \mathbf{P}(\xi_i = 0) = 0.40; \quad \mathbf{P}(\xi_i = -1) = 0.19; \quad \mathbf{P}(\xi_i = -2) = 0.01.$$

For each i, define

$$\xi_i' = \max(\xi_i, -1).$$

Then each ξ_i' has distribution

$$\mathbf{P}(\xi_i' = 1) = 0.40; \quad \mathbf{P}(\xi_i' = 0) = 0.40; \quad \mathbf{P}(\xi_i' = -1) = 0.20.$$

The random walk $S_n' = ca + \sum_{i=1}^{n} \xi_i'$, $n \geq 0$, satisfies $S_n' \geq S_n$ for all n. Hence

$$\begin{aligned} \mathbf{P}(S_n \leq 0 \text{ for some } n) &\geq \mathbf{P}(S_n' \leq 0 \text{ for some } n) \\ &= \left(\frac{0.20}{0.40}\right)^a = \left(\frac{1}{2}\right)^a. \end{aligned}$$

3.3.3 Scaling

Let $c > 0$ be a constant. Consider a random walk $\{S_n\}_{n\geq 0}$ defined on the multiples of c by $S_0 = ca$ (where a is a positive integer) and, for $n \geq 1$, $S_n = a + \sum_{i=1}^{n} \xi_i$ where $\xi_1, \xi_2, \ldots$ are independent and identically distribution random variables with

$$\begin{aligned} \mathbf{P}(\xi_i = c) &= p, \\ \mathbf{P}(\xi_i = 0) &= 1 - (p+q), \\ \mathbf{P}(\xi_i = -c) &= q, \end{aligned}$$

and $\mathbf{P}(\xi_i = kc) = 0$, for all other integers k. Here $p > 0, q > 0, p + q \leq 1$.

Then the ruin probability is the same for all $c > 0$.

Examples.

3.3.4 Combination of coupling and scaling arguments

Example. Consider a random walk $\{S_n\}_{n\geq 0}$ with initial value $S_0 = 4$ and with i.i.d. increments having the following distribution:

$$\begin{aligned} \mathbf{P}(\xi_i = 2) &= p, \\ \mathbf{P}(\xi_i = 0) &= 1 - (p+q), \\ \mathbf{P}(\xi_i = -1) &= q, \end{aligned}$$

Provide two upper bounds for the ruin probability, using two different couplings.

3.3.5 Use of simulation

Sometimes probabilities of complex events, e.g. *ruin*, need to be estimated by simulation.

Suppose that, in a large number n of simulations of a given process, an event of interest occurs a total of k times. Then k has a binomial distribution with parameters n and p where p is the true (unknown) probability of the event.

We thus estimate p by

$$\hat{p} = k/n.$$

The *standard error* (i.e. *standard deviation*) of our estimate is

$$\begin{aligned} \mathrm{SE}(\hat{p}) &= \frac{1}{n}\mathrm{SD}(k) \\ &= \frac{1}{n}\sqrt{np(1-p)} \\ &\approx \frac{1}{n}\sqrt{n\hat{p}(1-\hat{p})} \\ &= \sqrt{\frac{k(n-k)}{n^3}}. \end{aligned}$$

Using the normal approximation to the binomial distribution, an approximate 95% confidence interval for p is given by

$$(\hat{p} - 2\,\mathrm{SE}(\hat{p}),\ \hat{p} + 2\,\mathrm{SE}(\hat{p})).$$

4 Markov chains

4.1 Introduction

4.1.1 Revision of basic properties of matrices

In this lecture course, we deal with square matrices only.
Recall that if $A = (a_{i,j})_{1 \le i,j \le n}$ and $B = (b_{j,k})_{1 \le j,k \le n}$ are two $n \times n$ matrices, the matrix $C = A \times B$ is defined as

$$C = (c_{i,k})_{1 \le i,k \le n} \quad \text{where} \quad c_{i,k} = \sum_{j=1}^{n} a_{ij} b_{jk}.$$

Further, if $\mathbf{x} = (x_1, \ldots, x_n)$ is a vector and $A = (a_{i,j})_{1 \le i,j \le n}$ is a matrix, then

$$\mathbf{x}A = \mathbf{y} = (y_1, \ldots, y_n) \quad \text{where} \quad y_j = \sum_{i=1}^{n} x_i a_{i,j}, \quad \text{for} \quad j = 1, \ldots, n$$

and

$$A\mathbf{x} = \mathbf{z} = (z_1, \ldots, z_n) \quad \text{where} \quad z_i = \sum_{j=1}^{n} a_{i,j} x_j, \quad \text{for} \quad i = 1, \ldots, n.$$

(For short, we use the same notation for vectors written in the horizontal and vertical forms).

4.1.2 Definitions

A **discrete time** stochastic process $\{X_n\}_{n\geq 0}$ taking values in a **countable** state space $\mathcal{S}$ (e.g. $\mathcal{S}$ equal to the integers or some subset of the integers) is a **Markov chain** if and only if

$$\mathbf{P}(X_{n+1} = j \,|\, X_n = i,\, X_{n-1} = i_{n-1}, \ldots, X_0 = i_0) = \mathbf{P}(X_{n+1} = j \,|\, X_n = i)$$

for all $n \geq 0$ and for all $j, i, i_{n-1}, \ldots, i_1 \in \mathcal{S}$.

Thus, **given** the evolution of the process $\{X_n\}_{n\geq 0}$ up to any "current" time n, the probabilistic description of its behaviour at time $n+1$ (and, by induction, the probabilistic description of all its subsequent behaviour) depends only on the current state $\{X_n = i\}$, and not on the previous history of the process. This is the **Markov property**.

Introduce three events: event $F = \{X_{n+1} = j\}$ (event "Future"), event $N = \{X_n = i\}$ (event "Now", or "Present") and event $H = \{X_{n-1} = i_{n-1}, \ldots, X_0 = i_0\}$ (event "History", or "Past"). Then the Markov property means:

$$\mathbf{P}(F \,|\, NH) = \mathbf{P}(F \,|\, N) \tag{4}$$

where $NH = N \cap H$ is the intersection of events N and H.

Remark. One can rewrite property (4) in a symmetric form:

$$\mathbf{P}(HF \,|\, N) = \mathbf{P}(H \,|\, N) \cdot \mathbf{P}(F \,|\, N) \tag{5}$$

where $HF = H \cap F$ is the intersection of events H and F. Indeed,

$$\begin{aligned} \mathbf{P}(NFH) &= \mathbf{P}(N) \cdot \mathbf{P}(FH \,|\, N) \quad \text{and} \\ \mathbf{P}(NFH) &= \mathbf{P}(N) \cdot \mathbf{P}(H \,|\, N) \cdot \mathbf{P}(F \,|\, NH) \\ &= \mathbf{P}(N) \cdot \mathbf{P}(H \,|\, N) \cdot \mathbf{P}(F \,|\, N), \end{aligned}$$

by (4). Then (5) follows.

Exercise. Let events N and H be given as before, and let event F have a more general form: for some $k \geq 1$ and for $j_{n+1}, \ldots, j_{n+k} \in \mathcal{S}$,

$$F = \{X_{n+1} = j_{n+1}, X_{n+2} = j_{n+2}, \ldots, X_{n+k} = j_{n+k}\}.$$

Show that formulae (4) and (5) continue to hold.

Now we introduce the *time-homogeneity* property. The **Markov chain** $\{X_n\}_{n\geq 0}$ is **time homogeneous**, if, for all $i, j \in \mathcal{S}$, there exists p_{ij} such that

$$\mathbf{P}(X_{n+1} = j \,|\, X_n = i) = p_{ij}$$

independently of n.

The matrix

$$P = \begin{pmatrix} p_{11} & p_{12} & p_{13} & \cdots \\ p_{21} & p_{22} & p_{23} & \cdots \\ p_{31} & p_{32} & p_{33} & \cdots \\ \vdots & \vdots & \vdots & \vdots \end{pmatrix}$$

is then referred to at the **transition matrix** of the Markov chain. We will also use a shorter notation for this matrix, $P = (p_{ij})_{i,j\in\mathcal{S}}$.

Note that necessarily

$$p_{ij} \geq 0 \quad \text{for all } i, j \in \mathcal{S}, \qquad \sum_{j\in\mathcal{S}} p_{ij} = 1 \quad \text{for all } i \in \mathcal{S},$$

and we say that P is a **stochastic matrix** since it satisfies these properties.

For a **time homogeneous Markov chain** $\{X_n\}_{n\geq 0}$, given its evolution up to any "current" time n, the probabilistic description of its behaviour at time $n+1$ (and, by extension, the probabilistic description of all its subsequent behaviour) depends only on the current state $\{X_n = i\}$, and not on the previous history of the process **nor** on the time n itself.

Throughout this course *all* Markov chains are assumed time homogeneous.

4.1.3 Examples

Random walk. Perhaps the simplest example of a time homogeneous Markov chain is *any* random walk $\{S_n\}_{n\geq 0}$ taking values in a countable state space—typically the integers. To see this let

$$S_0 = a, \qquad S_n = a + \sum_{i=1}^{n} \xi_i, \quad n \geq 1.$$

where, as usual, $\xi_1.\xi_2, \ldots$ is a sequence of independent identically distributed random variables. Then, for any n and any set of states $i_0, \ldots, i_{n-1}, i, j$

$$\begin{aligned}\mathbf{P}(S_{n+1} = j \,|\, S_n = i,\, S_{n-1} = i_{n-1}, \ldots, S_0 = i_0) &= \mathbf{P}(\xi_{n+1} = j - i) \qquad \text{(by independence)} \\ &= \mathbf{P}(S_{n+1} = j \,|\, S_n = i).\end{aligned}$$

This is useful both for the study of random walks (we can apply the more general theory of Markov chains) and of Markov chains (random walks provide nice examples).

Random walk with absorbing/reflecting barrier(s). Consider the *simple* random walk $\{S_n\}_{n\geq 0}$ defined as in Section 3.2 and with the initial state $a > 0$, but suppose now that when this process hits state 0 it remains in that state for all future times. Then it is easy(!) to see that this modified process is again a Markov chain. The state 0 is referred to as an **absorbing state** or (in the specific context of the simple random walk) as an **absorbing barrier** for the chain. Routine methods (see below) enable us to calculate the absorption probability. We have already done this for this case in Theorem 3.2.1 (for two absorbing barriers at 0 and b), and the general technique is similar in spirit.

Suppose now that the *simple* random walk $\{S_n\}_{n\geq 0}$ is *differently* modified so that whenever the process is in state 0, its next step is to remain in state 0 with probability q or go to state 1 with probability p (where p and q are as usual), *independently of all previous moves*. Then it is again easy(!) to see that this modified process is a Markov chain. The state 0 is referred to as a **reflecting barrier** for the chain. This process might model the fortune of an individual whose negative equity is always reset to zero.

A more general scheme of modifying at 0 is: whenever the process is in state 0, its next step is to remain in state 0 with probability $\widetilde{q}$ or go to state 1 with probability $\widetilde{p}$ where $\widetilde{p}$ and $\widetilde{q}$ are any non-negative numbers with $\widetilde{p} + \widetilde{q} = 1$. Then, in particular, we get **absorption** at 0 is $\widetilde{p} = 0$ and **reflection** at 0 if $\widetilde{p} = p$.

There are also other, more realistic, examples in queueing theory.

Bonus-malus system. This is an insurance no-claims discount system.

As a simple example suppose that a motorist purchases an insurance policy each year. There are three levels of discount: level 0 (0% discount), level 1 (25% discount), or level 2 (50% discount). In the following year (s)he moves up a level (or remains at level 2 if already there) provided there has been no claim during the year; otherwise (i.e. if there has been at least one claim) (s)he moves back to (or stays at) level 0.

Suppose also that the probability of at least one claim in any year is p, *independently of all other years*. Let X_n be the level in year n. Then $\{X_n\}_{n\geq 0}$ is a time homogeneous Markov chain, with state space $\mathcal{S} = \{0, 1, 2\}$ and transition matrix $P = (p_{ij})_{i,j\in\mathcal{S}}$ given by

$$P = \begin{pmatrix} p & 1-p & 0 \\ p & 0 & 1-p \\ p & 0 & 1-p \end{pmatrix}.$$

Note that in this example, the state names 0, 1, and 2 are simply **labels**: we could, for example, relabel the states as 0%, 25%, and 50% without changing the problem!

Interesting questions:

1. Find (as a function of p) $\mathbf{P}(X_n = j)$, $j = 0, 1, 2$, given that $\mathbf{P}(X_0 = 0) = 1$, i.e. given that the process starts in level 0.
2. Find (as a function of p) $\lim_{n\to\infty} \mathbf{P}(X_n = j)$, $j = 0, 1, 2$, given that $\mathbf{P}(X_0 = 0) = 1$. (It will turn out that these limiting probabilities correspond to the long-term proportions of time spent in each of the three states.)

Health insurance system. Here the state space might be $\{1, 2, 3\}$ (here 1 means *healthy*, 2 means *sick*, and 3 means *dead*), and the states of an individual in successive years might form a time homogeneous Markov chain with transition matrix $P = (p_{ij})_{i,j\in\mathcal{S}}$ given by

$$P = \begin{pmatrix} p_{11} & p_{12} & p_{13} \\ p_{21} & p_{22} & p_{23} \\ 0 & 0 & 1 \end{pmatrix}$$

since 3 (*dead*) is an absorbing state! (Note that the sums of the probabilities in each of the first two rows must also be 1.)

Interesting question: Suppose that the individual starts in state 1. Find the expected time spent in each of the states 1 and 2 before finally arriving at the state 3.

4.1.4 Simple probability calculations

Suppose that $\{X_n\}_{n\geq 0}$ is a time homogeneous Markov chain with transition matrix $P = (p_{ij})_{i,j\in\mathcal{S}}$.
Further, let $\boldsymbol{\mu} = (\mu_i)_{i\in\mathcal{S}}$ be the distribution of the initial random variable X_0, i.e. $\mu_i = \mathbf{P}(X_0 = i)$.
Then the probabilistic description of the entire process $\{X_n\}_{n\geq 0}$ is determined by $\boldsymbol{\mu}$ and P. We may think of $\boldsymbol{\mu}$ as representing (the probability of) the initial state of the process, and P as representing its subsequent dynamics.
In particular we have, for any sequence of states $i_0, \ldots, i_n$,

$$\mathbf{P}(X_0 = i_0,\, X_1 = i_1, \ldots, X_n = i_n) = \mu_{i_0} p_{i_0 i_1} \ldots p_{i_{n-1} i_n}, \tag{6}$$

and so more general probabilities can also be calculated by summing the probabilities of elementary events of this form.
For any $n \geq 0$, define the **n-step transition probabilities**

$$p_{ij}^{(n)} = \mathbf{P}(X_{m+n} = j \,|\, X_m = i), \qquad\qquad i, j \in \mathcal{S}.$$

These are independent of m by *time homogeneity*. Note also that $p_{ij}^{(0)} = \delta_{ij}$ where $\delta_{ij} = 1$ if $i = j$ and $\delta_{ij} = 0$ if $i \neq j$. Then, from (6), writing i for i_0, j for i_n, summing probabilities over all intermediate states $i_1, \ldots, i_{n-1}$, and finally conditioning on $X_0 = i$ (or equivalently taking $\mathbf{P}(X_0 = i) = 1$), we have

$$p_{ij}^{(n)} = (P^n)_{ij}, \qquad\qquad i, j \in \mathcal{S},$$

i.e. the *n-step transition probabilities* $p_{ij}^{(n)}$, are given by (the components of) the matrix P^n. This is also necessarily a stochastic matrix. Further it should also be possible to see the above result directly.
Similarly, from (6), writing j for i_n, and summing probabilities over all intermediate states $i_0, i_1, \ldots, i_{n-1}$, we have

$$\mathbf{P}(X_n = j) = (\boldsymbol{\mu} P^n)_j, \qquad\qquad j \in \mathcal{S},$$

i.e. the distribution of X_n is given by the vector $\boldsymbol{\mu} P^n$.
Finally we have the **Chapman-Kolmogorov equations**: for any $m > 0$, $n > 0$, and $i, j \in \mathcal{S}$,

$$\begin{aligned}
p_{ij}^{(m+n)} &= \mathbf{P}(X_{m+n} = j \,|\, X_0 = i) \\
&= \sum_{k\in\mathcal{S}} \mathbf{P}(X_n = k,\, X_{m+n} = j \,|\, X_0 = i) \\
&= \sum_{k\in\mathcal{S}} \mathbf{P}(X_n = k \,|\, X_0 = i)\mathbf{P}(X_{m+n} = j \,|\, X_0 = i, X_n = k) \\
&= \sum_{k\in\mathcal{S}} \mathbf{P}(X_n = k \,|\, X_0 = i)\mathbf{P}(X_{m+n} = j \,|\, X_n = k) \qquad \text{(by the Markov property)} \\
&= \sum_{k\in\mathcal{S}} p_{ik}^{(n)} p_{kj}^{(m)}
\end{aligned}$$

and the **Chapman-Kolmogorov inequalities**: for any $m > 0$, $n > 0$, and $i, j, k \in \mathcal{S}$,

$$p_{ij}^{(m+n)} \geq p_{ik}^{(m)} p_{kj}^{(n)},$$

which follow immediately from the Chapman-Kolmogorov equations.

4.2 Classification of states

Throughout this section $\{X_n\}_{n\geq 0}$ is a time homogeneous Markov chain with state space $\mathcal{S}$ and transition matrix $P = (p_{ij})_{i,j\in\mathcal{S}}$.

Henceforth we (usually) write, for any event E,

$$\mathbf{P}_i(E) = \mathbf{P}(E \,|\, X_0 = i)$$

4.2.1 Classes

Definitions. For any $i, j \in \mathcal{S}$, we write $i \to j$ if there is some $n \geq 0$ such that $p_{ij}^{(n)} > 0$. (Note in particular that $i \to i$ for all states i.) We also write $i \leftrightarrow j$ if $i \to j$ and $j \to i$. If $i \leftrightarrow j$ we say that the states i and j **intercommunicate**. (Note that they do not necessarily intercommunicate directly, i.e. in single jumps.)

Property 1. Note that $i \to j$ if and only if

$$p_{ik_1}p_{k_1k_2}\dots p_{k_rj} > 0$$

for some sequence (possibly empty) of intermediate states $k_1, \dots, k_r$.

Property 2. It follows that we may determine whether $i \to j$ (and so also whether $i \leftrightarrow j$) simply by noting which of the one-step transition probabilities p_{ij} are strictly positive. It further follows that if $i \leftrightarrow j$ and $j \leftrightarrow k$, then $i \leftrightarrow k$.

Hence $\leftrightarrow$ is an equivalence relation and so partitions the state space into equivalence **classes** of **intercommunicating states**. Thus in any **class** all the states **intercommunicate**, but none of them *intercommunicates* with any state outside the class.

However it may be possible to have **one-way communication** between states in different classes. If a class C is such that $i \in C$, $i \to j$ implies $j \in C$ (i.e. no one-way communication is possible from states within C to states outside C – escape from C is impossible) then the class C is said to be **closed**. Otherwise the class C is said to be **nonclosed**, or **open** – escape from C is possible.

If the entire state space $\mathcal{S}$ is a single class (all states intercommunicate), then it is necessarily closed.

A closed class consisting of a single state is an **absorbing state**.

Examples

Simple random walk. For the simple random walk with $0 < p < 1$, all states intercommunicate, and so the state space (the integers) is a single, necessarily closed, class.

Suppose now that the state space is restricted to the nonnegative integers and that the state 0 is modified to be *absorbing*. Then the classes and their types are as follows:

Class	Type
$\{1, 2, 3, \dots\}$	nonclosed (open)
$\{0\}$	closed (absorbing)

If instead the state 0 is instead *reflecting*, then once more all states intercommunicate, and so the state space is a single, necessarily closed, class.

Bonus-malus system. The Bonus-malus system introduced earlier had state space $\mathcal{S} = \{0, 1, 2\}$ and transition matrix

$$P = \begin{pmatrix} p & 1-p & 0 \\ p & 0 & 1-p \\ p & 0 & 1-p \end{pmatrix}.$$

It is easier to see that, provided $0 < p < 1$, all states intercommunicate, and so the state space $\mathcal{S}$ is a single, necessarily closed, class.

Health insurance system. The health insurance system introduced earlier had state space $\{H, S, D\}$ and transition matrix

$$P = \begin{pmatrix} p_{11} & p_{12} & p_{13} \\ p_{21} & p_{22} & p_{23} \\ 0 & 0 & 1 \end{pmatrix}.$$

Assume that $0 < p_{ij} < 1$ for $i = 1, 2$, $j = 1, 2, 3$. Then the classes and their types are as follows:

Class	Type
$\{H, S\}$	nonclosed (open)
$\{D\}$	closed (absorbing)

Further example. Take $\mathcal{S} = \{1, 2, 3, 4, 5\}$ and the transition matrix to be given by

$$P = \begin{pmatrix} 1/2 & 1/2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 1/5 & 1/5 & 1/5 & 1/5 & 1/5 \\ 1/3 & 0 & 1/3 & 0 & 1/3 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

Then the classes and their types are as follows:

Class	Type
$\{1, 2\}$	closed
$\{3, 4\}$	nonclosed (open)
$\{5\}$	closed (absorbing)

4.2.2 Calculation of absorption probabilities for closed classes

It is useful to be able to calculate the probabilities of eventual absorption in each of the closed classes (whether or not these consist of single states).

For any closed class C define

$$\begin{aligned}\alpha_i^C &= \mathbf{P}_i(\text{Markov chain } \{X_n\} \text{ eventually absorbed in } C) \\ &= \mathbf{P}(\text{Markov chain } \{X_n\} \text{ eventually absorbed in } C \,|\, X_0 = i).\end{aligned}$$

Write also $\boldsymbol{\alpha}^C = (\alpha_i^C)_{i \in \mathcal{S}}$.

Theorem 4.2.1

1. The vector $\boldsymbol{\alpha}^C$ satisfies the following system of equations:
$$\begin{aligned}\boldsymbol{\alpha}^C &= P\boldsymbol{\alpha}^C \\ \alpha_i^C &= 1 \qquad \text{for all } i \in C \\ \alpha_i^C &= 0 \qquad \text{for all } i \text{ in any other closed classes.}\end{aligned}$$
2. The vector $\boldsymbol{\alpha}^C$ is the *minimum* solution of the above system of equations.
3. In the case where the set of states in *nonclosed* classes is *finite* (in particular when $\mathcal{S}$ is *finite*) the vector $\boldsymbol{\alpha}^C$ is the *unique* solution of the above system of equations. Further in this case, for any initial state i, the sum of the absorption probabilities over all closed classes is 1.

Proof: For the present, we prove 1 only. By conditioning on the first step from state i and using the Markov property, we have
$$\begin{aligned}\alpha_i^C &= \mathbf{P}_i(\text{Markov chain } \{X_n\} \text{ eventually absorbed in } C) \\ &= \sum_{j\in\mathcal{S}} p_{ij}\mathbf{P}_j(\text{Markov chain } \{X_n\} \text{ eventually absorbed in } C) \\ &= \sum_{j\in\mathcal{S}} p_{ij}\alpha_j^C.\end{aligned}$$
Clearly $\alpha_i^C = 1$ for any $i \in C$ and $\alpha_i^C = 0$ for any i in any other closed class. □

Examples. We consider again some of the earlier examples.

Simple random walk on $\mathcal{S} = \{0, 1, 2, \dots\}$ with 0 an absorbing state. The state 0 forms the single closed class $C = \{0\}$, and all other states form an open class $O = \{1, 2, \dots\}$. Then the probability of absorption in the class C is just the probability of ruin. It is a straightforward exercise to use the above theorem, including the *minimality* condition of part 2, to give a direct proof of the earlier Corollary 3.2.2, i.e.
$$\mathbf{P}_a(R) = \alpha_a^C \equiv \alpha_a = \begin{cases}\left(\frac{q}{p}\right)^a & p > \frac{1}{2} \\ 1 & p \le \frac{1}{2}.\end{cases}$$

Health insurance system. The health insurance system considered above has a single closed class consisting of the absorbing state D. Since $\mathcal{S}$ is finite it follows from part 3 of the above theorem (or by direct calculation using part 1)that eventual absorption in this state is certain!

Further example. Consider again the example of the previous section with $\mathcal{S} = \{1, 2, 3, 4, 5\}$. Let $\boldsymbol{\alpha}$ be the vector of absorption probabilities in the closed class consisting of the single absorbing state 5. We have $\alpha_1 = \alpha_2 = 0$, $\alpha_5 = 1$, and
$$\begin{aligned}\alpha_3 &= \frac{1}{5}\alpha_3 + \frac{1}{5}\alpha_4 + \frac{1}{5} \\ \alpha_4 &= \frac{1}{3}\alpha_3 + \frac{1}{3},\end{aligned}$$
giving $\alpha_3 = 4/11$, $\alpha_4 = 5/11$.
It follows directly from part 3 of the above theorem that the vector of absorption probabilities in the closed class $\{1, 2\}$ is $\mathbf{1} - \boldsymbol{\alpha}$.

4.3 Recurrence and transience

Throughout this section $\{X_n\}_{n\geq 0}$ continues to be a time homogeneous Markov chain with state space $\mathcal{S}$ and transition matrix $P = (p_{ij})_{i,j\in\mathcal{S}}$.

4.3.1 Recurrent and transient states

For each $i, j \in \mathcal{S}$ define

$$F_{ij} = \mathbf{P}_i(X_n = j \text{ for } \textit{some } n > 0)$$
$$[= \mathbf{P}_i(\text{process } \{X_n\} \text{ hits state } j \text{ at some time } n > 0)]$$

Clearly, for all i, j, we have $0 \leq F_{ij} \leq 1$ and $F_{ij} > 0$ if and only if $i \to j$.
In particular we have

$$F_{ii} = \mathbf{P}_i(\text{process } \{X_n\} \text{ ever returns to state } i).$$

Let N be the number of visits of state i, or equivalently,

$$N = Number\{n \geq 1 \ : \ X_n = i\}. \qquad \text{Here } \ 0 \leq N \leq \infty.$$

Let τ be the time of the first visit to state i,

$$\tau = \min\{n \geq 1 \ : \ X_n = i\} \ \text{ if such times exists,}$$

and $\tau = \infty$ if $X_n \neq i$, for all $n = 1, 2, \ldots$.
Then

$$F_{ii} = \mathbf{P}_i(N \geq 1) = \mathbf{P}_i(\tau < \infty).$$

Definitions. The state $i \in \mathcal{S}$ is called **recurrent** if $F_{ii} = 1$, and **transient** if $F_{ii} < 1$.
Examples: simple random walk with one or two barriers (that may be either absorbing or reflecting), no-claim discount scheme, healthy-sick-dead model.
We further list properties of random variable N with their proofs.

1. For $k \geq 2$

$$\mathbf{P}_i(N \geq k) = \sum_{n=1}^{\infty} \mathbf{P}_i(N \geq k, \tau = n)$$
$$= \sum_{n=1}^{\infty} \mathbf{P}_i(N \geq k | \tau = n)\mathbf{P}_i(\tau = n)$$
$$= \sum_{n=1}^{\infty} \mathbf{P}_i(\tau = n)\mathbf{P}(N \geq k | X_n = i, X_{n-1} \neq i, \ldots, X_1 \neq i, X_0 = i)$$
$$\text{(Markov property)} = \sum_{n=1}^{\infty} \mathbf{P}_i(\tau = n)\mathbf{P}(\text{ there are at least } k-1 \text{ visits to } i \text{ after time } n | X_n = i)$$
$$\text{(Time homogeneity)} = \sum_{n=1}^{\infty} \mathbf{P}_i(\tau = n)\mathbf{P}_i(N \geq k-1)$$
$$= \mathbf{P}_i(N \geq k-1) \sum_{n=1}^{\infty} \mathbf{P}_i(\tau = n)$$
$$= \mathbf{P}_i(N \geq k-1)\mathbf{P}_i(\tau < \infty)$$
$$= \mathbf{P}_i(N \geq k-1)F_{ii}.$$

By induction we get

$$\mathbf{P}_i(N \geq k) = F_{ii}^k.$$

2. $\mathbf{P}_i(N = k) = \mathbf{P}_i(N \geq k) - \mathbf{P}_i(N \geq k+1) = F_{ii}^k - F_{ii}^{k+1} = F_{ii}^k(1 - F_{ii})$.

3. $\mathbf{P}_i(N = \infty) = \lim_{k\to\infty} \mathbf{P}_i(N \geq k) = \lim_{k\to\infty} F_{ii}^k$. If $F_{ii} = 1$ then the limit is equal to 1 otherwise it is 0. Therefore

$$\mathbf{P}_i(N = \infty) = \begin{cases} 1 & \text{if } F_{ii} = 1 \\ 0 & \text{if } F_{ii} < 1 \end{cases}$$

4. For a random variable X that takes only non-negative integer values, $\mathbf{E}\, X = \sum_{k=1}^{\infty} \mathbf{P}(X \geq k)$. Therefore, if $F_{ii} < 1$, then we have

$$\mathbf{E}_i N = \sum_{k=1}^{\infty} \mathbf{P}_i(N \geq k) = \sum_{k=1}^{\infty} F_{ii}^k = \frac{F_{ii}}{1 - F_{ii}},$$

and if $F_{ii} = 1$, then we have $\mathbf{E}_i N = \infty$. With the convention that $\frac{1}{0} = \infty$ we can again write $\mathbf{E}_i N = \frac{F_{ii}}{1-F_{ii}}$.

5. Alternatively, by considering **indicator random variables** for successive possible hits on state i, we have

$$N = \mathbb{I}(X_1 = i) + \mathbb{I}(X_2 = i) + \ldots + \mathbb{I}(X_n = i) + \ldots$$

and

$$\begin{aligned} \mathbf{E}_i N = \mathbf{E}_i \left(\sum_{n=1}^{\infty} \mathbb{I}(X_n = i) \right) = \sum_{n=1}^{\infty} \mathbf{E}_i \left(\mathbb{I}(X_n = i) \right) = \\ = \sum_{n=1}^{\infty} \mathbf{P}_i(X_n = i) = \sum_{n=1}^{\infty} \mathbf{P}(X_n = i | X_0 = i) = \sum_{n=1}^{\infty} p_{ii}^{(n)}, \end{aligned}$$

since $\mathbf{E}\, \mathbb{I}(A) = \mathbf{P}(A)$ for any event A.

The following theorem is immediate from the above results.

Theorem 4.3.1 For any state $i \in \mathcal{S}$, we have

$$\begin{aligned} i \text{ recurrent} &\Leftrightarrow F_{ii} = 1 \\ &\Leftrightarrow \mathbf{P}_i(\text{process } \{X_n\} \text{ ever returns to state } i) = 1 \\ &\Leftrightarrow \mathbf{P}_i(\text{process } \{X_n\} \text{ returns to state } i \text{ infinitely often}) = 1 \\ &\Leftrightarrow \mathbf{E}_i(\text{number of returns by process } \{X_n\} \text{ to state } i) = \infty \\ &\Leftrightarrow \sum_{n=1}^{\infty} p_{ii}^{(n)} = \infty. \end{aligned}$$

and also

$$\begin{aligned} i \text{ transient} &\Leftrightarrow F_{ii} < 1 \\ &\Leftrightarrow \mathbf{P}_i(\text{process } \{X_n\} \text{ ever returns to state } i) < 1 \\ &\Leftrightarrow \mathbf{P}_i(\text{process } \{X_n\} \text{ returns to state } i \text{ infinitely often}) = 0 \\ &\Leftrightarrow \mathbf{E}_i(\text{number of returns by process } \{X_n\} \text{ to state } i) < \infty \\ &\Leftrightarrow \sum_{n=1}^{\infty} p_{ii}^{(n)} < \infty. \end{aligned}$$

Corollary 4.3.2 If $i \in \mathcal{S}$ is **transient**, then $\lim_{n\to\infty} p_{ii}^{(n)} = 0$.

The significance of a **transient** state is that, with probability 1, the process $\{X_n\}$ eventually leaves it forever. A **recurrent** state, if visited at all, is returned to infinitely often. It is possible to further classify recurrent states, In the case where $i \in \mathcal{S}$ is **recurrent** (and so, starting in state i, return to state i is certain), define

$$\mu_i = \mathbf{E}_i(\text{time to first return to state } i) \equiv \mathbf{E}_i\tau.$$

Definitions. The **recurrent** state $i \in \mathcal{S}$ is **positive recurrent** if $\mu_i < \infty$ and **null recurrent** if $\mu_i = \infty$.

Comment: Here $1/\mu_i$ is the *intensity to return* to i.

Comment: In this lecture course, we will not deal with *positive/null* recurrence.

4.3.2 Recurrent and transient classes

We now relate **recurrence/transience** to the idea of **classes** of intercommunicating states defined in Section 4.2. In particular we have the following very important theorem.

Theorem 4.3.3 In any class (of intercommunicating states), *either* all states are **recurrent**, *or* all states are **transient**.

Proof: Suppose that i and j belong to the same class. It is sufficient to show that if j is recurrent, then i is recurrent. Since i and j intercommunicate we can find $m > 0$, $n > 0$ such that $p_{ij}^{(m)} > 0$, $p_{ji}^{(n)} > 0$. Further, since j is recurrent, $\sum_{k\geq 0} p_{jj}^{(k)} = \infty$. Hence, from a double application of the Chapman-Kolmogorov inequalities, for any $k \geq 0$,

$$\begin{aligned}
\sum_{l\geq 0} p_{ii}^{(l)} &\geq \sum_{l\geq m+n} p_{ii}^{(l)} = \sum_{k\geq 0} p_{ii}^{(m+k+n)} \\
&\geq \sum_{k\geq 0} p_{ij}^{(m)} p_{jj}^{(k)} p_{ji}^{(n)} \\
&= p_{ij}^{(m)} p_{ji}^{(n)} \sum_{k\geq 0} p_{jj}^{(k)} \\
&= \infty,
\end{aligned}$$

and so, by Theorem 4.3.1, i is also recurrent. □

It follows that we may speak of **recurrent** or **transient classes**. It is also true, though

we do not prove it here, that, in any **recurrent** class, *either* all states are **positive recurrent**, *or* all states are **null recurrent**.

Recall that (by definition) within any recurrent class C, $F_{ii} = 1$ for all $i \in C$. The following theorem extends this idea.

Theorem 4.3.4 Suppose that the class C is **recurrent**. Then C is **closed**. Further $F_{ij} = 1$ for all $i, j \in C$.

Proof: Suppose that $j \in C$ and that $i \in \mathcal{S}$ is such that $j \to i$. We show that $F_{ij} = 1$. This implies that $i \to j$. Hence both assertions of the theorem follow immediately.

Since C is recurrent, we have

$$\begin{aligned}
1 &= \mathbf{P}_j(\text{process } \{X_n\} \text{ returns to state } j \text{ infinitely often}) \\
&= \mathbf{P}_j(\{X_n\} \text{ hits state } i \text{ at some time and hits state } j \text{ infinitely often}) \\
&\qquad + \mathbf{P}_j(\{X_n\} \text{ never hits state } i \text{ and hits state } j \text{ infinitely often}) \\
&\le \mathbf{P}_j(\{X_n\} \text{ hits state } i \text{ at some time and then state } j \text{ at some later time}) \\
&\qquad + \mathbf{P}_j(\{X_n\} \text{ never hits state } i) \\
&= F_{ji}F_{ij} + 1 - F_{ji}.
\end{aligned}$$

Hence

$$0 \le F_{ji}(F_{ij} - 1).$$

Since $j \to i$ we have $F_{ji} > 0$ and so $F_{ij} \ge 1$. But F_{ij} is a probability that cannot exceed 1. Therefore $F_{ij} = 1$.

(The intuitive idea is the following: suppose that the process $\{X_n\}$ starts in state j; return to state j infinitely often is certain (recurrence); but also the process visits state i with strictly positive probability; hence from that state return to state j is certain.) □

It follows that any **nonclosed (open)** class is **transient**.

We finally have the following theorem.

Theorem 4.3.5 Any **finite closed** class C is **recurrent**.

Proof: Fix $i \in C$. Then, since C is finite and closed,

$$\mathbf{P}_i(\text{process } \{X_n\} \text{ hits } \textit{some} \text{ state in } C \text{ infinitely often}) = 1$$

and so necessarily there exists some $j \in C$ such that

$$\begin{aligned}
0 &< \mathbf{P}_i(\{X_n\} \text{ hits state } j \text{ infinitely often}) \\
&= F_{ij}\mathbf{P}_j(\{X_n\} \text{ hits state } j \text{ infinitely often}).
\end{aligned}$$

Hence

$$\mathbf{P}_j(\{X_n\} \text{ hits state } j \text{ infinitely often}) > 0,$$

and so, by Theorem 4.3.1, j is recurrent, and so all states in C are recurrent. □

Examples. We revisit some of the earlier examples (though not in the same order).

Bonus-malus system. Recall that, provided $0 < p < 1$, the state space $\mathcal{S}$ is here a single, necessarily closed, class. Hence it is recurrent.

Health insurance system. Recall that here we obtained the earlier classification:

Class	Type
$\{H, S\}$	nonclosed
$\{D\}$	closed (absorbing)

It follows that the states H and S (which form a nonclosed class) are transient, while the state D (which forms a finite closed class) is recurrent.

Simple random walk. For the simple random walk with $0 < p < 1$ (and $q = 1 - p$), we have already observed that the state space is a single, necessarily closed, class. To check recurrence/transience, it is sufficient to compute F_{00}.

Solution 1. By conditioning as usual on the first step of the random walk, we have

$$F_{00} = pF_{10} + qF_{-10}$$

F_{10} and F_{-10} may be obtained from our earlier results on ruin.

For $p > 1/2$, we can use Corollary 3.2.2. to conclude that $F_{10} = q/p$ and $F_{-10} = 1$. Hence $F_{00} = 2q < 1$ and so state 0 (and hence all states) are transient.

For $p = 1/2$, we have $F_{10} = 1$ and $F_{-10} = 1$. Hence $F_{00} = 1$ and so state 0 (and hence all states) are recurrent.

For $p < 1/2$, it follows by Corollary 3.2.2. and by symmetry (interchanging p and q) that $F_{00} = 2p < 1$ and so state 0 (and hence all states) are again transient.

Solution 2. By Theorem 4.3.1, state 0 is recurrent if and only if $\sum_{n=1}^{\infty} p_{00}^{(n)} = \infty$. Earlier (see equation (2)) we obtained that $p_{00}^{(2n)} \sim \frac{1}{\sqrt{\pi n}} \cdot (4pq)^n$ and that $p_{00}^{(2n+1)} = 0$. Then, by basic results from calculus, $\sum p_{00}^{(n)} = \infty$ if and only if $4pq = 1$, and if and only if $p = q = 1/2$.
Indeed, if $4pq = 1$, then

$$\sum_{n \geq 1} p_{00}^{(2n)} \sim \frac{1}{\sqrt{\pi}} \sum_{n \geq 1} \frac{1}{n^{1/2}} = \infty$$

since $\sum_n n^{-\alpha} = \infty$ if and only if $\alpha \leq 1$ (see the calculus course),
and if $r := 4pq < 1$, then

$$\sum_{n \geq 1} p_{00}^{(2n)} \sim \frac{1}{\sqrt{\pi}} \sum_{n \geq 1} r^n = \frac{1}{\sqrt{\pi}} \frac{r}{1 - r} < \infty.$$

Exercise. Show that in the simple random walk on the nonnegative integers, modified so that state 0 is an absorbing state, this latter state is recurrent while all other states are transient.

4.4 Stationary and limiting distributions for a single closed class

Throughout this section $\{X_n\}_{n\geq 0}$ continues to be a time homogeneous Markov chain with state space $\mathcal{S}$ and transition matrix $P = (p_{ij})_{i,j\in\mathcal{S}}$. We further assume that the entire state space $\mathcal{S}$ is a **single, necessarily closed, class**, i.e. all states intercommunicate. The Markov chain is then said to be **irreducible**. This is the case in many applications.

4.4.1 Stationary distributions

Recall that the Markov chain $\{X_n\}_{n\geq 0}$ is such that $\mathcal{S}$ is a single closed class.

Recall that a vector $\boldsymbol{\pi} = (\pi_i)_{i\in\mathcal{S}}$ on $\mathcal{S}$ is a **(probability) distribution** if and only if $\pi_i \geq 0$ for all i and $\sum_{i\in\mathcal{S}} \pi_i = 1$.

Definition. A (probability) distribution $\boldsymbol{\pi}$ on $\mathcal{S}$ is **stationary** if and only if

$$\boldsymbol{\pi}P = \boldsymbol{\pi} \qquad \text{or equivalently} \qquad \sum_{i\in\mathcal{S}} \pi_i p_{ij} = \pi_j, \quad j \in \mathcal{S}. \tag{7}$$

We then have also, for all $n \geq 1$,

$$\boldsymbol{\pi}P^n = \boldsymbol{\pi} \qquad \text{or equivalently} \qquad \sum_{i\in\mathcal{S}} \pi_i p_{ij}^{(n)} = \pi_j, \quad j \in \mathcal{S}. \tag{8}$$

i.e. if the distribution of X_0 is $\boldsymbol{\pi}$, then, for all $n \geq 1$, the distribution of X_n is also $\boldsymbol{\pi}$.

A **stationary** distribution is also known as an **equilibrium** distribution for the Markov chain. An informal interpretation (which may be made formal) is that it gives the **long-term proportion of time** spent in each of the states.

Theorem 4.4.1 Assume the Markov chain to be irreducible. Suppose that the distribution $\boldsymbol{\pi}$ is **stationary**. Then $\pi_j > 0$ for all $j \in \mathcal{S}$.

Proof: Since $\sum_{i\in\mathcal{S}} \pi_i = 1$ we have $\pi_i > 0$ for *some* $i \in \mathcal{S}$. Then, since $\mathcal{S}$ is a single class, for any other $j \in \mathcal{S}$, there exists n such that $p_{ij}^{(n)} > 0$, and so, from (8), $\pi_j > 0$. □

Results

1. A **stationary** distribution **need not exist**.
2. If a **stationary** distribution **does exist** then it is **unique** and $\mathcal{S}$ is **positive recurrent**. This is always the case when $\mathcal{S}$ is **finite**.
3. When $\mathcal{S}$ is **infinite** and **no stationary distribution exists**, $\mathcal{S}$ is either **null recurrent** or **transient**.

Thus, if $\mathcal{S}$ is known to be **recurrent**, we may distinguish **positive recurrence** and **null recurrence** by checking whether or not a **stationary** distribution exists.

That a stationary distribution always exists when $\mathcal{S}$ is finite follows from the fact that, since P is a stochastic matrix, i.e. $P\mathbf{1} = \mathbf{1}$ (where $\mathbf{1} = (1, 1, \ldots, 1)$), 1 is an eigenvalue of P and so also there exists a left eigenvector $\boldsymbol{\pi}$ such that $\boldsymbol{\pi}P = \boldsymbol{\pi}$. However, in the next section we *outline* a probabilistic proof which also gives the asserted uniqueness.

Finding the stationary distribution. *Either*

1. Solve the equations (7) (note that any one of them can always be omitted—see below), together with the condition $\sum_{i\in\mathcal{S}} \pi_i = 1$, *or*
2. Guess the right answer $\boldsymbol{\pi}$ (you may be given this). If it satisfies the equations (7) and $\sum_{i\in\mathcal{S}} \pi_i = 1$ then it is indeed the unique stationary distribution.

Example: bonus-malus system. Consider the earlier example with the transition matrix

$$P = \begin{pmatrix} p & 1-p & 0 \\ p & 0 & 1-p \\ p & 0 & 1-p \end{pmatrix}.$$

The equations (7) for a stationary distribution $\boldsymbol{\pi}$ here become

$$\begin{aligned} p\pi_1 + p\pi_2 \quad + p\pi_3 \quad &= \pi_1 \\ (1-p)\pi_1 \qquad\qquad\quad &= \pi_2 \\ (1-p)\pi_2 + (1-p)\pi_3 \quad &= \pi_3 \end{aligned}$$

While this looks like 3 equations in 3 unknowns, any given one of these equations is implied by the remainder (as is always the case with the equations (7)—add them up to see this), and we must add the further requirement that $\boldsymbol{\pi}$ be a distribution, i.e.

$$\pi_1 + \pi_2 + \pi_3 = 1.$$

We thus obtain that the Markov chain corresponding to the transition matrix P has the unique stationary distribution

$$\boldsymbol{\pi} = (p,\, p(1-p),\, (1-p)^2).$$

Example: simple random walk. For the simple random walk we can check directly from the equations (7) that there is **no** stationary distribution (*exercise, if you like—you will need to remember that the components of a stationary distribution sum to* 1). However, suppose the contrary, i.e. that a, necessarily unique, stationary distribution π does exist. Then by the (spatial) translation invariance of the random walk, π_i is constant for all $i \in \mathcal{S}$ and also $\sum_{i\in\mathcal{S}} \pi_i = 1$, which is impossible since $\mathcal{S}$ is infinite. Indeed this argument shows that *any* random walk fails to have a stationary distribution.

4.4.2 Limiting distributions

We continue to assume that the Markov chain $\{X_n\}_{n\geq 0}$ is such that $\mathcal{S}$ is a single closed class.

Definition. The Markov chain $\{X_n\}_{n\geq 0}$ is **periodic with period** $d \geq 2$ if and only if (a) the state space $\mathcal{S}$ may be represented as a union of d disjoint sets,

$$\mathcal{S} = \mathcal{S}_1 \bigcup \mathcal{S}_2 \bigcup \ldots \bigcup \mathcal{S}_d$$

such that, for any $i = 1, 2, \ldots, d$ and for any $x \in \mathcal{S}_i$,

$$\mathbf{P}(X_1 \in \mathcal{S}_{i+1} \mid X_0 = x) = 1$$

(here $\mathcal{S}_{d+1} = \mathcal{S}_1$), and
(b) d is the biggest number for which such a splitting exists.

Example

$$P = \begin{pmatrix} 0 & 1/3 & 2/3 & 0 \\ 1/2 & 0 & 0 & 1/2 \\ 1/2 & 0 & 0 & 1/2 \\ 0 & 1/3 & 2/3 & 0 \end{pmatrix}.$$

Here $d = 2$ and there are two corresponding subclasses $\mathcal{S}_1 = \{1, 4\}$ and $\mathcal{S}_2 = \{2, 3\}$ such that $\mathbf{P}(X_1 \in \mathcal{S}_2 \mid X_0 \in \mathcal{S}_1) = 1$ and $\mathbf{P}(X_1 \in \mathcal{S}_1 \mid X_0 \in \mathcal{S}_2) = 1$.

Definition. The Markov chain $\{X_n\}_{n\geq 0}$ is **aperiodic** if it is not periodic with any $d \geq 2$.

Remark. Most chains arising in practical applications are aperiodic.

Here are two sufficient conditions for aperiodicity.

Condition 1. If there exists $j \in \mathcal{S}$ such that $p_{jj} > 0$, then the Markov Chain is aperiodic. *Indeed*, if $d > 1$, then j should belong to more than one of subsets $\mathcal{S}_i$, but this cannot be true. Therefore, $d = 1$.

Condition 2. If there exists $j \in \mathcal{S}$ such that, for some $n \geq 1$, both $p_{jj}^{(n)} > 0$ and $p_{jj}^{(n+1)} > 0$, then the Markov Chain is aperiodic.
Indeed, if $d > 1$, then $j \in \mathcal{S}_k$, for some $1 \leq k \leq d$. Then both n and $n + 1$ must be the multiples of d. But this cannot be true.

The following characterisation result holds:

Theorem 4.4.2 The Markov chain $\{X_n\}_{n\geq 0}$ is **aperiodic** if and only if there exists some $n > 0$ such that $p_{ij}^{(n)} > 0$ for all $i, j \in \mathcal{S}$.

We provide the Proof in one direction only. Assume that $p_{ij}^{(n_0)} > 0$ for some $n_0 \geq 1$ and for all $i, j \in \mathcal{S}$.

First, we show that then

$$p_{ij}^{(n)} > 0 \quad \text{for all} \quad i, j \in \mathcal{S} \quad \text{and all} \quad n > n_0. \tag{9}$$

Indeed, represent $n > n_0$ as $n = m + n_0$. Then

$$p_{ij}^{(n)} = \sum_k p_{ik}^{(m)} p_{kj}^{(n_0)}.$$

Since $\sum_k p_{ik}^{(m)} = 1$, there is k_0 such that $p_{i,k_0}^{(m)} > 0$. Then $p_{ij}^{(n)} \geq p_{i,k_0}^{(m)} p_{k_0,j}^{(n_0)} > 0.$, Therefore, we have proved (9).

Then, in particular, we have $p_{ii}^{(n_0)} > 0$ and $p_{ii}^{(n_0+1)} > 0$, for all $i \in \mathcal{S}$. Then Condition 2 implied the aperiodicity.

Definition. Fix any $i \in \mathcal{S}$. If, for any $j \in \mathcal{S}$, there exists a limit

$$\omega_{ij} = \lim_{n\to\infty} p_{i,j}^{(n)}$$

and if

$$\sum_{j\in\mathcal{S}} \omega_{ij} = 1,$$

then $\{\omega_{ij}\}_{j\in\mathcal{S}}$ is called a **limiting distribution** for initial state i.
Similarly, for any initial distribution $\mu = \{\mu_i\}$ on $\mathcal{S}$, a *limiting distribution* for this initial distribution is

$$\omega_{\mu,j} = \lim_{n\to\infty} \sum_{i\in\mathcal{S}} \mu_i p_{i,j}^{(n)}$$

The following theorem, which we do not prove, is crucial. It implies that the chain "eventually forgets its initial state".

Theorem 4.4.3 Suppose that the Markov chain $\{X_n\}_{n\geq 0}$ (is such that $\mathcal{S}$ is a **single closed class** and) is **aperiodic**. Then there exists a vector $\boldsymbol{\omega} = (\omega_j)_{j\in\mathcal{S}}$ such that, for all $i, j \in \mathcal{S}$,

$$\lim_{n\to\infty} p_{ij}^{(n)} = \omega_j.$$

independently of i.

Given this result, we can prove the following theorem for the case where $\mathcal{S}$ is **finite**. The theorem also justifies some of our earlier assertions.

Theorem 4.4.4 Suppose that $\mathcal{S}$ is **finite** (and a **single closed class**) and that the Markov chain $\{X_n\}_{n\geq 0}$ is **aperiodic**. Then $\boldsymbol{\omega}$ as defined above is the **unique stationary distribution** of the chain.

Proof: Since, for any i and for all n, $\sum_{j\in\mathcal{S}} p_{ij}^{(n)} = 1$, letting $n \to \infty$ and using Theorem 4.4.3, we obtain

$$\sum_{j\in\mathcal{S}} \omega_j = 1.$$

Therefore, $\{\omega_j\}$ is a distribution.
Further, for any $n \geq 1$ and for any $i, j \in \mathcal{S}$, we have, by the Chapman-Kolmogorov equations (or just see it directly)

$$p_{ij}^{(n+1)} = \sum_{k\in\mathcal{S}} p_{ik}^{(n)} p_{kj}.$$

Again letting $n \to \infty$ and using Theorem 4.4.3, we obtain, for all j,

$$\omega_j = \sum_{k\in\mathcal{S}} \omega_k p_{kj}.$$

Hence $\boldsymbol{\omega}$ is a stationary distribution.
To prove uniqueness, suppose that $\boldsymbol{\pi}$ is *any* stationary distribution for the Markov chain. Then, letting $n \to \infty$ in equation (8), we obtain, for all j,

$$\begin{aligned}\pi_j &= \sum_{i\in\mathcal{S}} \pi_i \omega_j \\ &= \omega_j \sum_{i\in\mathcal{S}} \pi_i \\ &= \omega_j\end{aligned}$$

since $\sum_{i\in\mathcal{S}} \pi_i = 1$. Thus, the stationary distribution is unique. □

Remark. In the case where $\mathcal{S}$ is **infinite** (and a **single closed class**) and the Markov chain $\{X_n\}_{n\geq 0}$ is **aperiodic**, it can be shown that *either*

1. there exists a **unique** stationary distribution which is again the vector $\boldsymbol{\omega}$ of limiting probabilities as above, *or*
2. there is **no** stationary distribution and $\omega_j = 0$ for all $j \in \mathcal{S}$.

Example: bonus-malus system. In the above example it is clear that the Markov chain is aperiodic, and we conclude that, for all $i, j \in \mathcal{S}$, $\lim_{n\to\infty} p_{ij}^{(n)} = \omega_j$, where

$$\boldsymbol{\omega} = (p,\, p(1-p),\, (1-p)^2).$$

4.4.3 More about Markov Chains with a finite state space

We already know:
if $\mathcal{S}$ is a finite closed class and **if** $\{X_n\}$ is aperiodic, **then** there exists a unique stationary distribution satisfying equations, for all $j \in \mathcal{S}$,

$$\sum_i \pi_i p_{i,j} = \pi_j \quad \text{and} \quad \sum_j \pi_j = 1,$$

and there exists a unique limiting distribution (which HAS TO COINCIDE with the stationary distribution), i.e., for any $i, j \in \mathcal{S}$,

$$p_{ij}^{(n)} \to \pi_j.$$

Therefore, for any initial distribution $\mu = \{\mu_i\}$, for any $j \in \mathcal{S}$,

$$\begin{aligned} \mathbf{P}(X_n = j) &= \sum_{i \in \mathcal{S}} \mathbf{P}(X_0 = i) p_{i,j}^{(n)} \\ &= \sum_i \mu_i p_{i,j}^{(n)} \\ &\to \sum_i \mu_i \pi_j = \pi_j. \end{aligned}$$

Assume further that $\mathcal{S}$ is finite. What one can say about EXISTENCE and UNIQUENESS of stationary and limiting distributions when $\{X_n\}$ is periodic or/and $\mathcal{S}$ is not a single closed class ?

Should they always (a) exist ? (b) coincide ? (c) be unique ?

Here are the answers.

I. A STATIONARY distribution always exists (at least one), but is non-unique, in general.

II. A LIMITING distribution may exist may not (depending on initial value). If exists, it is not unique, in general.

III. Any LIMITING is STATIONARY (in more detail: if a limiting distribution exists, it is stationary).

IV. Any STATIONARY is LIMITING (in more detail: if a stationary distribution exists, it is limiting for at least one initial (in general, random) value).

Consider now some particular cases.

(1). Assume there exists (exactly) one closed class C and within this class a Markov Chain is aperiodic.

Then there exists a unique stationary distribution $\{\pi_j\}$ such that

$$\sum_{j\in C}\pi_j = 1, \quad \pi_j > 0 \ \forall j \in C, \quad \text{and} \quad \pi_j = 0 \ \forall j \in S \setminus C.$$

Therefore, there exists a unique limiting distribution which coincides with the stationary one.

Example

$$P = \begin{pmatrix} 1/4 & 1/4 & 1/4 & 1/4 \\ 1/3 & 1/3 & 1/3 & 0 \\ 0 & 0 & 1/3 & 2/3 \\ 0 & 0 & 1/2 & 1/2 \end{pmatrix}.$$

There are two classes: $\{1,2\}$ is non-closed; $\{3,4\}$ is closed. Within the closed class, the transition matrix is

$$\tilde{P} = \begin{pmatrix} 1/3 & 2/3 \\ 1/2 & 1/2 \end{pmatrix}.$$

and the stationary distribution has to satisfy the following equations:

$$\frac{1}{3}\pi_3 + \frac{1}{2}\pi_4 = \pi_3,$$

$$\frac{2}{3}\pi_3 + \frac{1}{2}\pi_4 = \pi_4,$$

$$\pi_3 + \pi_4 = 1.$$

Then

$$\pi = \left(0, 0, \frac{3}{7}, \frac{4}{7}\right).$$

(2). Assume that $\mathcal{S}$ is a single closed class and $\{X_n\}$ is periodic with period $d \geq 2$ (this means that the state space $\mathcal{S}$ may be divided into d subsets $\mathcal{S}_1, \ldots, \mathcal{S}_d$ such that $\mathbf{P}(X_1 \in \mathcal{S}_{i+1} \mid X_0 = x \in \mathcal{S}_i) = 1$, for all x and for all i).

Then there exists a unique stationary distribution $\{\pi_j\}_{j\in\mathcal{S}}$ where

$$\sum_{j\in\mathcal{S}_1} \pi_j = \sum_{j\in\mathcal{S}_2} \pi_j = \ldots = \sum_{j\in\mathcal{S}_d} \pi_j = \frac{1}{d}.$$

This stationary distribution is limiting for an initial distribution $\mu = \{\mu\}$ **if** and **only if** the initial distribution $\mu = \{\mu_j\}$ is such that

$$\sum_{j\in\mathcal{S}_1} \mu_j = \sum_{j\in\mathcal{S}_2} \mu_j = \ldots = \sum_{j\in\mathcal{S}_d} \mu_j = \frac{1}{d}.$$

Comment. Recall that a limiting distribution for an initial distribution $\mu = \{\mu\}$ is

$$\omega_{\mu,j} = \lim_{n\to\infty} \sum_{i\in\mathcal{S}} \mu_i p_{i,j}^{(n)}.$$

Example

$$P = \begin{pmatrix} 0 & 1/3 & 2/3 & 0 \\ 1/2 & 0 & 0 & 1/2 \\ 1/2 & 0 & 0 & 1/2 \\ 0 & 1/3 & 2/3 & 0 \end{pmatrix}.$$

Recall that here $d = 2$ and there are two corresponding subclasses $\mathcal{S}_1 = \{1, 4\}$ and $\mathcal{S}_2 = \{2, 3\}$ such that $\mathbf{P}(X_1 \in \mathcal{S}_2 \mid X_0 \in \mathcal{S}_1) = 1$ and $\mathbf{P}(X_1 \in \mathcal{S}_1 \mid X_0 \in \mathcal{S}_2) = 1$.

There are two ways to find the stationary distribution.

The *first* way (direct) is: to solve a family of equations

$$\pi_1 \cdot 0 + \pi_2 \cdot \frac{1}{2} + \pi_3 \cdot \frac{1}{2} + \pi_4 \cdot 0 = \pi_1,$$

$$\pi_1 \cdot \frac{1}{3} + \pi_2 \cdot 0 + \pi_3 \cdot 0 + \pi_4 \cdot \frac{1}{3} = \pi_2,$$

$$\pi_1 \cdot \frac{2}{3} + \pi_2 \cdot 0 + \pi_3 \cdot 0 + \pi_4 \cdot \frac{2}{3} = \pi_1,$$

$$\pi_1 \cdot 0 + \pi_2 \cdot \frac{1}{2} + \pi_3 \cdot \frac{1}{2} + \pi_4 \cdot 0 = \pi_1,$$

$$\pi_1 + \pi_2 + \pi_3 + \pi_4 = 1.$$

The *second* way (by reduction) is:

Consider the Markov Chain with transition matrix P^2. For this Markov Chain, the sets $\mathcal{S}_1$ and $\mathcal{S}_2$ form two closed classes.

Consider first one of closed classes (say, the class $\mathcal{S}_1$ its reduced matrix (the matrix of transition probabilities within the class) is

$$\tilde{P} = \begin{pmatrix} p_{11}^{(2)} & p_{14}^{(2)} \\ p_{41}^{(2)} & p_{44}^{(2)} \end{pmatrix} = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix}$$

and solve the set of equations

$$\frac{1}{2}\pi_1 + \frac{1}{2}\pi_4 = \pi_1, \quad \pi_1 + \pi_4 = \frac{1}{2}.$$

Then $\pi_1 = \pi_4 = \frac{1}{4}$, and then

$$\pi_2 = \frac{1}{3}\pi_1 + \frac{1}{3}\pi_4 = \frac{1}{6}, \quad \pi_3 = \frac{2}{3}\pi_1 + \frac{2}{3}\pi_4 = \frac{1}{3}.$$

(3). Assume there are two or more (say r) closed classes and, within each class, a Markov Chain is aperiodic.

In this case, there are many *stationary* distributions.

Let P_1 be a reduction of a matrix P to the class C_1 (i.e. we delete all rows and columns for j from other classes).

Within class C_1, there exists a unique stationary distribution $\{\pi_j^{(1)}\}_{j\in C_1}$ which satisfies the following equations:

$$\sum_{j\in C_1} \pi_i^{(1)} p_{ij} = \pi_j^{(1)} \quad \forall j \in C_1.$$

Similarly, let $\{\pi_j^{(2)}\}_{j\in C_2}, \ldots, \{\pi_j^{(r)}\}_{j\in C_r}$ be stationary distributions within other classes. Let $\gamma_1, \ldots, \gamma_r \geq 0$ be any numbers such that $\sum_{m=1}^{r} \gamma_m = 1$.

Let $\pi_j = \pi_j^{(k)} \gamma_k$ for $j \in C_k$, $k = 1, \ldots, r$ and $\pi_j = 0$ if j is a transient state. $(*)$

Then

(i) $\{\pi_j\}$ *is a stationary distribution*,

(ii) *any stationary distribution satisfies equations* $(*)$.

Turn to **limiting** distributions.

(a) If $X_0 = i$ and $i \in C_k$ for some closed class C_k, then the limitiing distribution is the stationary distribution $\{\pi_h^{(k)}\}_{j\in C_k}$ within that class.

(b) If $X_0 = i$ and i is transient, then let $\alpha_i^{(k)}$ be the probability of absorption in class C_k. Then the limiting distribution is:

for any $k = 1, \ldots, r$,

$$\omega_j = \pi_j^{(k)} \cdot \alpha_j^{(k)} \quad \text{for} \quad j \in C_k$$

and $\omega_j = 0$ for any transient state j.

c) If X_0 is random ... —- For you !

Example

$$P = \begin{pmatrix} 1/4 & 1/4 & 1/4 & 1/4 \\ 0 & 1/3 & 2/3 & 0 \\ 0 & 1/3 & 2/3 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Here $\{1\}$ is a non-closed class, and there are two closed classes: $\{2,3\}$ is closed and aperiodic, and $\{4\}$ is absorbing (and, therefore, aperiodic too !).

The *stationary distribution within class* $C_1 = \{2,3\}$ is

$$\left(\pi_2^{(1)}, \pi_3^{(1)}\right) = \left(\frac{1}{3}, \frac{2}{3}\right).$$

The *stationary distribution within class* $C_2 = \{4\}$ is

$$\pi_2^{(4)} = 1.$$

Thus, a general form of a stationary distribution is:

$$\pi = \left(0, \frac{\gamma_1}{3}, \frac{2\gamma_1}{3}, \gamma_2\right)$$

where $\gamma_1, \gamma_2 \geq 0$ and $\gamma_1 + \gamma_2 = 1$.

Now turn to *limiting distributions*.

If either $X_0 = 2$ or $X_0 = 3$, then the limiting distribution is

$$\omega = \left(0, \frac{1}{3}, \frac{2}{3}, 0\right).$$

If $X_0 = 4$, then

$$\omega = (0, 0, 0, 1).$$

If $X_0 = 1$, then

$$\alpha_1^{(1)} = \frac{1}{3} + \frac{1}{3} = \frac{2}{3}, \quad \alpha_1^{(2)} = \frac{1}{3},$$

and

$$\omega = \left(0, \frac{2}{9}, \frac{4}{9}, \frac{1}{3}\right).$$

If X_0 is random... —- FOR YOU !!!

5 Properties of Some Standard Distributions and Poisson Processes

Here we consider properties of standard families of distributions: Bernoulli, Binomial, Poisson, Geometric, Exponential, Gamma, and Uniform. We revise some known properties and prove new ones. Then we introduce and study Poisson Processes.

Recall that a random variable Y has a Bernoulli distribution with parameter p, i.e. $Y \sim B(1,p)$ if $\mathbf{P}(Y=1)=p$ and $\mathbf{P}(Y=0)=1-p\equiv q$.

5.1 Properties of Binomial distribution

Definition. Let $n \geq 1$ be an integer and $p \in (0,1)$. We say that $X \sim B(n,p)$ if, for any $k=0,\ldots,n$,

$$\mathbf{P}(X=k)=\frac{n!}{k!(n-k)!}p^k q^{n-k}$$

where $q=1-p$. For $n=0$, $X\sim B(0,p)$ means $X\equiv 0$.

Informally: we consider n independent tosses of a coin with $\mathbf{P}(\text{"head"})=p$ and $\mathbf{P}(\text{"tail"})=q=1-p$. Then $X\equiv X_n=$ "total number of heads in n trials" and $X_n=Y_1+Y_2+\ldots+Y_n$ where $Y_i=1$ if there is "head" in the ith trial and $Y_i=0$, otherwise (this means that Y_i ias a Bernoulli distribution).

Property B1. $\mathbf{E}X_n=np$ and $\mathbf{Var}X_n=npq$.
Indeed, $\mathbf{E}\,X_n=\mathbf{E}\,Y_1+\ldots+\mathbf{E}\,Y_n=n\mathbf{E}\,Y_1=np$ and $\mathbf{Var}X_n=n\mathbf{Var}Y_1=npq$.

Property B2. If $X_n\sim B(n,p)$, $X_m\sim B(m,p)$ (where p is the same), and X_n and X_m are independent, then

$$X_n+X_m\sim B(n+m,p)$$

. Indeed, represent X_n as the sum $X_n=Y_1+\ldots Y_n$ and X_m as the sum $X_m=Y_{n+1}+\ldots+Y_{n+m}$ where all Y_i are independent and Bernoulli with parameter p.

5.2 From Binomial to Poisson

In the previous subsection, we fix p and consider various n. Now we allow both n and p to vary. Therefore, we use notation: $X_{n,p}\sim B(n,p)$ instead of $X_n\sim B(n,p)$.

Definition. Let $\lambda>0$. We say $N\sim\Pi(\lambda)$ if, for any $k=0,1,2,\ldots$,

$$\mathbf{P}(N=k)=\frac{\lambda^k}{k!}e^{-\lambda}.$$

Here is the famous Poisson theorem.

Theorem 5.2.1 Let $n\to\infty$ and $p\equiv p_n\to 0$ in such a way that $np_n\to\lambda\in(0,\infty)$. Then

$$B(n,p_n)\to\Pi(\lambda).$$

More precisely, for any k,

$$\mathbf{P}(X_{n,p_n}=k)=\frac{n!}{k!(n-k)!}p^k(1-p)^{n-k}\to\frac{\lambda^k}{k!}e^{-\lambda}.$$

Proof: For simplicity, assume that $p = p_n = \lambda/n$. Then

$$\begin{aligned}\mathbf{P}(X_{n,p_n} = k) &= \frac{1}{k!} \cdot \frac{n(n-1)\cdot \ldots \cdot (n-k+1)}{n^k} \cdot (np_n)^k \cdot (1-\lambda/n)^n \cdot (1-\lambda/n)^{-k} \\ &\to \frac{1}{k!} \cdot 1 \cdot \lambda^k \cdot e^{-\lambda} \cdot 1 \\ &= \frac{\lambda^k}{k!} e^{-\lambda}.\end{aligned}$$

□

5.3 Properties of Poisson Distribution

Let $\lambda_1 > 0$ and $\lambda_2 > 0$.

Property Π1. If $N_1 \sim \Pi(\lambda_1)$, $N_2 \sim \Pi(\lambda_2)$, and N_1 and N_2 are independent, then $N_1 + N_2 \sim \Pi(\lambda_1 + \lambda_2)$.

There are various proofs of this property: 1) "intuitive" (verbal explanation); 2) by use of probability generating functions; 3) by use of "partition rule". Here is the third proof: For $k = 0$,

$$\mathbf{P}(N_1 + N_2 = 0) = \mathbf{P}(N_1 = 0; N_2 = 0) =^{indep.} \mathbf{P}(N_1 = 0)\mathbf{P}(N_2 = 0) = e^{-\lambda_1 - \lambda_2}.$$

Then for $k \geq 1$:

$$\begin{aligned}\mathbf{P}(N_1 + N_2 = k) &= \sum_{j=0}^{k} \mathbf{P}(N_1 = j, N_2 = k - j) \\ &= \sum_{j=0}^{k} \frac{\lambda_1^j}{j!} e^{-\lambda_1} \frac{\lambda_2^{k-j}}{(k-j)!} e^{-\lambda_2} \quad \text{(by independence)} \\ &= \frac{e^{-\lambda_1-\lambda_2}}{k!} \sum_{j=0}^{k} \frac{k!}{j!(k-j)!} \lambda_1^j \lambda_2^{k-j} \\ &= \frac{e^{-\lambda_1-\lambda_2}}{k!} (\lambda_1 + \lambda_2)^k \quad \text{(Newton's binom).}\end{aligned}$$

Property Π1′. Let $\lambda_1, \ldots, \lambda_l > 0$. If $N_1 \sim \Pi(\lambda_1), \ldots N_l \sim \Pi(\lambda_l)$, and r.v.'s are mutually independent, then $N_1 + \ldots + N_l \sim \Pi(\lambda_1 + \ldots \lambda_l)$.

Property Π2. If $N_1 \sim \Pi(\lambda_1)$, $N_2 \sim \Pi(\lambda_2)$, and N_1 and N_2 are independent, then, for any $k = 0, 1, 2, \ldots$ and for any $j = 0, \ldots, k$,

$$\mathbf{P}(N_1 = j \mid N_1 + N_2 = k) = \frac{k!}{j!(k-j)!} \left(\frac{\lambda_1}{\lambda_1 + \lambda_2}\right)^j \left(\frac{\lambda_2}{\lambda_1 + \lambda_2}\right)^{k-j}.$$

Comment. Property Π2 means:
given $N_1 + N_2 = k$, a r.v. N_1 has *conditionally* Binomial distribution $B\left(k, \frac{\lambda_1}{\lambda_1+\lambda_2}\right)$.

Proof of Π2. Recall the following notation

$$(a) \qquad \mathbf{P}(A \mid B) = \frac{\mathbf{P}(A \bigcap B)}{\mathbf{P}(B)}$$

and relation

$$(b) \qquad C = D \quad \text{implies} \quad \mathbf{P}(C) = \mathbf{P}(D).$$

Then

$$\begin{aligned}
\mathbf{P}(N_1 = j \mid N_1 + N_2 = k) &= \frac{\mathbf{P}(N_1 = j, N_1 + N_2 = k)}{\mathbf{P}(N_1 + N_2 = k)} \quad (\text{ by (a)}) \\
&= \frac{\mathbf{P}(N_1 = j, N_2 = k - j)}{\mathbf{P}(N_1 + N_2 = k)} \quad (\text{by (b)}) \\
&= \frac{\mathbf{P}(N_1 = j)\mathbf{P}(N_2 = k - j)}{\mathbf{P}(N_1 + N_2 = k)} \quad (\text{by independence}) \\
&= \frac{\frac{\lambda_1^j}{j!} e^{-\lambda_1} \frac{\lambda_2^{k-j}}{(k-j)!} e^{-\lambda_2}}{\frac{(\lambda_1+\lambda_2)^k}{k!} e^{-\lambda_1-\lambda_2}} \quad (\text{by Definition and } \Pi 1) \\
&= \frac{k!}{j!(k-j)!} \left(\frac{\lambda_1}{\lambda_1 + \lambda_2}\right)^j \left(\frac{\lambda_2}{\lambda_1 + \lambda_2}\right)^{k-j}.
\end{aligned}$$

Property $\Pi 2'$. Let $\lambda_1, \ldots, \lambda_l > 0$. If $N_1 \sim \Pi(\lambda_1), \ldots N_l \sim \Pi(\lambda_l)$, and r.v.'s are mutually independent, then, for any $k = 0, 1, 2, \ldots$ and for any $j = 0, \ldots, k$,

$$\mathbf{P}(N_1 = j \mid N_1 + \ldots + N_l = k) = \frac{k!}{j!(k-j)!} \left(\frac{\lambda_1}{\lambda_1 + \ldots + \lambda_l}\right)^j \left(1 - \frac{\lambda_1}{\lambda_1 + \ldots + \lambda_l}\right)^{k-j}.$$

Property $\Pi 3$. Let $N \sim \Pi(\lambda)$ where $\lambda > 0$ (say, there are N balls). Independently of everything else, colour each ball "red" with probability $p \in (0, 1)$ and "green" with probability $q = 1 - p$. Let R be a total number of red balls and G a total number of green ones. Then
(i) $R + G = N$;
(ii) $R \sim \Pi(\lambda p)$;
(iii) $G \sim \Pi(\lambda q)$;
(iv) R and G are independent.

Proof: Proof of (ii). For any $k = 0, 1, \ldots$,

$$\begin{aligned}
\mathbf{P}(R = k) &= \sum_{n=k}^{\infty} \mathbf{P}(R = k; N = n) = \sum_{n=k}^{\infty} \mathbf{P}(N = n) \cdot \mathbf{P}(R = k \mid N = n) \\
&= \sum_{n=k}^{\infty} \frac{\lambda^n}{n!} e^{-\lambda} \cdot \frac{n!}{k!(n-k)!} p^k q^{n-k} = \frac{\lambda^k p^k}{k!} e^{-\lambda} \sum_{n=k}^{\infty} \frac{\lambda^{n-k} q^{n-k}}{(n-k)!} \\
&= \frac{(\lambda p)^k}{k!} e^{-\lambda} \sum_{m=0}^{\infty} \frac{(\lambda q)^m}{m!} = \frac{(\lambda p)^k}{k!} e^{-\lambda} e^{\lambda q} = \frac{(\lambda p)^k}{k!} e^{-\lambda p}
\end{aligned}$$

Proof of (iii) – by the symmetry.
Proof of (iv). For any $k, i = 0, 1, \ldots$,

$$\begin{aligned}
\mathbf{P}(R = k; G = i) &= \mathbf{P}(R = k; N = k + i) \\
&= \mathbf{P}(N = k + i) \cdot \mathbf{P}(R = k \mid N = k + i) \\
&= \frac{\lambda^{k+i}}{(k+i)!} e^{-\lambda} \cdot \frac{(k+i)!}{k!i!} p^k q^i \\
&= \frac{(\lambda p)^k}{k!} e^{-\lambda p} \cdot \frac{(\lambda q)^i}{i!} e^{-\lambda q} \\
&= \mathbf{P}(R = k) \cdot \mathbf{P}(G = i) \quad \text{from (ii) and (iii).}
\end{aligned}$$

□

Property $\Pi 4$. If $N \sim \Pi(\lambda)$, then $\mathbf{E}N = \mathbf{Var}N = \lambda$.
(See Introduction II).
[*Question to you:* could one use the Poisson theorem to obtain property $\Pi 4$?]

5.4 Geometric distribution

Definition. Let $p \in (0,1)$ and $q = 1-p$. $X \sim G(p)$ if, for any $k = 1, 2, \ldots$,

$$\mathbf{P}(X = k) = pq^{k-1}.$$

Informally: we toss a coin until the time instant when the head falls out first time.
Note that the tail of geometric distribution is: for any $k = 0, 1, \ldots$,

$$\mathbf{P}(X > k) = \sum_{j=0}^{\infty} \mathbf{P}(X = k + j + 1)$$

$$= \sum_{j=0}^{\infty} pq^{k+j} = pq^k \sum_{j=0}^{\infty} q^j = pq^k \cdot \frac{1}{1-q} = q^k.$$

And here is an alternative (and even shorter) version of the proof:
The event $\{X > k\}$ means that there are only tails$T_1, T_2, \ldots, T_k$ in the first k tosses of the coin, and the probability of that is

$$\mathbf{P}(T_1 T_2 \ldots T_k) = (\mathbf{P}(T_1))^k = q^k.$$

Property G 1 (*memoryless* property). If $X \sim G(p)$, then, for any $k, m = 0, 1, \ldots$,

$$\mathbf{P}(X > k + m \mid X > k) = \mathbf{P}(X > m).$$

Proof:

$$\mathbf{P}(X > k + m \mid X > k) = \frac{\mathbf{P}(X > k + m; X > k)}{\mathbf{P}(X > k)} = \frac{\mathbf{P}(X > k + m)}{\mathbf{P}(X > k)}$$

$$= \frac{q^{k+m}}{q^k} = q^m.$$

□

Property G 2. If $X \sim G(p)$, then $\mathbf{E}X = \frac{1}{p}$ and $\mathbf{Var}X = \frac{q}{p^2}$.
(Exercise for you.)

5.5 From Geometric to Exponential

Definition. Let $\alpha > 0$. We write $Y \sim Exp(\alpha)$ if $\mathbf{P}(Y > x) = e^{-\alpha x}$ for $x \geq 0$.
Then $\mathbf{P}(Y > 0) = 1$ and $f_Y(x) = \alpha e^{-\alpha x}$ for $x \geq 0$.
Here is a simple version of the famous Renyi theorem (Renyi was a Hungarian mathematician).

Theorem 5.5.1 Let $p \to 0$. Then

$$pG(p) \to Exp(1).$$

More precisely, if $X \sim G(p)$, then, for any $t > 0$

$$\mathbf{P}(pX > t) \to e^{-t}.$$

Proof: For simplicity, assume that t/p is an integer. Then

$$\mathbf{P}(pX > t) = \mathbf{P}(X > t/p) = (1-p)^{t/p} = \left((1-p)^{1/p}\right)^t \to \left(e^{-1}\right)^t = e^{-t}.$$

□

5.6 Properties of Exponential Distribution

Property E 1 (*memoryless* property). If $Y \sim Exp(\alpha)$, then, for any $x, y \geq 0$,

$$\mathbf{P}(Y > x + y \mid Y > x) = \mathbf{P}(Y > y).$$

Property E 2. If $Y \sim Exp(\alpha)$ and $c > 0$, then $Z = cY \sim Exp\left(\frac{\alpha}{c}\right)$.
Property E 3. If $Y \sim Exp(\alpha)$, then $\mathbf{E}Y = \frac{1}{\alpha}$ and $\mathbf{Var}Y = \frac{1}{\alpha^2}$.
Property E 4. If $Y_1 \sim Exp(\alpha_1)$, $Y_2 \sim Exp(\alpha_2)$, and Y_1 and Y_2 are independent, then $\min(Y_1, Y_2) \sim Exp(\alpha_1 + \alpha_2)$.

Proof: For any $x \geq 0$,

$$\mathbf{P}(\min(Y_1, Y_2) > x) = \mathbf{P}(Y_1 > x; Y_2 > x) =^{\mathbf{indep}} \mathbf{P}(Y_1 > x) \cdot \mathbf{P}(Y_2 > x)$$
$$= e^{-\alpha_1 x} e^{-\alpha_2 x} = e^{-(\alpha_1+\alpha_2)x}.$$

□

Exercise. Prove the following natural generalization of Property E4. Let $m \geq 2$ be an integer and let $Y_1 \sim Exp(\alpha_1), \ldots, Y_m \sim Exp(\alpha_m)$ be m mutually independent exponential random variables. Then

$$\min(Y_1, \ldots, Y_m) \sim Exp(\alpha),$$

where $\alpha = \alpha_1 + \ldots + \alpha_m$.

5.7 Gamma distribution

Definition. Let $\alpha > 0$ and let n be a positive integer. We say that $X \sim \Gamma(\alpha, n)$ if X is a non-negative absolutely continuous random variable with density

$$f_X(x) = \frac{x^{n-1}}{(n-1)!}\alpha^n e^{-\alpha x}, \quad x \geq 0$$

and $f_X(x) = 0$, for $x < 0$.
Comment 1. $Exp(\alpha) = \Gamma(\alpha, 1)$.
Comment 2. Note that $f_X(x)$ is indeed a density function, i.e. it is non-negative and its integral $\int_0^\infty f_X(x)dx$ is equal to 1.
Comment 3. There exists a more general notion of Gamma distribution, where n may take a real value.
Property Γ1. If $Y_1, \ldots, Y_n$ are i.i.d. having exponential distribution $Exp(\alpha)$, then

$$S_n \equiv Y_1 + \ldots + Y_n \sim \Gamma(\alpha, n).$$

Proof: Proceed with induction arguments. For $n = 1$, the result holds – see Comment 1. Assume it holds for $n = m$. Then prove it for $n = m + 1$. Represent $S_{m+1} = S_m + Y_{m+1}$. Recall the following convolution formula: if X and Y are two independent absolutely continuous random variables with corresponding densities f_X and f_Y, then

$$f_{X+Y}(t) = \int_{-\infty}^{\infty} f_X(x) f_Y(t - x)dx.$$

(intuitively, the sum $X + Y$ takes value t when X takes any value x and then Y takes value $t - x$).

Apply the formula with $X = S_m$ and $Y = Y_{m+1}$. Since both random variables are non-negative, $f_{S_m}(x) = 0$ for $x < 0$ and $f_{Y_{m+1}}(t-x) = 0$ for $x > t$. Therefore, $f_{S_{m+1}}(t) = 0$ for $t < 0$ and, for $t \geq 0$,

$$\begin{aligned} f_{S_{m+1}}(t) &= \int_0^t f_{S_m}(x) f_{Y_{m+1}}(t-x)dx \\ &= \int_0^t \frac{x^{m-1}}{(m-1)!}\alpha^m e^{-\alpha x} \cdot \alpha e^{-\alpha(t-x)}dx \\ &= \frac{1}{(m-1)!} \cdot \alpha^{m+1} e^{-\alpha t} \int_0^t x^{m-1}dx \\ &= \frac{1}{(m-1)!} \cdot \alpha^{m+1} e^{-\alpha t} \cdot \frac{t^m}{m} \\ &= \frac{t^m}{m!}\alpha^{m+1} e^{-\alpha t}. \end{aligned}$$

□

Corollary If $X_1 \sim \Gamma(\alpha, n)$ and $X_2 \sim \Gamma(\alpha, m)$, and if X_1 and X_2 are independent, then $X_1 + X_2 \sim \Gamma(\alpha, n+m)$.

Proof: From Property $\Gamma 1$, one can represent X_1 as $X_1 = Y_1 + \ldots + Y_n$ where $Y_1, \ldots, Y_n$ are i.i.d. exponential. Similarly, $X_2 = Y_{n+1} + \ldots + Y_{n+m}$, and $X_1 + X_2 = Y_1 + \ldots + Y_{n+m}$ is a sum of $n+m$ i.i.d. exponential r.v.'s. □

Property Γ2. If $S_n \sim \Gamma(\alpha, n)$, then $\mathbf{E}S_n = \frac{n}{\alpha}$ and $\mathbf{Var}S_n = \frac{n}{\alpha^2}$.

Proof: follows from Property $\Gamma 1$, Property $E3$, and properties of expectation. □

We turn now to the main construction. Assume $Y_1, Y_2, \ldots$ are i.i.d. exponentially distributed r.v.'s with parameter α. For $n = 1, 2, \ldots$, let $S_n = Y_1 + \ldots + Y_n$. Let $t > 0$.

Let

$$N(t) = \max\{n \ : \ S_n \leq t\}$$

be a "counting" random variable (it counts the number of points in $(0, t]$) and let

$$\psi(t) = S_{N(t)+1} - t$$

be the "overshoot" (or "excess") at t.

Property Γ3. The following results hold: for any t,
(i) $\psi(t) \sim Exp(\alpha)$;
(ii) $N(t) \sim \Pi(\alpha t)$;
(iii) $\psi(t)$ and $N(t)$ are independent.
Comments !!!

Proof: Start with preliminary calculations. For any $n = 0, 1, \ldots$ and $x \geq 0$, let

$$P_t(n, x) = \mathbf{P}(N(t) = n; \psi(t) > x) = \mathbf{P}(S_n \leq t; S_n + Y_{n+1} > t + x).$$

Then, for $n = 0$,

$$P_t(0, x) = \mathbf{P}(S_1 > t + x) = \mathbf{P}(Y_1 > t + x) = e^{-\alpha(t+x)} = e^{-\alpha t} \cdot e^{-\alpha x}.$$

Then, for $n > 0$,

$$\begin{aligned} P_t(n,x) &= \int_0^t f_{S_n}(z)\mathbf{P}(z + Y_{n+1} > t + x)dz \\ &= \int_0^t \frac{z^{n-1}}{(n-1)!}\alpha^n e^{-\alpha z} \cdot e^{-\alpha(t+x-z)}dz \\ &= e^{-\alpha(t+x)} \int_0^t \frac{z^{n-1}}{(n-1)!}\alpha^n dz \\ &= e^{-\alpha(t+x)} \frac{(\alpha t)^n}{n!} \\ &= \frac{(\alpha t)^n}{n!} e^{-\alpha t} \cdot e^{-\alpha x}. \end{aligned}$$

Proof of (i):

$$\begin{aligned} \mathbf{P}(\psi(t) > x) &= \sum_{n=0}^{\infty} \mathbf{P}(N(t) = n; \psi(t) > x) \\ &= e^{-\alpha x} \cdot e^{-\alpha t} \sum_{n=0}^{\infty} \frac{(\alpha t)^n}{n!} \\ &= e^{-\alpha x} \cdot e^{-\alpha t} \cdot e^{\alpha t} \\ &= e^{-\alpha x}. \end{aligned}$$

Proof of (ii): since $\mathbf{P}(\psi(t) > 0) = 1$, we have

$$\mathbf{P}(N(t) = n) = \mathbf{P}(N(t) = n; \psi(t) > 0) = \frac{(\alpha t)^n}{n!} e^{-\alpha t}$$

Proof of (iii): For any $n = 0, 1, \ldots$ and $x > 0$,

$$\mathbf{P}(N(t) = n; \psi(t) > x) = \frac{(\alpha t)^n}{n!} e^{-\alpha t} \cdot e^{-\alpha x} = \mathbf{P}(N(t) = n) \cdot \mathbf{P}(\psi(t) > x).$$

□

Paradox Buses arrive to a bus stop at random times. Assume that inter-arrival times between consecutive buses are i.i.d. Exp (1) random variables. Somebody comes to the bus stop at time $t > 0$. His waiting time to the next bus has Exp (1) distribution (thanks to Property Γ3). On the other hand, the waiting time is only a part of the inter-arrival time between the bus he is waiting for and the previous bus, and this time seems to have Exp (1) distribution, too. Then we have two random variables, and one of them is strictly smaller than the other, while both have the same Exp (1) distribution. This cannot be true. What is wrong?

5.8 Uniform distribution

Let $-\infty < a < b < \infty$.

Definition. A continuous r.v. X has a uniform distribution on the interval (a, b), $X \sim U(a, b)$ if its density function is given by

$$f_X(x) = \frac{1}{b-a} \quad \text{for} \quad a < x < b \quad \text{and} \quad f_X(x) = 0 \quad \text{otherwise.}$$

Property $U1$**.** If $X \sim U(a, b)$ and $c > 0$, $-\infty < d < \infty$, then
$Z = cX + d \sim U(ac + d, bc + d)$. (Exercise for you.)

Property $U2$**.** If $X \sim U(a, b)$, then $\mathbf{E}X = \frac{a+b}{2}$ and $\mathbf{Var}X = \frac{(b-a)^2}{12}$. (Exercise for you.)

Property $U3$**.** Let $X_1, X_2, \ldots, X_n$ be i.i.d. r.v.'s with $U(a, b)$ distribution and $a < c < b$. Let

$$Z = Number\{i \ : \ X_i \in (a, c)\}.$$

Then

$$Z \sim B\left(n, \frac{c-a}{b-a}\right).$$

Vice versa, if n random variables $X_1, \ldots, X_n$ are such that $Z \sim B\left(n, \frac{c-a}{b-a}\right)$ for any $c \in (a, b)$, then $X_1, X_2, \ldots, X_n$ are i.i.d. r.v.'s with $U(a, b)$ distribution.

Proof: (only in one direction). Assume that $X_1, X_2, \ldots, X_n$ are i.i.d. r.v.'s with $U(a, b)$ distribution and $a < c < b$. Let

$$Y_i = 1 \quad \text{if} \quad X_i \in (a, c) \quad \text{and} \quad Y_i = 0 \quad \text{otherwise.}$$

Then

$$Z = Y_1 + \ldots + Y_n$$

where $\mathbf{P}(Y_i = 1) = \frac{c-a}{b-a}$ and $\mathbf{P}(Y_i = 0) = \frac{b-c}{b-a} = 1 - \frac{c-a}{b-a}$. □

5.9 First Definition and some Properties of a Poisson Process.

Remark. Earlier we wrote $Exp(\alpha)$ and $\Pi(\lambda)$. From now, we use notation λ for both parameters.

First Definition of a Poisson Process. Let $\{Y_i\}$ be independent and identically distributed (i.i.d.) random variables (r.v.'s) having an $Exp(\lambda)$ distribution. Let $S_0 = 0$ and $S_n = Y_1 + \ldots + Y_n$ for $n = 1, 2, \ldots$. For $t \geq 0$, put

$$N(t) = \max\{n \ : \ S_n \leq t\}.$$

Then $\{N(t), t \geq 0\}$ is a (homogeneous) *Poisson Process.*
This is a *counting* process which *counts* points.

For $0 \leq t_1 < t_2$, denote

$$N(t_1, t_2) = N(t_2) - N(t_1)$$

(thus, $N(0, t) = N(t)$).

Property $PP1$. Assume $\{N(t), t \geq 0\}$ is a PP. Then
(i) for any $0 \leq t_1 < t_2$,

$$N(t_1, t_2) \sim \Pi(\lambda(t_2 - t_1)).$$

(ii) for any $l = 2, 3, \ldots$ and for any $0 < t_1 < t_2 < \ldots < t_l$, random variables $N(t_1), N(t_1, t_2), \ldots N(t_{l-1}, t_l)$ are mutually independent.

Proof: Start with proof of (i). If $t_1 = 0$, then take $t = t_2$ and apply property $\Gamma 3$ with $\alpha = \lambda$ and $t = t_2$. If $t_1 > 0$, then, for any $k = 0, 1, 2, \ldots$,

$$\mathbf{P}(N(t_1, t_2) = k) = \sum_{j=0}^{\infty} \mathbf{P}(N(t_1) = j; N(t_1, t_2) = k). \tag{10}$$

Apply property $\Gamma 3$ with $t = t_1$ and $\alpha = \lambda$. Denote $\tilde{Y}_1 = \psi(t_1), \tilde{Y}_2 = Y_{j+2}, \tilde{Y}_3 = Y_{j+3}$, etc. Then let $\tilde{S}_0 = 0$ and $\tilde{S}_n = \tilde{Y}_1 + \ldots + \tilde{Y}_n$ for $n \geq 1$. Then, on the event $\{N(t_1) = j\}$,

$$N(t_1, t_2) = \max\{n \ : \ \tilde{S}_n \leq t_2 - t_1\}.$$

Therefore, given $N(t_1) = j$, r.v.'s $\{\tilde{Y}_n\}$ are mutually independent and all have $Exp(\lambda)$ distribution. Again, by property $\Gamma 3$, for any $j = 0, 1, \ldots$,

$$\begin{aligned}\mathbf{P}(N(t_1) = j; N(t_1, t_2) = k) &= \mathbf{P}(N(t_1) = j)\mathbf{P}(N(t_1, t_2) = k \mid N(t_1) = j) \\ &= \mathbf{P}(N(t_1) = j) \cdot \frac{(\lambda(t_2 - t_1))^k}{k!} e^{-\lambda(t_2 - t_1)}\end{aligned}$$

and, from (10),

$$\mathbf{P}(N(t_1, t_2) = k) = \sum_{j=0}^{\infty} \mathbf{P}(N(t_1) = j) \cdot \frac{(\lambda(t_2 - t_1))^k}{k!} e^{-\lambda(t_2 - t_1)} = \frac{(\lambda(t_2 - t_1))^k}{k!} e^{-\lambda(t_2 - t_1)}.$$

Now – about the proof of (ii). For $l = 2$, the independence has been already established in the previous equality ! For $l > 2$, the proof is omitted (follows from induction arguments !) □

Remark. We also use the following notation: for $I = (t_1, t_2)$, $N(I) = N(t_1, t_2)$.

Corollary. Let $\{N(t), t \geq 0\}$ be a PP. Let $t > 0$ and $n = 1, 2, \ldots$ be fixed. Given $N(t) = n$, all n points of the PP on $(0, t)$ "behave like" mutually independent and uniformly distributed.

Comment. Assume we only know that there are n points on $(0, t)$. How to "simulate" locations of these points ?
(a) We take n points "at random"; (b) then order these points.

Proof: of Corollary.
(1) Apply property $U3$ with $a = 0$ and $b = t$: we have to show that, for any $0 < c < t$, the conditional distribution of a number of point in the interval $(0, c)$ is $B\left(n, \frac{c}{t}\right)$.
(2) From property $PP1$,

$$N(t) = N(c) + N(c, t)$$

where $N(c) \sim \Pi(\lambda c)$, $N(c, t) \sim \Pi(\lambda(t - c))$, and $N(c)$ and $N(c, t)$ are independent.
(3) From property $\Pi 2$, with $k = n$,

$$\mathbf{P}(N(c) = j \mid N(t) = n) = \mathbf{P}(N(c) = j \mid N(c) + N(c, t) = n) = \frac{n!}{j!(n-j)!} \left(\frac{c}{t}\right)^j \left(1 - \frac{c}{t}\right)^{n-j}.$$

□

5.10 Second Definition of a Poisson Process. Superposition and Splitting Theorems

Second Definition of a Poisson Process. Let

$$0 \le S_1 \le S_2 \le \dots$$

be a non-decreasing sequence of random variables ("random points") on the non-negative half-line. For $t \ge 0$, let

$$N(t) = \max\{n \ge 0 \ : \ S_n \le t\}$$

be a "counting" random process (i.e. a process which "counts" points). We say that $\{N(t).t \ge 0\}$ is a homogeneous *Poisson Process* if it satisfies Property PP1, i.e.
(i) for any $0 \le t_1 < t_2$,

$$N(t_1, t_2) \sim \Pi(\lambda(t_2 - t_1)).$$

(ii) for any $l = 2, 3, \dots$ and for any $0 < t_1 < t_2 < \dots < t_l$, random variables $N(t_1), N(t_1, t_2), \dots N(t_{l-1}, t_l)$ are mutually independent.

We say also that λ is a *rate* of a Poisson Process.

Theorem 5.10.1 First and Second Definitions of a Poisson Process are equivalent.

Remark. The implication "First" $\rightarrow$ "Second" has been proved in the previous subsection. A proof of the converse result is more technical and is omitted here.

We give now two results which together assert that k **independent** Poisson processes are together **equivalent** to a single combined Poisson process in which points are **independently assigned** to the individual processes.

Theorem 5.10.2 (Superposition Theorem) Let $N_1(\cdot), \dots, N_k(\cdot)$ be **independent** Poisson processes with rates $\lambda_1, \dots, \lambda_k$ respectively. Let the process $N(\cdot) = N_1(\cdot) + \dots + N_k(\cdot)$ be the **superposition** of these processes (that is, for all intervals I, $N(I) = \sum_{j=1}^{k} N_j(I)$). Then

(i) $N(\cdot)$ is a Poisson process with rate $\lambda = \sum_{j=1}^{k} \lambda_j$.

(ii) Given the occurrence of a point of the process $N(\cdot)$ at time t it belongs to any given original process $N_i(\cdot)$ with probability $p_i = \lambda_i/\lambda$, **independently** of all else.

To see the first part of this result, note that it is easy to verify that the conditions *PP1* of the Second Definition of a Poisson Process are satisfied for the combined process $N(\cdot)$ from the fact that they are satisfied for the individual processes $N_1(\cdot), \dots, N_k(\cdot)$ and also from **Property** $\Pi 1'$ of Poisson distributions. The second part follows from **Property** $\Pi 2'$ of Poisson distributions.

The converse of the Superposition Theorem is the Splitting Theorem.

Theorem 5.10.3 (Splitting Theorem) Let $N(\cdot)$ be a Poisson process with rate λ. Suppose that each point of $N(\cdot)$ is assigned to a new point process $N_i(\cdot)$ $(1 \leq i \leq k)$ with probability p_i, **independently** of all else (where $p_1 + \cdots + p_k = 1$). Then $N_1(\cdot), \ldots, N_k(\cdot)$ are **independent** Poisson processes with rates $p_1\lambda, \ldots, p_k\lambda$.

A *proof* of the splitting theorem follows **directly** from **Property** Π3 of Poisson distributions and from the Second Definition of a Poisson Process. To verify this is an *Exercise* for you. You may try to do it just for $k = 2$.

Example 5.10.1 A factory opens at a certain time and thereafter accidents to male employees happen as a Poisson process $N_m(\cdot)$ with rate 1 per year, while accidents to female employees happen as an *independent* Poisson process $N_f(\cdot)$ with rate 3 per year (maybe there are more females).

Note that, by the Superposition Theorem, these two processes are equivalent to a single Poisson process $N(\cdot)$ $(= N_m(\cdot) + N_f(\cdot))$ of accidents with rate 4 per year, in which each accident, independently of all others, has probability 0.25 of being "male" and 0.75 of being "female".

1. Find the probability that, in any given period of length 2 years, (a) no accidents occur, (b) at least 2 accidents occur.

   Solution: since, for any t, $N[t, t+2]$ has a Poisson distribution with mean 8, the required probabilities are (a) $e^{-8} \approx 0.0003$, (b) $1 - 9e^{-8} \approx 0.9970$.

2. Find the probability that, after the factory opens, 4 "female" accidents have happened before 2 "male" accidents have happened. *Solution:* consider again the combined process $N(\cdot)$; the required event happens if and only if, of the first 5 accidents, at least 4 are "female"; since each accident is independently "male" or "female", we require the probability $\mathbf{P}(X \geq 4)$, where X has a binomial distribution with parameters 5 and 3/4; this is $5(3/4)^4(1/4) + (3/4)^5 \approx 0.6328$.

5.11 Third definition of a Poisson process

Definition. A *(homogeneous) Poisson process* $N(\cdot)$ with *rate* (intensity) λ is defined by the following two properties.

(PP3(i)) For any t and $h > 0$ (h small),

$$\begin{aligned}
&\mathbf{P}(N[t, t+h] = 0) = 1 - \lambda h + o(h) && \text{as } h \to 0,\\
&\mathbf{P}(N[t, t+h] = 1) = \lambda h + o(h) && \text{as } h \to 0,\\
&\mathbf{P}(N[t, t+h] \geq 2) = o(h) && \text{as } h \to 0;
\end{aligned}$$

(PP3(ii)) The numbers of points $N(I_1), \ldots, N(I_k)$ occurring in *disjoint* finite intervals $I_1, \ldots, I_k$ are *independent* random variables.

Here $o(h)$ is any function of h such that $o(h)/h \to 0$ as $h \to 0$, so that $o(h)$ is very small even in comparison to h. Examples are h^2, $3h^3$, etc.

The condition (PP3(i)) *essentially* requires that, in a *very small* interval of length h, we have either a *single point*—with probability λh, or *no point*—with probability $1 - \lambda h$.

Note. Any two of the three statements in the condition (PP3(i)) *necessarily* imply the third (since the probability that the random variable $N[t, t+h]$ takes some value is 1).

Let the condition (PP1(i)) be given by

(PP1(i)) For any finite interval $I = [t_1, t_2]$, the random variable $N(I) = N[t_1, t_2]$ has a Poisson distribution with mean $\lambda(t_2 - t_1)$.

Theorem 5.11.1 The Second and the Third definitions of a Poisson process are equivalent. In other words, given the condition (PP3(ii)), the conditions (PP3(i)) and (PP1(i)) are equivalent.

Proof: (a) Suppose first that (PP3(ii)) and (PP3(i)) hold; we show that (PP1(i)) holds. For $n = 0, 1, \ldots$ and $t_1, t \geq 0$, let $p_n(t) = \mathbf{P}(N[t_1, t_1 + t] = n)$ be the probability of exactly n points in any interval of length t. We require to show that, for all n and for all t,

$$p_n(t) = e^{-\lambda t} \frac{(\lambda t)^n}{n!}. \tag{11}$$

We establish this result by induction on n.

First consider $n = 0$. For any $h > 0$,

$$\begin{aligned}
p_0(t+h) &= p_0(t) p_0(h) && \text{(independence in disjoint intervals)}\\
&= p_0(t)[1 - \lambda h + o(h)] && \text{as } h \to 0.
\end{aligned}$$

Hence

$$\frac{p_0(t+h) - p_0(t)}{h} = -\lambda p_0(t) + \frac{o(h)}{h} \qquad \text{as } h \to 0,$$

and so

$$p_0'(t) = -\lambda p_0(t)$$

giving $p_0(t) = ce^{-\lambda t}$. The initial condition $p_0(0) = 1$ gives $c = 1$, and so finally

$$p_0(t) = e^{-\lambda t},$$

as required.

Now suppose that the result (11) holds for $n = k-1$ where $k \geq 1$. We show that it also holds for $n = k$. Since $\mathbf{P}(N[t, t+h] \geq 2) = o(h)$ as $h \to 0$, it follows that, again as $h \to 0$,

$$\begin{aligned} p_k(t+h) &= \mathbf{P}(N[0, t+h] = k) \\ &= \mathbf{P}(N[0,t] = k)\mathbf{P}(N[t, t+h] = 0) \\ &\quad + \mathbf{P}(N[0,t] = k-1)\mathbf{P}(N[t, t+h] = 1) + o(h) \\ &= p_k(t)p_0(h) + p_{k-1}(t)p_1(h) + o(h) \\ &= p_k(t)(1-\lambda h) + p_{k-1}(t)\lambda h + o(h). \end{aligned}$$

Hence

$$\frac{p_k(t+h) - p_k(t)}{h} = -\lambda p_k(t) + \lambda p_{k-1}(t) + \frac{o(h)}{h} \qquad \text{as } h \to 0,$$

and so

$$\begin{aligned} p_k'(t) &= -\lambda p_k(t) + \lambda p_{k-1}(t) \\ &= -\lambda p_k(t) + e^{-\lambda t}\frac{\lambda^k t^{k-1}}{(k-1)!}, \end{aligned}$$

where the last line follows since the result (11) is assumed true for $n = k-1$. Multiply the above result by $e^{\lambda t}$ to obtain

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(p_k(t)e^{\lambda t}\right) = \frac{\lambda^k t^{k-1}}{(k-1)!},$$

and so, using also the initial condition $p_k(0) = 0$ for $k \geq 1$, we finally obtain

$$p_k(t)e^{\lambda t} = \frac{(\lambda t)^k}{k!},$$

giving the result (11) for $n = k$ as required, and completing the proof by induction.
(b) Now suppose instead that the condition (PP1(i)) holds. Then, for any t,

$$\begin{aligned} \mathbf{P}(N[t, t+h] = 0) &= e^{-\lambda h} \\ &= 1 - \lambda h + o(h) \qquad \text{as } h \to 0, \end{aligned}$$

and

$$\begin{aligned} \mathbf{P}(N[t, t+h] = 1) &= e^{-\lambda h}\lambda h \\ &= \lambda h + o(h) \qquad \text{as } h \to 0. \end{aligned}$$

Since the probability that the random variable $N[t, t+h]$ takes some value is 1, we also have

$$\mathbf{P}(N[t, t+h] \geq 2) = o(h) \qquad \text{as } h \to 0.$$

Hence (PP3(i)) follows. □

5.12 Heterogeneous one-dimensional Poisson processes

We can also consider Poisson processes in which events or points occur at a *time-dependent* rate. Let $\lambda(t)$ be a positive *function* on $\mathbb{R}$ (or $\mathbb{R}_+$ as appropriate).

Definition. A *(heterogeneous or non-homogeneous) Poisson process* with *rate function* $\lambda(t)$ is again a point process $N(\cdot)$ such that (PP3(ii)) and (PP3(i)) hold, except that (PP3(i)) is now *generalised* to read

(PP3(i)) For any t and $h > 0$ (h small),

$$\begin{aligned} &\mathbf{P}(N[t, t+h] = 0) = 1 - \lambda(t)h + o(h) && \text{as } h \to 0, \\ &\mathbf{P}(N[t, t+h] = 1) = \lambda(t)h + o(h) && \text{as } h \to 0, \\ &\mathbf{P}(N[t, t+h] \geq 2) = o(h) && \text{as } h \to 0. \end{aligned}$$

The property (PP1(i)) then becomes

(PP1(i)) For any finite interval $I = [t_1, t_2]$, the random variable $N(I) = N[t_1, t_2]$ has a Poisson distribution with mean $\int_{t_1}^{t_2} \lambda(t)\, dt$.

We now have the obvious generalisations of Theorem 5.11.1 and also the Superposition and Splitting Theorems. For example, Theorem 5.11.1 becomes:

Theorem 5.12.1 Given the condition (PP3(ii)), the (revised) conditions (PP3(i)) and (PP1(i)) are equivalent.

Note, however, that there is no straightforward alternative characterisation of a heterogeneous Poisson process via exponential inter-arrival times.

Example. Calls to the fire service arrive as a Poisson process with rate 1 per hour between 6.00 am and midnight and 2 per hour between midnight and 6.00 am. What is the distribution of the total number of calls per day?

Solution: This is a Poisson distribution with parameter 30.

5.13 Compound Poisson process

Let $N(t), t \geq 0$ be a time-homogeneous Poisson process with rate λ,
and let $\{X_n\}$ be an i.i.d. sequence of random variables which does not depend on the Poisson process.
Assume that $a = \mathbf{E}X_1$ and $b = \mathbf{E}X_1^2$ are finite. Let also $\sigma^2 = \mathbf{Var}X_1$. Then $b = \sigma^2 + a^2$.
We assume that $N(t)$ represents the process of claims arrivals, and that X_n is the size of the nth claim.
Let $W(t)$ be the total of all claims within a time interval of length t.

Lemma 5.13.1 The mean and the variance of $W(t)$ are given by

$$\mathbf{E}W(t) = a\lambda t \quad \text{and} \quad \mathbf{Var}W(t) = b\lambda t.$$

Proof: Without loss of generality, we may consider time interval $(0, t)$. We have $W(t) = \sum_{i=1}^{N(t)} X_i$. Then

$$\begin{aligned}
\mathbf{E}W(t) &= \sum_{n\geq 1} \mathbf{P}(N(t)=n)\mathbf{E}\left(\sum_{i=1}^{N(t)} X_i \mid N(t)=n\right) \\
&= \sum_{n\geq 1} \mathbf{P}(N(t)=n)\mathbf{E}\left(\sum_{i=1}^{n} X_i\right) \\
&= \sum_{n\geq 1} \mathbf{P}(N(t)=n)\cdot na \\
&= a\mathbf{E}N(t) = a\lambda t
\end{aligned}$$

and

$$\begin{aligned}
\mathbf{E}W^2(t) &= \mathbf{E}\left(\sum_{i=1}^{N(t)} X_i\right)^2 \\
&= \sum_{n\geq 1} \mathbf{P}(N(t)=n)\mathbf{E}\left(\sum_{i=1}^{n} X_i\right)^2 \\
&= \sum_{n\geq 1} \mathbf{P}(N(t)=n)\left(n\mathbf{E}X_1^2 + n(n-1)(\mathbf{E}X_1)^2\right) \\
&= \sum_{n\geq 1} \mathbf{P}(N(t)=n)(n^2a^2 + n\sigma^2) \\
&= a^2\mathbf{E}(N(t))^2 + \sigma^2\mathbf{E}N(t) = a^2(\lambda t)^2 + a^2\lambda t + \sigma^2\lambda t.
\end{aligned}$$

Then

$$\mathbf{Var}W(t) = \mathbf{E}W^2(t) - (\mathbf{E}W(t))^2 = (a^2+\sigma^2)\lambda t = b\lambda t.$$

□

Exercise. Assume that claims arrive in a Poisson process with rate 4 per day, and each claim has distribution F (measured in some units).
Let W be the total claim size within a typical week. Find the mean and the variance of W if
(a) F is a discrete distribution which has mass $1/3$ at level 2 and mass $2/3$ at level 5;
(b) F is a uniform distribution in the interval $(0, 2)$.

Solution.

5.14 Insurance applications

The Poisson process is generally accepted as a good model for claims from most types of general insurance policy: motor, household, accident, hurricane. This is because the postulates for the process are in many circumstances very natural, at least as a first approximation.

Note that, by the superposition theorem, if claims from individual policies are well-modelled by Poisson processes (not necessarily with the same means), then claims arising from the aggregate (portfolio) of such polices are also well-modelled by a Poisson process.

Some data. The table below gives numbers of claims during the year 1976 under each of a total of 106,974 motor policies operated by a Belgian company (*source:* Lemaine, 1995, p 25).

No. of claims	No. of policies	Expt no. of policies
0	96,978	96,690.0
1	9,240	9,773.5
2	704	493.9
3	43	16.6
4	9	0.4
≥ 5	0	0.0
Total	106,974	106,974

The mean and variance of number of claims per policy are 0.1011 and 0.1074.

We can examine whether it is reasonable to model the observations as arising from independent Poisson random variables (one for each policy), each with the same mean. Under this hypothesis, the maximum likelihood estimate of the mean is the sample mean of 0.1011. The table also shows the expected number of policies with each number of claims, calculated under this hypothesis and with the estimated mean. A χ^2 test rejects this model—there are many more policies with a high number of claims than the model predicts.

A more appropriate model for the data might be that claims under individual policies (and hence also the total number of claims for the entire portfolio) are indeed reasonably modelled as Poisson random variables, but with means which vary according to the policyholder. The Poisson assumption is reasonable on *a priori* grounds, but there is in fact no way to check it from these data.

However, a model with many Poisson random variables having different means/rates may be too complicated, and frequently modelling by a *mixture* of two Poisson random variables may help. Namely, let $X \sim \Pi(\lambda_1)$ and $Y \sim \Pi(\lambda_2)$ be two independent Poisson random variables, and let ψ be an independent random variable taking two values, 1 and 0, with probabilities $\mathbf{P}(\psi = 1) = p = 1 - \mathbf{P}(\psi = 0)$, where $0 < p < 1$. Then a random variable $Z = \psi X + (1 - \psi)Y$ is a *mixture* of X and Y. Here, for any $n = 0, 1, 2, \ldots$,

$$\begin{aligned}\mathbf{P}(Z = n) &= \mathbf{P}(\psi = 1)\mathbf{P}(X = n) + \mathbf{P}(\psi = 0)\mathbf{P}(Y = n) \\ &= p \cdot \frac{\lambda_1^n}{n!} e^{-\lambda_1} + (1 - p) \cdot \frac{\lambda_2^n}{n!} e^{-\lambda_2}.\end{aligned}$$

In particular, the distribution of Z is not Poisson.

6 Markov processes

6.1 Introduction

A **continuous time** stochastic process $\{X(t)\}_{t\geq 0}$ taking values in a **countable** state space S (e.g. S equal to the integers or some subset of the integers) is a **Markov process** if and only if, for all $i, j \in S$, $s, t > 0$,

$$\begin{aligned}\mathbf{P}(X(s+t) = j \,|\, X(s) = i \text{ and the history of the process prior to } s) \\ = \mathbf{P}(X(s+t) = j \,|\, X(s) = i).\end{aligned}$$

Thus, **given** the evolution of the process $\{X(t)\}_{t\geq 0}$ up to any "current" time s, the probabilistic description of its behaviour at all future times depends only on the current state $\{X(s) = i\}$, and not on the previous history of the process. This is the continuous time version of the **Markov property**.

Additionally the **Markov process** $\{X(t)\}_{t\geq 0}$ is **time homogeneous**, or has **stationary transition rates**, if, for all $i, j \in S$, there exists a **probability function** $p_{ij}(\cdot)$ such that

$$\mathbf{P}(X(s+t) = j \,|\, X(s) = i) = p_{ij}(t), \qquad \text{for all } s, t > 0,$$

independently of s.

Thus, for a **time homogeneous Markov process** $\{X(t)\}_{t\geq 0}$, given its evolution up to any "current" time s, the probabilistic description of its behaviour at at all future times depends only on the current state $\{X(s) = i\}$, and not on the previous history of the process **nor** on the time s itself.

Until further notice we shall consider only a time homogeneous Markov process $\{X(t)\}_{t\geq 0}$

For each $t \geq 0$, define $P(t) = (p_{ij}(t))_{i,j\in S}$. Note that, for each t, $P(t)$ is a **stochastic matrix**, i.e.

$$p_{ij}(t) \geq 0 \text{ for all } i, j \in S, \qquad \sum_{j\in S} p_{ij}(t) = 1 \text{ for all } i \in S.$$

Analogously to the discrete time case, the **distribution of the entire process** $\{X(t)\}_{t\geq 0}$ is determined by the **initial distribution** $\boldsymbol{\mu}^{(0)} = (\mu_i^{(0)})_{i\in S}$, where $\mu_i^{(0)} = \mathbf{P}(X(0) = i)$ for each i, and by the collection of **probability functions** $(p_{ij}(\cdot))_{i,j\in S}$.

We also have, for each $i, j \in S$,

$$p_{ij}(0) = \delta_{ij} \tag{12}$$

where

$$\delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j, \end{cases}$$

and also, for any $s, t > 0$, the **Chapman-Kolmogorov equations**:

$$p_{ij}(s+t) = \sum_{k \in S} p_{ik}(s) p_{kj}(t) \tag{13}$$

(proved exactly as in the discrete time case), together with the **Chapman-Kolmogorov inequalities**.

Examples. Most of the processes considered in Section 4.1.3 become more realistic in a continuous time setting, when transitions from one state to another can occur at any instant in time (rather than just at multiples of some discrete unit time interval). Thus random walks reflecting the fortunes of an insurance company, and, e.g., "healthy-sick-dead" models are more naturally formulated as continuous time models.

Further example: the Poisson process. Let $N(\cdot)$ be a homogeneous Poisson process with rate λ, which we may consider to start at some time 0. For $t \geq 0$, write $N(t) = N[0, t]$, so that $N(t)$ simply counts the number of points which have occurred by time t. This is the **counting process** formulation of a Poisson process.

Then the process $\{N(t)\}_{t \geq 0}$ is a time homogeneous Markov process taking values in the state space $S = \{0, 1, 2, \dots\}$. We have, for any $t \geq 0$,

$$p_{ij}(t) = \begin{cases} e^{-\lambda t} \frac{(\lambda t)^{j-i}}{(j-i)!}, & j \geq i \\ 0, & j < i. \end{cases}$$

6.2 The matrix of transition rates

Suppose that $\{X(t)\}_{t\geq 0}$ is a **time homogeneous Markov process**.

We henceforth assume **continuity** of the probability functions $p_{ij}(\cdot)$. It can be shown that this implies also **differentiability**.

For all $i, j \in S$, define $q_{ij} = p'_{ij}(0)$. Then for all $t, h \geq 0$,

$$\begin{aligned}\mathbf{P}(X(t+h) = j | X(t) = i) &= p_{ij}(h) \\ &= p_{ij}(0) + q_{ij}h + o(h) \quad \text{as } h \to 0 \\ &= \delta_{ij} + q_{ij}h + o(h) \quad \text{as } h \to 0. \end{aligned} \tag{14}$$

Hence q_{ij} is the **(instantaneous) transition rate** of the process from state i to state j. The matrix $Q = (q_{ij})_{i,j\in S}$ is the **transition rate matrix**, or **generator**, of the process $\{X(t)\}_{t\geq 0}$.

In applications the distribution of the process $\{X(t)\}$ is usually defined via the transition rate matrix Q, which determines the dynamics of the process, together with the initial distribution of the process.

Example: birth-and-death process. Here $S = \{0, 1, 2, \dots\}$ and $X(t)$ may be thought of as, for example, a population size at time t. We have

$$\begin{aligned} q_{i,i+1} &= \lambda_i, \qquad i \geq 0 \qquad \text{(birth rate in state } i\text{)}, \\ q_{i,i-1} &= \mu_i, \qquad i \geq 1 \qquad \text{(death rate in state } i\text{)}, \\ q_{ij} &= 0 \qquad \text{for all other } j \neq i. \end{aligned}$$

We show below that then $q_{00} = -\lambda_0$ and, for each $i \geq 1$, $q_{ii} = -(\lambda_i + \mu_i)$.

For a *linear* birth-and-death process we would take $\lambda_i = \lambda i$ (so λ can be thought of as a birth rate per individual) and $\mu_i = \mu i$ (so μ can be thought of as a death rate per individual).

Properties of the transition rate matrix $Q = (q_{ij})$ We shall *assume* the following:

$$0 \leq q_{ij} < \infty \qquad \text{for all } i, j \text{ with } j \neq i, \tag{Q1}$$

$$0 \leq -q_{ii} < \infty \qquad \text{for all } i, \tag{Q2}$$

$$\sum_{j\in S} q_{ij} = 0 \qquad \text{for all } i. \tag{Q3}$$

These conditions are satisfied for all realistic processes (for example, for S finite the condition (Q3) follows since $\sum_{j\in S} p_{ij}(t) = 1$ for all t implies $\sum_{j\in S} p'_{ij}(0) = 0$).

Conversely, given a matrix $Q = (q_{ij})_{i,j\in S)}$ satisfying (Q1), (Q2), (Q3) above, there exists a homogeneous Markov process with Q as transition rate matrix.

6.3 The forward and backward differential equations

How, given the matrix $Q = (q_{ij})_{i,j\in S}$ of transition rates, may we recover the probability functions $(p_{ij}(\cdot))_{i,j\in S}$?

The answer is via one of two possible systems of differential equations.

Theorem 6.3.1 Given Q (satisfying (Q1), (Q2), (Q3)) we have

$$p'_{ij}(t) = \sum_{k\in S} p_{ik}(t)q_{kj} \qquad \text{for all } t \text{ and all } i,j \in S \tag{FDE}$$

—the **(Kolmogorov) forward differential equations**, and also

$$p'_{ij}(t) = \sum_{k\in S} q_{ik}p_{kj}(t) \qquad \text{for all } t \text{ and all } i,j \in S \tag{BDE}$$

—the **(Kolmogorov) backward differential equations**.

Proof: **(S finite.)** Recall the Chapman-Kolmogorov equations:

$$p_{ij}(s+t) = \sum_{k\in S} p_{ik}(s)p_{kj}(t), \qquad i,j \in S, \quad s,t > 0.$$

Differentiate with respect to t to obtain

$$p'_{ij}(s+t) = \sum_{k\in S} p_{ik}(s)p'_{kj}(t),$$

and put $t = 0$ to obtain (FDE) (with s instead of t).

Similarly, differentiate with respect to s to obtain

$$p'_{ij}(s+t) = \sum_{k\in S} p'_{ik}(s)p_{kj}(t),$$

and put $s = 0$ to obtain (BDE). □

Note. Usually, and always when S is finite, given Q, the transition probabilities $(p_{ij}(\cdot))_{i,j\in S}$ are **uniquely** determined by either (FDE) or (BDE), together with the initial condition $p_{ij}(0) = \delta_{ij}$.

In general the **forward differential equations** (FDE) are more useful in practice for determining the functions $(p_{ij}(\cdot))_{i,j\in S}$. The **backward differential equations** (BDE) are mostly used to prove theorems.

Example: general two-state process. Here we have $S = \{0, 1\}$, where, for example 0 might be **healthy** and 1 might be **sick**—in this very simple model nobody ever dies! The transition rate matrix Q is given by

$$Q = \begin{pmatrix} -\lambda & \lambda \\ \mu & -\mu \end{pmatrix}$$

for some $\lambda, \mu > 0$.

The forward differential equations (FDE) for the initial state i become

$$\begin{aligned} p'_{i0}(t) &= p_{i0}(t)(-\lambda) + p_{i1}(t)(\mu) \\ p'_{i1}(t) &= p_{i0}(t)(\lambda) + p_{i1}(t)(-\mu) \end{aligned}$$

and we also have the initial condition $p_{ij}(0) = \delta_{ij}$.

Take $i = 0$: it is easy to verify that

$$\begin{aligned} p_{00}(t) &= \frac{\mu}{\lambda+\mu} + \frac{\lambda}{\lambda+\mu} e^{-(\lambda+\mu)t} \\ p_{01}(t) &= \frac{\lambda}{\lambda+\mu} - \frac{\lambda}{\lambda+\mu} e^{-(\lambda+\mu)t} \qquad (= 1 - p_{00}(t)) \end{aligned}$$

satisfy the forward differential equations and initial condition, and hence give the transition probabilities.

Take $i = 1$: we similarly obtain (e.g. by symmetry)

$$\begin{aligned} p_{10}(t) &= \frac{\mu}{\lambda+\mu} - \frac{\mu}{\lambda+\mu} e^{-(\lambda+\mu)t} \\ p_{11}(t) &= \frac{\lambda}{\lambda+\mu} + \frac{\mu}{\lambda+\mu} e^{-(\lambda+\mu)t} \qquad (= 1 - p_{10}(t)). \end{aligned}$$

Note also that

$$P(t) \to \begin{pmatrix} \frac{\mu}{\lambda+\mu} & \frac{\lambda}{\lambda+\mu} \\ \frac{\mu}{\lambda+\mu} & \frac{\lambda}{\lambda+\mu} \end{pmatrix} \quad \text{as } t \to \infty,$$

so that the limiting distribution is independent of the initial state.

6.4 The jumping chain

Suppose that $\{X(t)\}_{t\geq 0}$ is a **time homogeneous Markov process** with transition rate matrix $Q = (q_{ij})_{i,j\in S}$.

Then, for any t and for any states $i, j \in S$, we have

$$\begin{aligned}\mathbf{P}(X(t+h) \neq i \,|\, X(t) = i) &= 1 - p_{ii}(h) \\ &= -q_{ii}h + o(h) \qquad\qquad \text{as } h \to 0,\end{aligned}$$

and, for $j \neq i$,

$$\begin{aligned}\mathbf{P}(X(t+h) = j \,|\, X(t) = i) &= p_{ij}(h) \\ &= q_{ij}h + o(h) \qquad\qquad \text{as } h \to 0,\end{aligned}$$

so that

$$\begin{aligned}\mathbf{P}(X(t+h) = j \,|\, X(t) = i,\, X(t+h) \neq i) &= \frac{q_{ij}h + o(h)}{-q_{ii}h + o(h)} \\ &\to \frac{q_{ij}}{-q_{ii}} \qquad\qquad \text{as } h \to 0.\end{aligned}$$

Thus, using also the Markov property, if at any time t the process $\{X(\cdot)\}$ is in state i, the distribution of the further time to the next jump is $\text{Exp}(-q_{ii})$ (mean $-q_{ii}^{-1}$) independently of the history of the process prior to time t, and, when the jump occurs, it is to state $j \neq i$ with probability $-q_{ij}/q_{ii}$, independently of the time of the jump and the history of the process prior to the jump.

Define

$$\begin{aligned}T_0 &= 0 \\ T_n &= \text{time of } n\text{th jump of } \{X(\cdot)\} \text{ to a different state,}\end{aligned}$$

and, for all $n \geq 0$, let $X_n^* = X(T_n)$, where $X_n^* = X(T_n)$ is the *new* state of the process at time T_n. Then, conditional on $X(T_n) = i$, the distribution of $T_{n+1} - T_n$ is $\text{Exp}(-q_{ii})$, independently of the history of the process prior to time T_n, and the *discrete time* process $\{X_n^*\}_{n\geq 0}$ is a **Markov chain**, called the **jumping chain** or the **embedded Markov chain**, with transition matrix $P^* = (p_{ij}^*)_{i,j\in S}$ given by:

$$\begin{array}{llllll}\text{if } q_{ii} < 0, & \text{then } p_{ij}^* = -\dfrac{q_{ij}}{q_{ii}} & \text{for } j \neq i, & \text{and } p_{ii}^* = 0; & & \\ \text{if } q_{ii} = 0, & \text{then } p_{ij}^* = 0 & \text{for } j \neq i, & \text{and } p_{ii}^* = 1 & \text{(state } i \text{ absorbing).}\end{array}$$

We can now use the discrete time theory of the **jumping chain** $\{X_n^*\}_{n\geq 0}$ to answer many questions about the original **process** $\{X(t)\}_{t\geq 0}$.

First note that, as in the discrete time case, a state $i \in S$ is defined to be **recurrent** if and only if, starting at any time from that state, the process $\{X(t)\}$ returns to it with probability 1; otherwise it is **transient**.

Result. The division of the state space into **classes of intercommunicating states**, the characterisation of these classes as **closed** or **nonclosed**, the **absorption probabilities** associated with the closed classes, and the characterisation of classes as **recurrent** or **transient**, are all the same for the **process** $\{X(t)\}_{t\geq 0}$ and the associated **jumping chain** $\{X_n^*\}_{n\geq 0}$.

Further, for any $i, j \in S$, $i \to j$ is equivalent to $p_{ij}(t) > 0$ for **all** $t > 0$, so there are no problems of periodicity in the process (though there may be in the associated jumping chain).

Note, however, that the classification of **recurrent** states as **positive recurrent** or **null recurrent** may be different for the process from that for the jumping chain—see Section 6.5.2.

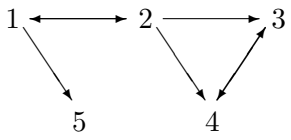
Example. Take $S = \{1, 2, 3, 4, 5\}$, and

$$Q = \begin{pmatrix} -4 & 1 & 0 & 0 & 3 \\ 2 & -6 & 3 & 1 & 0 \\ 0 & 0 & -5 & 5 & 0 \\ 0 & 0 & 4 & -4 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Then the transition matrix P^* of the jumping chain is given by

$$P^* = \begin{pmatrix} 0 & 1/4 & 0 & 0 & 3/4 \\ 1/3 & 0 & 1/2 & 1/6 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

Hence we have the following *one* jump communications:



Note that $1 \to 3$ and $1 \to 4$ also.
Hence the **classes** are

$\{1, 2\}$	non-closed, hence transient
$\{3, 4\}$	closed finite, hence recurrent
$\{5\}$	closed finite (absorbing state), hence recurrent

For the **absorption probabilities**, let

$$y_i = \mathbf{P}(X(\cdot) \text{ eventually absorbed in state } 5 \,|\, X(0) = i).$$

Then, letting $P^* = (p^*_{ij})$ be the transition matrix of jumping chain, we obtain

$$\begin{aligned} y_i &= \sum_j p^*_{ij} y_j \qquad\qquad j = 1, 2 \\ y_3 &= y_4 = 0 \\ y_5 &= 1 \end{aligned}$$

Hence we obtain $y_1 = 9/11$, $y_2 = 3/11$ (*check*).
Further, since the set of states in nonclosed (open) classes is finite and there are just two closed classes, we have, for all i,

$$\mathbf{P}(X(\cdot) \text{ eventually absorbed in class } \{3, 4\} \,|\, X(0) = i) = 1 - y_i.$$

6.5 Stationary and limiting distributions for a single closed class

Throughout this section $\{X(t)\}_{t\geq 0}$ continues to be a **time homogeneous Markov process** with with state space S and transition rate matrix $Q = (q_{ij})_{i,j\in S}$. We further assume that the entire state space S is a **single, necessarily closed, class**, i.e. all states intercommunicate ($\Leftrightarrow$ all states intercommunicate in the jumping chain $\Leftrightarrow p_{ij}(t) > 0$ for all $i, j \in S$ and $t > 0$). The Markov process is then said to be **irreducible**.

6.5.1 Stationary distributions

Definition. A (probability) distribution $\boldsymbol{\pi}$ on S ($\pi_j \geq 0$ for all $j \in S$ and $\sum_{j\in S} \pi_j = 1$) is **stationary** (for the process $\{X(t)\}_{t\geq 0}$) if and only if, for all $t \geq 0$,

$$\boldsymbol{\pi} P(t) = \boldsymbol{\pi} \qquad \text{or equivalently} \qquad \sum_{i\in S} \pi_i p_{ij}(t) = \pi_j, \quad j \in S. \tag{15}$$

i.e. if the distribution of $X(0)$ is $\boldsymbol{\pi}$, then, for all $t \geq 0$, the distribution of $X(t)$ is also $\boldsymbol{\pi}$.

A **stationary** distribution is also known as an **equilibrium** distribution for the Markov process. As with Markov chains, an informal interpretation (which may be made formal) is that it gives the **long-term proportion of time** spent in each of the states.

Analogously to the corresponding result for Markov chains, we have the following theorem.

Theorem 6.5.1 Assume that the state space $\mathcal{S}$ is a single closed class for the Markov process. Suppose that the distribution $\boldsymbol{\pi}$ is **stationary**. Then $\pi_j > 0$ for all $j \in S$.

Proof: Since $\sum_{i\in S} \pi_i = 1$ we have $\pi_i > 0$ for *some* $i \in S$. Then, since S is a single class (all states intercommunicate), for any other $j \in S$ and for any $t > 0$, $p_{ij}(t) > 0$, and so, from (15), $\pi_j > 0$. □

Unlike the discrete time analogue, the definition (15) is not very useful for finding stationary distributions. The dynamics of a process are usually given via the transition rate matrix Q, and it is helpful to be able to obtain the stationary distribution directly from Q. The following theorem shows how to do this.

Theorem 6.5.2 A distribution $\boldsymbol{\pi}$ is stationary if and only if

$$\boldsymbol{\pi}Q = \mathbf{0} \qquad (\sum_{i\in S} \pi_i q_{ij} = 0 \quad \text{for all } j).$$

Proof: We prove the result in the case where S is finite.

$$\begin{aligned}
\boldsymbol{\pi} \text{ stationary} &\Rightarrow \sum_{i\in S} \pi_i p_{ij}(t) = \pi_j \qquad \text{for all } j \text{ and for all } t \\
&\Rightarrow \sum_{i\in S} \pi_i p'_{ij}(t) = 0 \qquad \text{for all } j \text{ and for all } t \\
&\Rightarrow \sum_{i\in S} \pi_i q_{ij} = 0 \qquad \text{for all } j \qquad (\text{putting } t = 0)
\end{aligned}$$

Conversely, suppose that $\sum_{i\in S} \pi_i q_{ij} = 0$ for all $j \in S$. Recall the *backward differential equations* (BDE):

$$p'_{ij}(t) = \sum_{k\in S} q_{ik} p_{kj}(t) \qquad \text{for all } i, j \in S.$$

Then, for all $j \in S$,

$$\begin{aligned}
\sum_{i\in S} \pi_i p'_{ij}(t) &= \sum_{i\in S}\sum_{k\in S} \pi_i q_{ik} p_{kj}(t) \\
&= \sum_{k\in S} \left(\sum_{i\in S} \pi_i q_{ik} \right) p_{kj}(t) \\
&= 0 \qquad \text{for all } j.
\end{aligned}$$

Hence, for each j, $\sum_{i\in S} \pi_i p_{ij}(t)$ is constant in t; putting $t = 0$, it follows that this constant is π_j. □

Example. Let $S = \{1, 2, 3\}$ and let the matrix of transition rates of the Markov process be given by

$$Q = \begin{pmatrix} -5 & 2 & 3 \\ 5 & -6 & 1 \\ 1 & 2 & -3 \end{pmatrix}$$

Thus all states form a single closed class. The stationary distribution $\boldsymbol{\pi}$ of the process is given by the solution of $\boldsymbol{\pi}Q = \mathbf{0}$, i.e. by

$$\begin{aligned} -5\pi_1 + 5\pi_2 + \pi_3 &= 0 \\ 2\pi_1 - 6\pi_2 + 2\pi_3 &= 0 \\ 3\pi_1 + \pi_2 - 3\pi_3 &= 0. \end{aligned}$$

Note that one of these equations is redundant. However, we also have

$$\pi_1 + \pi_2 + \pi_3 = 1.$$

Hence we obtain that the stationary distribution is given by $\pi = (1/3, 1/4, 5/12)$.

Example: general two-state process. Consider again the (**healthy-sick**) example of Section 6.3, i.e. with the transition rate matrix Q is given by

$$Q = \begin{pmatrix} -\lambda & \lambda \\ \mu & -\mu \end{pmatrix}$$

for some $\lambda, \mu > 0$. Here it is easy to verify that the stationary distribution is given by

$$\boldsymbol{\pi} = \left(\frac{\mu}{\lambda + \mu}, \frac{\lambda}{\lambda + \mu} \right)$$

which coincides with the limiting distribution found earlier—see the next section.

Note. The equation $\boldsymbol{\pi}Q = \mathbf{0}$ may also be written as

$$\sum_{i \neq j} \pi_i q_{ij} = \sum_{i \neq j} \pi_j q_{ji} \qquad \text{for all } j \in S.$$

(*Exercise.*)

6.5.2 Stationary and limiting distributions

Unlike the discrete time case, there are here no problems of *periodicity*. We now have results entirely analogous to those at the end of Section 4.4.1 and in Section 4.4.2—with analogous proofs where these have been given (i.e. for the case where S is finite). We state all these results in a single summary theorem.

Theorem 6.5.3 Suppose that S is a **single closed class**. Then *either*

1. there is a **unique stationary distribution** $\boldsymbol{\pi}$ (the solution of $\boldsymbol{\pi}Q = \mathbf{0}$ and $\sum_{j\in S}\pi_j = 1$) and then
$$\lim_{t\to\infty} p_{ij}(t) = \pi_j, \qquad \text{for all } i,j \in S;$$
in this case S is **positive recurrent** with respect to $\{X(t)\}_{t\geq 0}$;
or

2. there is **no stationary distribution** and then
$$\lim_{t\to\infty} p_{ij}(t) = 0, \qquad \text{for all } i,j \in S;$$
in this case S is **null recurrent** or **transient** with respect to $\{X(t)\}_{t\geq 0}$.

If S is finite then the case 1 necessarily holds.

Stationary and limiting distributions for a Markov process and its embedded Markov chain.
We know that, for an *irreducible* Markov process with finite state space, there always exists a unique stationary distribution. The same holds for the embedded Markov chain. then the question is: when do these distributions coincide?

The answer is: if and only if all elements q_{ii} of the main diagonal of matrix Q do coincide. Here is the reasoning: $-q_{ii}$ are the parameters of the distributions of holding time at different states so the expectations of the holding times are $-1/q_{ii}$. We know that stationary probabilities are also the long-run time averages. Namely, let $N_j(t)$ be the number of visits to state $j \in S$ within time interval $(0,t)$ and Y_k the duration of the kth visit. Then all Y_k have $Exp(-q_{jj})$ distribution, and the stationary probability π_i for the process to be at state j may be defined as

$$\begin{aligned}
\pi_j &= \lim_{t\to\infty} \frac{\text{total length of time intervals where } \ X(u)=j}{t} \\
&= \lim_{t\to\infty} \frac{\sum_{k=1}^{N_j(t)} Y_k}{t} = \lim_{t\to\infty} \frac{\sum_{k=1}^{N_j(t)} Y_k}{N_j(t)} \cdot \frac{N_j(t)}{t} \\
\text{(by the SLLN)} \quad &= \frac{1}{-q_{jj}} \cdot \lim_{t\to\infty} \frac{N_j(t)}{t} \\
&= \frac{1}{-q_{jj}} \cdot \lim_{n\to\infty} \frac{N_j(T_n)}{T_n},
\end{aligned}$$

where T_n is the time of the n'th jump of the Markov process. In the last line, we use the fact that if the limit $N_j(t)/t$ exists, then it is equal to the limit along any subsequence of times, and choose this subsequence as $T_1, T_2, \ldots$.

Now let $N_j^*(n)$ be the number of visits to j after n jumps, for the embedded Markov chain. Then, clearly, $N_j(T_n) = N_j^*(n)$, and we may continue the series of equations above with

$$\pi_j = \frac{1}{-q_{jj}} \cdot \lim_{n\to\infty} \frac{N_j^*(n)}{T_n} = \frac{1}{-q_{jj}} \cdot \lim_{n\to\infty} \frac{N_j^*(n)/n}{T_n/n}.$$

If all q_{jj} are the same (and, say, equal to q), then T_n has a Gamma distribution $\Gamma(-q, n)$ and, by the SLLN,

$$\frac{T_n}{n} \to \frac{1}{-q}.$$

Then

$$\pi_j = \frac{1}{-q} \cdot \frac{\lim_{n\to\infty} N_j^*(n)/n}{\frac{1}{-q}} = \lim_{n\to\infty} \frac{N_j^*(n)}{n}$$

where the latter limit is nothing else than π_j^*, the stationary probability for the embedded Markov chain to be at j.

If, on the contrary, not all q_{jj} are the same, then we may take j such that q_{jj} is the smallest among the diagonal terms. Then one can show that

$$\lim_{n\to\infty} \frac{T_n}{n} < \frac{1}{-q_{jj}}$$

and, therefore,

$$\pi_j > \pi_j^*.$$

6.6 Stationary and limiting distributions in the case of several communication classes (finite state space).

Here the theory is very similar to that for discrete-time Markov chains.

Assume there is given a matrix Q of transition rates of a (time-homogeneous) Markov process $X(t)$ with a finite state space. Here is a collection of questions students may be asked to answer.

1. Verify that Q is indeed a transition rates matrix.
Solution: verify conditions (Q1) $q_{ij} \geq 0$ for all $i \neq j$; (Q2) $q_{ii} \leq 0$ for all i; (Q3) $\sum_{j \in S} q_{ij} = 0$, for all i.

2. Find the transition matrix P^* of the embedded Markov chain $X^*(n)$.

3. Classify the states of the Markov process using matrix P^* (we know that the classification is the same for the Markov process and for its embedded chain).

4. Assume there are k closed classes $C_1, \ldots, C_k$ and a number of open classes.
Then find absorption probabilities $\alpha_i^{(l)} \equiv \alpha_i^{(C_l)}$, for each initial value $X_0^* = i$ and for each class $C_l, l = 1, 2, \ldots, k$.

5. Find a family of all stationary distributions of Markov process $X(t)$.
Solution: for each closed class C_l, find the stationary distribution $\pi^{(l)}$ within that class. Then a general form of a stationary distribution is given by

$$\sum_{l=1}^{k} \gamma_l \pi^{(l)}$$

where $\gamma_1, \ldots, \gamma_k$ are non-negative constants with $\sum_{l=1}^{k} \gamma_l = 1$.

6. For any initial value $X(0) = i$ of the Markov process, find the limiting distribution (for each initial value there is only one limiting distribution).
Solution. If $i \in C_l$ for some l, then the limiting distribution coincides with the stationary distribution $\pi^{(l)}$ in the closed class C_l.
If i belongs to an open class, then the limiting distribution is

$$\sum_{l=1}^{k} \alpha_i^{(l)} \pi^{(l)}$$

where, as before, $\alpha_i^{(l)}$ are the absoprtion probabilities.

7. You may also be asked to find stationary and limiting distributions for the embedded Markov chain X_n^*.

8. When do the families of stationary distributions of the Markov process and of its embedded Markov chain coincide? An answer is again based on the numbers q_{ii} on the main diagonal of matrix Q. However, only states from closed classes are important.

The answer is: if the rates q_{ii} are the same within each closed class, then the families of distributions coincide. Otherwise they differ.

7 Branching Processes – an important subclass of Markov Chains and Markov Jump Processes

We describe a basic model for population growth known as the Galton-Watson process. We introduce and discuss the model in discrete time and, in particular, show that it is a Markov chain. We formulate a theorem on characterisation of its extinction probability and provide two its proofs. Then we introduce a continuous-time Galton-Watson process and show that it is a Markov Jump Process.

7.1 Branching Process in Discrete Time

Here is the model.
At each integer time n we have a number X_n of individuals each of which gives birth to a random number of offspring, independently from one another, and following the same distribution. The parent dies immediately upon giving birth.
Let p_j be the probability that an individual will bear j offspring, $j = 0, 1, 2, ...$, with $\sum_{j=0}^{\infty} p_j = 1$. Let further $\xi_k^{(n)}$ be the number of offspring of the kth individual in the nth generation. Then the size X_{n+1} of the $(n+1)$-st generation may be found by adding the number of offspring of each individual,

$$X_{n+1} = \xi_1^{(n)} + \xi_2^{(n)} + \ldots + \xi_{X_n}^{(n)} = \sum_{r=1}^{X_n} \xi_r^{(n)}.$$

Since we assume that all the random variables $\{\xi_i^{(n)}\}, n = 0, 1, 2, \ldots, i = 1, 2, \ldots$ are i.i.d., we can easily see that the Markov property holds:

$$\begin{aligned}
&\mathbf{P}(X_{n+1} = j \mid X_n = i, X_{n-1} = i_{n-1}, \ldots, X_0 = i_0) \\
&= \mathbf{P}(\sum_{r=1}^{X_n} \xi_r^{(n)} = j \mid X_n = i, X_{n-1} = i_{n-1}, \ldots, X_0 = i_0) \\
&= \mathbf{P}(\sum_{r=1}^{i} \xi_r^{(n)} = j \mid X_n = i, X_{n-1} = i_{n-1}, \ldots, X_0 = i_0) \\
&= \mathbf{P}(\sum_{r=1}^{i} \xi_r^{(n)} = j) =: p_{ij}
\end{aligned}$$

is a function of i and j only and does not depend on $i_{n-1}, \ldots, i_0$ and on n. So this is a time-homogeneous Markov chain.
For this Markov chain, computing the transition probabilities p_{ij} may be a hard problem, in general. So we will find a different method to study its properties. However, there are particular cases where direct computations are available, and here is one.

Example 7.1.1 Suppose that $p_1 = p$, $p_0 = 1 - p$ and $p_k = 0$, for $k \geq 2$. Thus, each individual either gives birth to one child (w.p. p) or has no children (w.p. $(1-p)$). If we know that $X_n = i$, then X_{n+1} is the number of successful births amongst i individuals, which has a binomial distribution $B(i, p)$, so

$$p_{ij} = \frac{i!}{j!(i-j)!} p^j (1-p)^{(i-j)}, \quad j = 0, 1, ..., i.$$

In this case, the population cannot grow. Moreover, it will eventually reach 0, and 0 is always an absorbing state. Why?
Hint: $X_0 = i$, τ_l = time to die for the lth branch, they have a Geometric distribution with

parameter $1-p$; $T = \max_{1\le l\le i} \tau_l$– time to extinct. An interesting question is: what is a distribution of the time to reach 0, given that at the beginning there were, say, $X_0 = i$ individuals.

Another special case is where $p_1 = P(\xi = 1) = 1$. Then $X_n = X_0$, for all n. If instead $p_2 = P(\xi = 2) = 1$, then $X_n = 2^n X_0$, so the population grows exponentially fast.

Back to the general case, one question of interest is whether the process will become extinct or not. We will not attempt to compute the probabilities p_{ij} neither draw the transition diagram because they are both futile. So we assume that $p_1 \neq 1$ or, equivalently, that

$$\mathbf{P}(\xi = 0) + \mathbf{P}(\xi \geq 2) > 0.$$

We distinguish two cases:
Case I: If $p_0 = \mathbf{P}(\xi = 0) = 0$, then the population can only grow. Hence all states $i \geq 1$ are transient and, for any pair of states $i < j$, state j may be achievable from i, $i \to j$, but i cannot be achievable from j. State 0 is absorbing, but it cannot be reached from anywhere else.
Case II: If $p_0 > 0$ then state 0 may be reached from any other state (even in one step): $p_{i0} = p_0^i > 0$. But 0 is an absorbing state. Therefore all states $i \geq 1$ are transient. This means that each of them may be visited only a finite number of times. So if the Markov chain never visits 0 then, for each k, one can choose finite (random) time T_k such that

$$X_n > k, \text{ for all } n > T_k.$$

The latter means that, with probability one, either the Markov chain arrives at 0 and stays there or $X_n \to \infty$.
Let D be the event of "death", or extinction:

$$D := \{X_n = 0 \text{ for some } n \geq 0\}.$$

We wish to compute the extinction probability

$$\varepsilon(i) = \mathbf{P}_i(D) = \mathbf{P}(D|X_0 = i).$$

Obviously, $\varepsilon(0) = 1$. Then, for $i \geq 1$, we can argue that

$$\varepsilon(i) = \varepsilon^i \text{ where } \varepsilon = \varepsilon(1) = \mathbf{P}_1(D).$$

Indeed, if we start with $X_0 = i \geq 1$ initial ancestors, each one behaves independently of one another. For each $l = 1, ..., i$, let D_l be the event that the branching process corresponding to the lth initial ancestor dies. We then have that

$$D = D_1 \cap D_2 \cap \cdots \cap D_i$$

so

$$\mathbf{P}_i(D) = \mathbf{P}(D|X_0 = i) = \mathbf{P}(D_1) \cdot \cdots \cdot \mathbf{P}(D_i) = \varepsilon^i.$$

Therefore the problem of funding the extinction probability reduces to the study of the branching process that starts from $X_0 = 1$. Here is the result.

Theorem 7.1.1 Consider a Galton-Watson process starting from $X_0 = 1$ initial ancestor. Let

$$\mu = \mathbf{E}\,\xi = \sum_{k=1}^{\infty} k p_k$$

be the mean number of offspring of an individual. Assume that $p_0 > 0$ and $p_0 + p_1 < 1$. Let

$$\varphi(z) = \mathbf{E}\, z^{\xi} = \sum_{k=0}^{\infty} z^k p_k$$

be the probability generating function of ξ. Let ε be the extinction probability of the process. Then

1. if $\mu \leq 1$ then the $\varepsilon = 1$ and
2. if $\mu > 1$ then the 'fixed-point equation'

$$\varphi(z) = z,$$

has only two solutions, $0 < z_1 < z_2 = 1$, and $\varepsilon = z_1$ is the minimal one.

Proof: Let

$$\varphi_n(z) = \mathbf{E}\, z^{X_n} = \sum_{k \geq 0} z^k \mathbf{P}(X_n = k)$$

be the probability generating function of X_n. This function is of interest because it gives an information about the extinction probability ε. Indeed,

$$\varphi_n(0) = \mathbf{P}(X_n = 0).$$

Since

$$X_n = 0 \text{ implies } X_m = 0 \text{ for all } m \geq n,$$

we have

$$D_1 = \{\text{there exists } n \text{ such that } X_m = 0 \text{ for all } m \geq n = \bigcup_{n \geq 0} \{X_n = 0\}$$

where the events $\{X_n = 0\}$ are increasing. Therefore, from the basic probability theory, we have the following continuity property:

$$\varepsilon = \mathbf{P}(D_1) = \lim_{n \to \infty} \mathbf{P}(X_n = 0) = \lim_{n \to \infty} \varphi_n(0).$$

Note also that

$$\varphi_0(z) = \mathbf{E}\, z^{X_0} = z^1 = z \text{ and } \varphi_1(z) = \varphi(z).$$

We then have

$$\begin{aligned}
\varphi_{n+1}(z) = \mathbf{E}\, z^{X_{n+1}} &= \mathbf{E}\left(z^{\xi_1^{(n)}} z^{\xi_2^{(n)}} \dots z^{\xi_{X_n}^{(n)}}\right) \\
&= \sum_{i=0}^{\infty} \mathbf{E}\left(z^{\xi_1^{(n)}} z^{\xi_2^{(n)}} \dots z^{\xi_{X_n}^{(n)}} \mathbb{I}(X_n = i)\right) \\
&= \sum_{i=0}^{\infty} \mathbf{E}\left(z^{\xi_1^{(n)}} z^{\xi_2^{(n)}} \dots z^{\xi_i^{(n)}} \mathbb{I}(X_n = i)\right) \\
&= \sum_{i=0}^{\infty} \mathbf{E}\left(z^{\xi_1^{(n)}}\right) \mathbf{E}\left(z^{\xi_2^{(n)}}\right) \cdots \mathbf{E}\left(z^{\xi_i^{(n)}}\right) \mathbf{E}\, \mathbb{I}(X_n = i) \\
&= \sum_{i=0}^{\infty} \varphi(z)^i \mathbf{P}(X_n = i) \\
&= \mathbf{E}\left(\varphi(z)^{X_n}\right) \\
&= \varphi_n(\varphi(z)).
\end{aligned}$$

Hence

$$\varphi_2(z) = \varphi_1(\varphi(z)) = \varphi(\varphi(z)), \varphi_3(z) = \varphi_2(\varphi(z)) = \varphi(\varphi(\varphi((z))),$$

and so on,

$$\varphi_{n+1}(z) = \varphi(\varphi_n(z)).$$

In particular,

$$\varphi_{n+1}(0) = \varphi(\varphi_n(0)).$$

But, as we already know, $\varphi_n(0) \to \varepsilon$ as $n \to \infty$. Therefore, $\varphi_{n+1}(0) \to \varepsilon$ too. Since φ is a continuous function, $\varphi(\varphi_n(0)) \to \varphi(\varepsilon)$ as $n \to \infty$. Therefore

$$\varepsilon = \varphi(\varepsilon).$$

Notice also that

$$\mu = \frac{d}{dz}\varphi(z)\Big|_{z=1}$$

Also, recall that φ is a strictly convex function with $\varphi(1) = 1$. Why?

Therefore if the slope μ at $z = 1$ is ≤ 1, the graph of $\varphi(z)$ lies above the diagonal (as the function of z) and so the equation $\varphi(z) = z$ has no solution other than $z = 1$. This means that $\varepsilon = 1$.
On the other hand, if $\mu > 1$, then the graph of $\varphi(z)$ does intersect the diagonal at some point $\varepsilon^* < 1$. Let us show that, in fact, $\varepsilon = \varepsilon^*$. Since both ε^* and ε satisfy equation $z = \varphi(z)$, and since the latter has only two solutions ($z = 1$ and $z = \varepsilon^*$), all we need to show is that $\varepsilon < 1$. Let us do this.
First, clearly $0 \leq \varepsilon^*$. Since φ is a monotone increasing function, $\varphi(0) \leq \varphi(\varepsilon^*) = \varepsilon^*$. Then, by induction, $\varphi_n(0) \leq \varepsilon^*$. But $\varphi_n(0) \to \varepsilon$. So $\varepsilon \leq \varepsilon^* < 1$. So $\varepsilon = \varepsilon^*$. This finishes the proof of the theorem. □

There is also an alternative version of the proof using theory of Markov chains.

Proof: We already know that $\varepsilon(i) = \varepsilon(1)^i = \varepsilon^i$. In other terms the extinction probablity is the absorption probability at $\{0\}$, here $C = \{0\}$ is the absorbing class (state). By theorem 4.2.1, with $\varepsilon = \alpha_1^C$ and $\varepsilon(i) = \alpha_i^C$ we have

$$\varepsilon = \sum_{i=0}^{\infty} p_{1i}\varepsilon(i) = \sum_{i=0}^{\infty} p_i\varepsilon^i,$$

since $p_{1i} = p_i$, according to earlier notation. So as before, ε is a solution to the fixed point problem

$$z = \varphi(z),$$

where $\phi(z)$ is the probability generating function of r.v. ξ. We can proceed further along the lines in the previous proof.
Alternatively, the other way to prove the result for $\mu > 1$ is to modify dynamics of the branching process by assuming that individuals bear offsprings one-by-one, within each generation. Then we arrive at a random walk with jumps having the distribution of ξ. Since $\mathbf{E}\,\xi - 1 = \mu - 1 > 0$ such a random walk tends to $+\infty$ with a positive probability, avoiding 0, therefore the extinction probability cannot be equal to 1. The next picture illustrates the dynamics. □

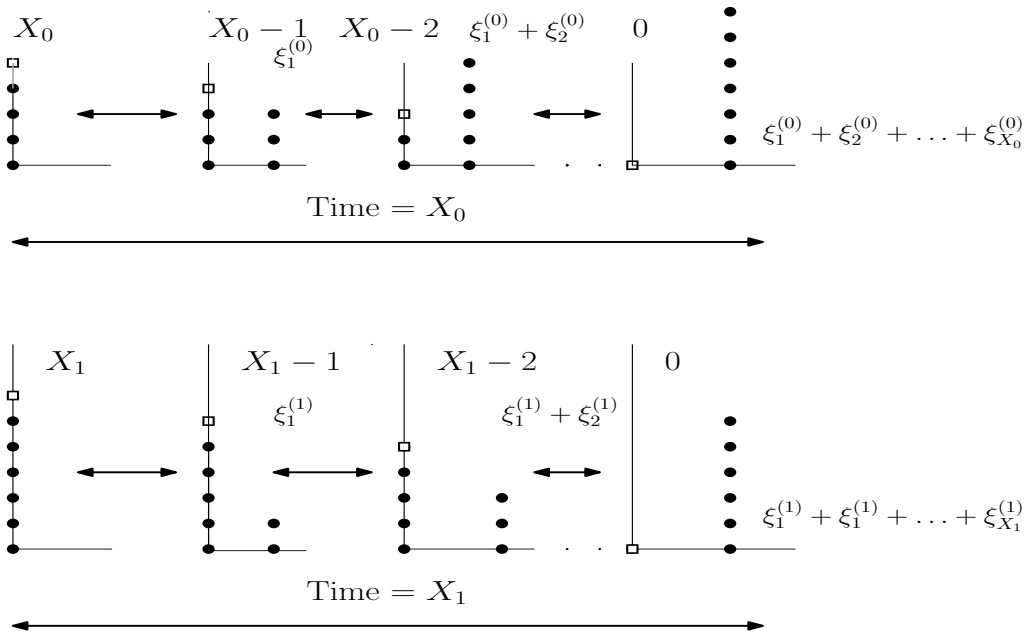


Figure 1: Branching process as a random walk

7.2 Branching Process in Continuous Time

In this short subsection, we introduce a branching process in continuous time (BPCT) and make a few comments.

In the branching process in discrete time the life time of each individual is equal to one. Assume instead that each individual has a random life time which has an exponential distribution with parameter $\lambda > 0$, and that all life times are i.i.d. and do not depend on the branching mechanism. Then we obtain a branching process $X(t)$ in continuous time which is an example of a Markov Process in continuous time. The fact that this is a Markov process follows from the memoryless property of exponential distributions and from mutual independence of the branching mechanism and the life times.

For any $t \geq 0$ and for $h > 0$ small, there may be either no jumps within time interval $(t, t+h)$ (this means that all individuals that were alive at time t stay alive at time $t+h$) or there may be only one jump (this means that all individuals but one stay alive, and one individuals dies within $(t, t+h)$ and produces a random number of offspring) or there many be two or more jumps (this event has probability of order $o(h)$). If there is no jumps, then $X(t+h) = X(t)$ and if there is one jump, then $X(t+h) = X(t) - 1 + Y_t$ where Y_t is the number of offspring of the individual that dies. Therefore, for any $i \geq 1$,

$$\mathbf{P}(X(t+h) = j \mid X(t) = i) = \lambda i p_{j+1-i} h + o(h) \quad for\ j = i-1 \text{ and for } j = i+1, i+2, \ldots,$$

$$\mathbf{P}(X(t+h) = i \mid X(t) = i) = 1 - \lambda i \sum_{r \neq 1} p_r h + o(h).$$

Therefore, the transition rates of the BPCT are given as $q_{ij} = \lambda i p_{j+1-i}$ for $i \geq 1$ and $j \neq i$, $j \geq i-1$, $q_{ii} = -\lambda i \sum_{r \neq 1} p_r$ for $i \geq 1$ and $j \neq i$, and $q_{ij} = 0$ for all other pairs of i and j.

Finally, the extinction probabilities for discrete-time and continuous-time branching processes are the same. You are supposed to provide reasons for that.

Remark. A linear Birth-and-Death process with the state space $\mathcal{S} = \{0, 1, 2, \ldots\}$ and transition rates $q_{i,i+1} = \lambda i p_2$ for $i = 0, 1, 2, \ldots$, $q_{i,i-1} = \lambda i p_0$ for $i = 1, 2, \ldots$ and $q_{i,j} = 0$ for all $|i-j| > 1$ is a particular case of the continuous-time branching process with $p_0 + p_2 = 1$ and $p_k = 0$, for $k \neq 0, 2$.

8 Time inhomogeneous Markov processes and semi-Markov processes: actuarial applications

8.1 Time inhomogeneous Markov processes

8.1.1 Introduction

Many actuarial applications need to be modelled as time inhomogeneous Markov processes.

We first generalise some ideas from the theory of time homogeneous processes.

Let $\{X(t)\}_{t\geq 0}$ be a Markov process taking values in a countable state space S. We no longer assume this process to be time homogeneous.

It is now necessary to define the probability functions, for all $i, j \in S$,

$$p_{ij}(t,u) = \mathbf{P}(X(u) = j \,|\, X(t) = i), \qquad 0 \leq t \leq u.$$

Note that these are functions of two variables, and that for any fixed t and u the matrix

$$P(t,u) = (p_{ij}(t,u))_{i,j\in S}$$

is stochastic. Note also that, for any t, and for all i, j,

$$p_{ij}(t,t) = \delta_{ij}, \tag{16}$$

where, as usual,

$$\delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j. \end{cases}$$

The **Chapman-Kolmogorov equations** now become: for any $i, j \in S$ and for any $t \leq s \leq u$,

$$p_{ij}(t,u) = \sum_{k\in S} p_{ik}(t,s)p_{kj}(s,u). \tag{17}$$

We now have time-dependent transition rates given, for any time t, by $Q(t) = (q_{ij}(t))_{i,j\in S}$, where, for all i, j,

$$q_{ij}(t) = \frac{\partial}{\partial u} p_{ij}(t,u)\Big|_{u=t} = \lim_{h\to 0} \frac{1}{h} \left(p_{ij}(t,t+h) - p_{ij}(t,t)\right) = \lim_{h\to 0} \frac{1}{h} \left(p_{ij}(t,t+h) - \delta_{ij}\right).$$

We *assume* that, for all t, these transition rates exist and, analogously to the time homogeneous case, satisfy the regularity conditions

$$\begin{aligned} 0 \leq q_{ij}(t) < \infty \qquad & \text{for all } i, j \text{ with } j \neq i, \\ 0 \leq -q_{ii}(t) < \infty \qquad & \text{for all } i, \\ \sum_{j\in S} q_{ij}(t) = 0 \qquad & \text{for all } i. \end{aligned}$$

These conditions are automatically satisfied when the state space S is finite, and are also satisfied in all applications of interest.

We have the usual *and very important* interpretation: for all $t, h \geq 0$,

$$\begin{aligned}\mathbf{P}(X(t+h) = j | X(t) = i) &= p_{ij}(t, t+h) \\ &= p_{ij}(t,t) + q_{ij}(t)h + o(h) \quad \text{as } h \to 0 \\ &= \delta_{ij} + q_{ij}(t)h + o(h) \quad \text{as } h \to 0. \end{aligned} \tag{18}$$

[Recall in particular that $q_{ii}(t) \leq 0$ for all $t \geq 0$ and for all i.]

Again the question arises as to how the functions $(p_{ij}(\cdot,\cdot))_{i,j\in S}$ may be recovered from the functions $(q_{ij}(\cdot))_{i,j\in S}$ and, analogously to the time homogeneous case, we have the following result.

Theorem 8.1.1 Given the functions $(q_{ij}(\cdot))_{i,j\in S}$ (satisfying the above conditions) we have, for any given t,

$$\frac{\partial}{\partial u} p_{ij}(t,u) = \sum_{k\in S} p_{ik}(t,u) q_{kj}(u) \qquad \text{for all } u \geq t \text{ and all } i, j \in S \tag{FDE2}$$

—the **(Kolmogorov) forward differential equations**.

Proof: (S finite.) The proof is essentially the same as in the time homogeneous case—we simply regard everything as starting at the fixed time t rather than at time 0. However it is convenient to give a slightly different presentation, in effect calculating the derivatives from first principles.

For $u \geq t$ and $h > 0$, we have from the Chapman-Kolmogorov equations (17)

$$\begin{aligned} p_{ij}(t, u+h) &= \sum_{k\in S} p_{ik}(t,u) p_{kj}(u, u+h) \\ &= \sum_{k\in S} p_{ik}(t,u) \left(\delta_{kj} + q_{kj}(u)h + o(h)\right) \qquad \text{as } h \to 0 \qquad \text{(using (18))} \\ &= p_{ij}(t,u) + \left(\sum_{k\in S} p_{ik}(t,u) q_{kj}(u) \right) h + o(h) \qquad \text{as } h \to 0. \end{aligned} \tag{19}$$

Now divide $p_{ij}(t, u+h) - p_{ij}(t,u)$ by h, and then let $h \to 0$ to obtain (FDE2). □

Again it is usually the case that, given $(q_{ij}(\cdot))_{i,j\in S}$, for any fixed t, the transition probabilities $(p_{ij}(t,u))_{i,j\in S}$ are **uniquely** determined by (FDE2), together with the initial condition $p_{ij}(t,t) = \delta_{ij}$.

Note that, as in the time homogeneous case, the forward differential equations (FDE2) "decouple" so that there is a separate set of simultaneous equations in $p_{ij}(\cdot,\cdot)$, $j \in S$, for each initial state i. Further, again for each i, by summing these equations over all j, we see that one of them is always redundant.

In the **time inhomogeneous** case, it is usually necessary to solve the forward differential equations (FDE2) **numerically**. One very simple possibility is to write them in the form (19) above; then we choose h sufficiently small that we may ignore $o(h)$ and, for fixed t, use (19) to update $p_{ij}(t,u)$ by incrementing u in successive steps of size h. Of course we start with the initial conditions $p_{ij}(t,t) = \delta_{ij}$. We consider some other possibilities later.

Note that it is also possible to formulate the (Kolmogorov) backward differential equations in the time inhomogeneous case, but it seems difficult to find any applications for these.

8.1.2 Probabilities of remaining in states for given time periods

It is also useful to define, for any $i \in S$, and for any $0 \leq t \leq u$,

$$p_{\overline{ii}}(t,u) = \mathbf{P}(X(s) = i \text{ for all } s \text{ such that } t \leq s \leq u \,|\, X(t) = i).$$

This is the probability, given that the process is in state i at time t, it *remains* in state i until at least time u; it is *not* equal to $p_{ii}(t,u)$, except in the special case where, on exiting from state i, return to state i is impossible.

Theorem 8.1.2 (Under our usual assumptions about $(q_{ij}(\cdot))_{i,j\in S}$), for $0 \leq t \leq u$,

$$p_{\overline{ii}}(t,u) = \exp\left(\int_t^u q_{ii}(s)\,ds\right). \tag{20}$$

[Recall yet again that $q_{ii}(s) \leq 0$ for all $s \geq 0$ and for all i.]

Proof: Since the probability of two or more transitions of the process in an interval of length h is $o(h)$ as $h \to 0$, we have, from (18),

$$\begin{aligned} p_{\overline{ii}}(u,u+h) &= p_{ii}(u,u+h) + o(h) \\ &= 1 + q_{ii}(u)h + o(h) \qquad \text{as } h \to 0, \end{aligned}$$

and so, by conditioning also on being in state i at at time u,

$$\begin{aligned} p_{\overline{ii}}(t,u+h) &= p_{\overline{ii}}(t,u)p_{\overline{ii}}(u,u+h) \\ &= p_{\overline{ii}}(t,u)(1 + q_{ii}(u)h) + o(h) \qquad \text{as } h \to 0. \end{aligned}$$

Hence we obtain

$$\frac{\partial}{\partial u} p_{\overline{ii}}(t,u) = p_{\overline{ii}}(t,u) q_{ii}(u)$$

(analogously to the forward differential equations). It is now straightforward to check that this differential equation is satisfied by (20), and that (20) also satisfies the initial condition $p_{\overline{ii}}(t,t) = 1$. □

The *usefulness* of the above result is that $p_{\overline{ii}}(t,u)$ is very easily calculated, e.g. by simple numerical integration, and is a useful adjunct to other probability calculations (discussed later).

8.1.3 The alive-dead model

Suppose that the lifetime of an individual is given by a random variable T which has an absolutely continuous distribution (i.e. a distribution with a density).

We may then define the **force of mortality** μ_x at age x by

$$\mu_x = \lim_{h \to 0} \frac{1}{h} \mathbf{P}(x < T \leq x + h \,|\, T > x)$$

Define also

$${}_np_x = \mathbf{P}(T > x + n \,|\, T > x).$$

We then have the following well-known result.

Result.

$${}_np_x = \exp\left(- \int_x^{x+n} \mu_y \, dy \right). \tag{21}$$

We can also obtain this result by modelling the life of the individual as a two-state time inhomogeneous Markov process in which state 1 is "alive" and state 2 is "dead". (*Why is this process Markov?*) The transition rate matrix $Q(x)$ at age x is given by

$$Q(x) = \begin{pmatrix} -\mu_x & \mu_x \\ 0 & 0 \end{pmatrix}.$$

We have ${}_np_x = p_{11}(x, x+n) = p_{\overline{11}}(x, x+n)$ (*why are these last two quantities equal in this case?*), and so the result (21) is now immediate from Theorem 8.1.2.

Alternatively we may verify (FDE2). There is here effectively just one forward differential equation (*why?*): hence we must check that

$$\frac{\partial}{\partial n} p_{11}(x, x+n) = p_{11}(x, x+n)(-\mu_x) + p_{12}(x, x+n) \times 0,$$

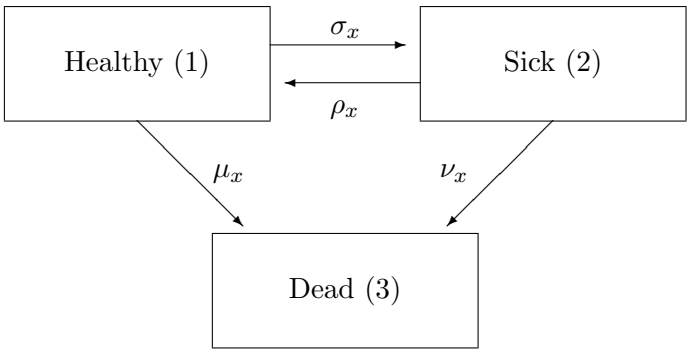
together with the initial condition $p_{11}(x, x) = 1$, where $p_{11}(x, x+n)$ is given by (21). However, this is really a repeat of the argument which proves Theorem 8.1.2.

We also have ${}_nq_x \equiv p_{12}(x, x+n) = 1 - p_{11}(x, x+n) = 1 - {}_np_x$.

8.2 The Markov sickness model

8.2.1 Introduction

A Markov model for sickness has states and transition rates at each age (time) x as illustrated in the following diagram:



Hence the time inhomogeneous transition rate matrix at each age x is given by

$$Q(x) = \begin{pmatrix} -(\sigma_x + \mu_x) & \sigma_x & \mu_x \\ \rho_x & -(\rho_x + \nu_x) & \nu_x \\ 0 & 0 & 0 \end{pmatrix}.$$

Note the symmetry of the model under the interchange of the states 1 and 2. Note also that state 3 ("Dead") is absorbing.

Our goal is to determine the various probability functions.

Note first that, by Theorem 8.1.2,

$$p_{\overline{11}}(x, x+t) = \exp\left(-\int_x^{x+t} (\sigma_y + \mu_y)\, dy\right)$$
$$p_{\overline{22}}(x, x+t) = \exp\left(-\int_x^{x+t} (\rho_y + \nu_y)\, dy\right).$$

The forward differential equations associated with the "initial" state 1 are

$$\begin{aligned} \frac{\partial}{\partial u} p_{11}(t,u) &= p_{11}(t,u)q_{11}(u) + p_{12}(t,u)q_{21}(u) + 0 \\ &= -p_{11}(t,u)(\sigma_u + \mu_u) + p_{12}(t,u)\rho_u \qquad (22) \\ \frac{\partial}{\partial u} p_{12}(t,u) &= p_{11}(t,u)\sigma_u - p_{12}(t,u)(\rho_u + \nu_u) \qquad (23) \\ \frac{\partial}{\partial u} p_{13}(t,u) &= p_{11}(t,u)\mu_u + p_{12}(t,u)\nu_u \end{aligned}$$

As usual, any one of these three equations may be omitted (add them together to check this), and we may ignore the third, regarding (22) and (23) as a pair of simultaneous differential equations in $p_{11}(t,\cdot)$ and $p_{12}(t,\cdot)$.

Note that we have a similar pair of equations for $p_{21}(t,\cdot)$ and $p_{22}(t,\cdot)$.

8.2.2 Numerical evaluation of probabilities

The quantities $p_{\overline{11}}(x, x+t)$ and $p_{\overline{22}}(x, x+t)$, for which expressions are given above, are easily calculated by direct (numerical) integration. These are of use in other calculations—see below.

We now consider solution of simultaneous differential equations (22) and (23) for $p_{11}(t, \cdot)$ and $p_{12}(t, \cdot)$.

One possibility is to use the simple approach outlined in Section 8.1.1, i.e to write the equations (22) and (23) in the alternative form

$$\begin{aligned} p_{11}(t, u+h) &= p_{11}(t,u) + h\left(-p_{11}(t,u)(\sigma_u + \mu_u) + p_{12}(t,u)\rho_u\right) + o(h) \qquad \text{as } h \to 0 \\ p_{12}(t, u+h) &= p_{12}(t,u) + h\left(p_{11}(t,u)\sigma_u - p_{12}(t,u)(\rho_u + \nu_u)\right) + o(h) \qquad \text{as } h \to 0. \end{aligned}$$

Thus, for fixed t and for small h, given $p_{11}(t,u)$ and $p_{12}(t,u)$ for any $u \geq t$, we may calculate $p_{11}(t, u+h)$ and $p_{12}(t, u+h)$ to a good approximation, and continue to increment u in steps of size h.

A further option is to eliminate $p_{12}(t,u)$ from the equations (22) and (23): we differentiate the equation (22) to give a third equation, and from these three equations we eliminate $p_{12}(t,u)$ and $\frac{\partial}{\partial u}p_{12}(t,u)$ to obtain an equation connecting $p_{11}(t,u)$, $\frac{\partial}{\partial u}p_{11}(t,u)$ and $\frac{\partial^2}{\partial u^2}p_{11}(t,u)$ (*exercise*). This is a second-order ordinary differential equation solvable by standard numerical methods. In particular there are good computer routines for accurate solution.

A pair of integral equations. Recall that, for $u \geq t$, $p_{\overline{11}}(t,u)$ and $p_{\overline{22}}(t,u)$ are relatively easily calculated. We now have the following pair of integral equations, valid for all $u \geq t$.

$$p_{11}(t,u) = p_{\overline{11}}(t,u) + \int_t^u p_{12}(t,y)\rho_y p_{\overline{11}}(y,u)\, dy \tag{24}$$

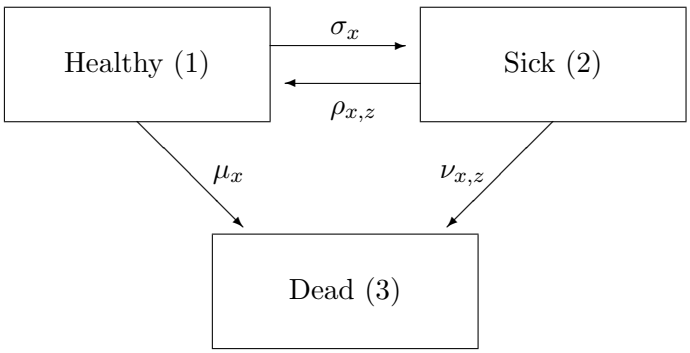
$$p_{12}(t,u) = \int_t^u p_{11}(t,y)\sigma_y p_{\overline{22}}(y,u)\, dy. \tag{25}$$

The explanation for the equation (24) is that, given the process is in state 1 at age t, in order for it to be in state 1 at age u, it must *either* remain in state 1 throughout the entire intervening period—giving the probability $p_{\overline{11}}(t,u)$, *or* there must be some unique intermediate age y such that the process is in state 2 at age y, moves back to state 1 in the interval $[y, y+dy]$, and then *remains* in state 1 until time u—giving the probability $p_{12}(t,y)\rho_y p_{\overline{11}}(y,u)\, dy$, which must then be integrated over all y. The explanation for the equation (25)is similar, but simpler.

These equations provide probabilistic insight, and can be used to provide improved numerical evaluation of $p_{11}(t,u)$ and $p_{12}(t,u)$.

8.3 A semi-Markov sickness model

It is perhaps more realistic to modify the previous model so as to allow the transition rates (intensities) from "Sick" to "Healthy" and from "Sick" to "Dead" to depend on the length of time z already spent in the "Sick" state, as well as on the age x of the individual period (see, e.g., CMIR **12**) This leads to a **semi-Markov** model in which transition rates depend also on the length of time already spent in the current state. The model is illustrated in the following diagram:



$x \equiv$ current age

$z \equiv$ duration to age x of current stay in Sick state

The transition rates ρ_x and ν_x of the process considered in Section 8.2 are replaced by transition rates $\rho_{x,z}$ and $\nu_{x,z}$ which depend also on z. We assume these to be well-defined.

Let $Y(x)$ be the state (1, 2, or 3) at time/age x. The $\{Y(x)\}_{x\geq 0}$ is **not** a Markov process (*why?*), but is a **semi-Markov** process.

Now, for any age x, define the random variable $Z(x)$ to be the length of time spend so far in the current state $Y(x)$ (since last entering that state). Then $\{Y(x),\, Z(x)\}_{x\geq 0}$ **is** a Markov process. We continue to focus on the process $\{Y(x)\}_{x\geq 0}$, adjoining $\{Z(x)\}_{x\geq 0}$ where necessary.

We have formally

$$\lim_{h\to 0}\frac{1}{h}\mathbf{P}(Y(x+h)=2\,|\,Y(x)=1,\,Z(x)=z)=\sigma_x \qquad \text{(independent of } z\text{)}$$

$$\lim_{h\to 0}\frac{1}{h}\mathbf{P}(Y(x+h)=1\,|\,Y(x)=2,\,Z(x)=z)=\rho_{x,z}$$

$$\lim_{h\to 0}\frac{1}{h}\mathbf{P}(Y(x+h)=3\,|\,Y(x)=1,\,Z(x)=z)=\mu_x \qquad \text{(independent of } z\text{)}$$

$$\lim_{h\to 0}\frac{1}{h}\mathbf{P}(Y(x+h)=3\,|\,Y(x)=2,\,Z(x)=z)=\nu_{x,z}$$

While the forward differential equations exist in principle for this model, they are here very complex. We therefore proceed somewhat differently.

Define

$$p_{ij}(x, u \,|\, z) = \mathbf{P}(Y(u) = j \,|\, Y(x) = i,\, Z(x) = z).$$

Note that in the case where $i =$"Healthy", this probability does not depend on z. Define also

$$p_{\overline{ii}}(x, u \,|\, z) = \mathbf{P}(Y(t) = i \text{ for } x \le t \le u \,|\, Y(x) = i,\, Z(x) = z).$$

We then have

$$p_{\overline{11}}(x, u \,|\, z) = \exp\left(-\int_x^u (\sigma_y + \mu_y)\, dy\right) \tag{26}$$

$$p_{\overline{22}}(x, u \,|\, z) = \exp\left(-\int_x^u (\rho_{y,z+y-x} + \nu_{y,z+y-x})\, dy\right) \tag{27}$$

The result (26) is immediate from Theorem 8.1.2, as in the case of the Markov model considered in Section 8.2. The result (27) is also immediate from Theorem 8.1.2—just note that, if the $Y(x) = 2$ (in "Sick" state at age x) and $Z(x) = z$, then, *while the process* $\{Y(\cdot)\}$ *remains in the "Sick" state*, at any age $y \ge x$, the total transition rate to the other two states is $\rho_{y,z+y-x} + \nu_{y,z+y-x}$, i.e we can treat z as fixed and the transition rate as just age dependent.

Now note that, $p_{11}(x, u \,|\, z)$, $p_{12}(x, u \,|\, z)$ and $p_{\overline{11}}(x, u \,|\, z)$ do not depend on z (*why?*) and so we can drop z from the notation for these quantities. We now have the following theorem, which gives a complete set of integral equations for the evolution of the process.

Theorem 8.3.1 For any $u \geq x$,

$$\frac{\partial}{\partial u} p_{11}(x,u) = -(\mu_u + \sigma_u) p_{11}(x,u) + \int_x^u p_{11}(x,y)\sigma_y \, p_{\overline{22}}(y,u \,|\, 0)\rho_{u,u-y}\, dy \tag{28}$$

$$p_{12}(x,u) = \int_x^u p_{11}(x,y)\sigma_y \, p_{\overline{22}}(y,u \,|\, 0)\, dy \tag{29}$$

$$p_{21}(x,u \,|\, z) = \int_x^u p_{\overline{22}}(x,y \,|\, z)\rho_{y,z+y-x}\, p_{11}(y,u)\, dy \tag{30}$$

$$p_{22}(x,u \,|\, z) = p_{\overline{22}}(x,u \,|\, z) + \int_x^u p_{\overline{22}}(x,y \,|\, z)\rho_{y,z+y-x}\, p_{12}(y,u)\, dy. \tag{31}$$

Proof: To prove (28), note that, for $h > 0$ (think h very small),

$$\begin{aligned}
p_{11}(x,u+h) &= \mathbf{P}(Y(u+h) = 1 \,|\, Y(x) = 1) \\
&= \mathbf{P}(Y(u) = 1,\, Y(u+h) = 1 \,|\, Y(x) = 1) \\
&\quad + \mathbf{P}(Y(u) = 2,\, Y(u+h) = 1 \,|\, Y(x) = 1) \\
&= \mathbf{P}(Y(u) = 1 \,|\, Y(x) = 1)\mathbf{P}(Y(u+h) = 1 \,|\, Y(x) = 1, Y(u) = 1) \\
&\quad + \int_x^u \mathbf{P}(Y(y) = 1,\, Y(t) = 2 \text{ for } y + dy \leq t \leq u,\, Y(u+h) = 1 \,|\, Y(x) = 1) \\
&= p_{11}(x,u)[1 - (\mu_u + \sigma_u)h] + \int_x^u p_{11}(x,y)\sigma_y p_{\overline{22}}(y,u \,|\, 0)\rho_{u,u-y} h\, dy + o(h),
\end{aligned}$$

where the integrals above are obtained by considering all possible intervals $[y, y+dy]$ at which the process $\{Y(\cdot)\}$ might move into state 2 for the *last* time prior to time u. Now, as usual, divide $p_{11}(x,u+h) - p_{11}(x,u)$ by h and let $h \to 0$ to obtain (28).

The remaining results are similarly, but more simply, proved. For (29), we again consider the possible intervals $[y, y+dy]$ at which the process $\{Y(\cdot)\}$ might move into state 2 for the last time prior to time u. For each of (30) and (31), we consider the possible intervals $[y, y+dy]$ at which the process $\{Y(\cdot)\}$ might move *out of* state 2 for the *first* time after time/age x. □

Numerical solution. Note first that, for all x, z, u, the quantities $p_{\overline{11}}(x,u)$ and $p_{\overline{22}}(x,u \,|\, z)$ are easily determined via (26) and (27).

For any initial age x, the equation (28) may then be solved numerically to give $p_{11}(x,u)$ for all $u \geq x$. (For example, if, for fixed x, $p_{11}(x,y)$ has been determined for all y such that $x \leq y \leq u$, then (28) may be used as usual—with numerical integration required—to estimate $p_{11}(x,u+h)$ for some sufficiently small increment h; hence $p_{11}(x,u)$ may be estimated for all u by incrementing u in small steps.) For fixed x and, where necessary, for fixed z, the equations (29), (30) and (31) then give expressions for $p_{12}(x,u)$, $p_{21}(x,u \,|\, z)$ and $p_{22}(x,u \,|\, z)$, which are easily evaluated by numerical integration.